Cosmology

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For Yiren, or anyone who is willing to listen to me ramble.

Hieu

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1

The Problems In Cosmology

"I'm dark and mysterious"

– Approximately 25% of the universe

To be completely honest, I am not entirely sure where to begin. In almost all of my other notes, it's clear that we should start at blah because of blah so that we can eventually do blah. But in this case, cosmology is such a vast subject with many different components that involve all corners of physics; it's hard to say that this is the definitive beginning. Most texts begin with a discussion on general relativity, and we will do GR shortly. There is no avoiding that. I believe that in order to make sense of cosmology, we need to discuss why one might be interested and what the big problems in cosmology are.

1.1 Dark Matter

Let us get the big one out of the way. "Dark matter" nowadays is a buzz word in pop science, and is probably the only thing that most people have heard about other than black holes. We will walk through a very simple argument for the existence of dark matter using purely Newtonian mechanics.

Suppose you have a star of mass m_s located at some radius r away from the galactic center. How would you go about calculating the rotation speed of the star? This is just basic first

year mechanics. We know that Newton's universal gravitation states

$$\boldsymbol{F_g} = -G\frac{M(r)\, m_s}{r^2}\hat{\boldsymbol{r}} \tag{1.1}$$

where $M(r)$ is the mass enclosed by a sphere of radius r (if you're wondering why we don't consider the interactions of the mass outside the enclosed sphere, look up Newton's shell theorem). If the orbit of such a star is circular, then we can say that the centripetal force is given by

$$\boldsymbol{F_c} = -\frac{m_s v^2(r)}{r}\hat{\boldsymbol{r}} \tag{1.2}$$

Setting the 2 equations above equal to each other, we find that

$$v^2(r) = G\frac{M(r)}{r}$$

$$\implies v(r) = \sqrt{G\frac{M(r)}{r}} \tag{1.3}$$

The question now is: what is the mass distribution $M(r)$ in a galaxy? Using data of starlight, we would expect that most of the visible matter in any particular galaxy would be clustered towards the center. In that case, for distances far outside the galactic center, we would expect that $v(r) \propto 1/\sqrt{r}$. However, this is NOT what we observe. Looking at figure 1.1, we

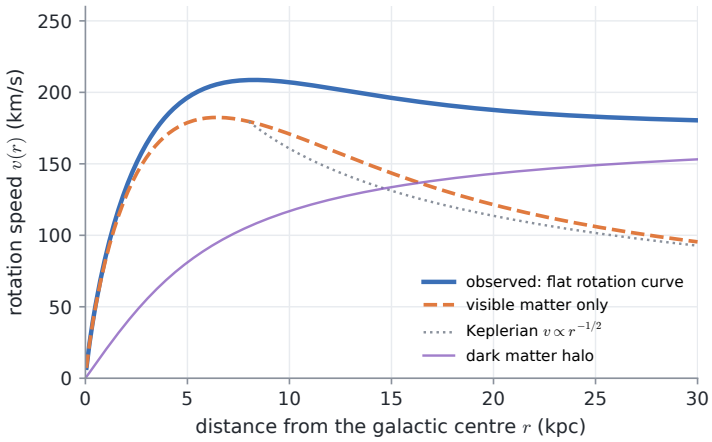


Figure 1.1: A model rotation curve for a Milky Way sized galaxy. If the visible disk were all there is, the curve would peak and then fall off (dashed), eventually like the Keplerian $v \propto r^{-1/2}$ (dotted). What we observe instead is a curve that stays flat far outside the visible disk, which requires a halo of dark matter.

expect to see the dashed line, which eventually falls like $v(r) \propto 1/\sqrt{r}$. But instead, we see a

flat rotation curve. The only way for us to have a flat rotation curve at radii far from the galactic center is if the mass distribution $M(r)$ didn't die out as a function of r but instead grew linearly. But that doesn't make sense because we don't *see* it. This abundance of mass that we can't see is called **dark matter.**

It's tempting to say that maybe our calculation was wrong, or that our observational data is faulty. However, there is one glaring issue: the light of galaxies is being gravitationally distorted. A detailed explanation would require some general relativity and optics, but the simple fact is this: gravity bends light, and mass generates gravity. We see the light of distant galaxies being distorted, implying the existence of an abundance of mass. Like with the rotation curves, we can't see this mass. This is more evidence that dark matter must exist.

1.2 Dark Energy And Inflation

In the early 1900s, astronomers (namely Vesto Slipher) found that the light from distant galaxies was redshifted as it arrived on Earth. This would imply that galaxies were moving away from us. This wouldn't be such a big deal except for the fact that all galaxies were moving away from us as well as each other. This means that the universe is *expanding* (or more accurately, the distances between objects are expanding). Alexander Friedmann was the first to mathematically explain this using Einstein's field equations, which are known today as the **Friedmann equations**. From those years, many astronomers have made observations of cosmological expansion, but the most famous of these would be Edwin Hubble, who found that the more distant galaxies were from us, the faster their speed of recession. Funny enough, Hubble's data at first was...optimistic but as we collected more data, we found that this relation fit snuggly to a line. The slope of this line would tell us the expansion rate of the universe today (around 70 km s^{-1} Mpc^{-1}, with different measurements disagreeing between about 67 and 73, which is its own open problem called the **Hubble tension**), and this would be known as **Hubble's constant** (not really a constant, it's just that the expansion on human scales is so long and large that it might as well be a constant).

Ok so what? The universe is expanding. Well, it turns out, not only is the universe expanding, but its expansion is *accelerating*. There is an analogy that I like, even though it might not be entirely accurate. Imagine you are throwing a ball in the air. In the absence of gravity, we would expect that the ball to just move upwards at constant speed. In the presence of gravity, we would expect that the ball would slow down, eventually stop, and then accelerate back towards the Earth. But now imagine you throw a ball in the presence of gravity, and instead of it going down, it continues to go up, and not only does it continue to go up, it *speeds up* too. This is analogous to the expansion of the universe. We would

expect that with the abundance of mass in the universe, massive objects would eventually clump up together to form massive mega-structures. On small scales, they do. But on the scale of the universe, space's expansion is accelerating, and objects are getting further and further apart. Where is the energy coming from to cause the expansion of the universe to accelerate? Well...We don't know. We call this energy **dark energy**.

1.2.1 The Horizon Problem

Ok, so the universe is expanding and its expansion is accelerating. But there's another problem. Shown in figure 1.2 is (a simulation of) the cosmic microwave background (CMB). It is a picture of the light emitted about 380,000 years after the Big Bang, when the universe first became transparent. Obviously, the temperature of the universe has cooled since then

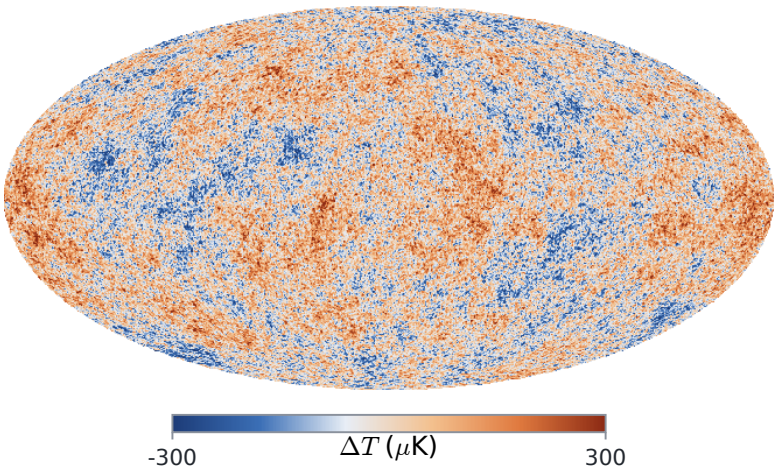


Figure 1.2: A simulated map of the cosmic microwave background over the whole sky, with fluctuations that have the same statistics as the real CMB. The average temperature is 2.7255 K, and the colors show deviations of a few hundred μK, about one part in 10^5.

and that's why the energy of the light has been redshifted to microwave. Here is the glaring issue: the temperature of these ancient photons is uniformly distributed across the universe to one part in 100,000. The reason that this is a problem is because the distances between these photons now are huge and are getting larger faster; therefore, they could not have been in thermal contact with each other to equilibrate. So how is it possible that the temperature of the universe is so uniform? Maybe universal expansion explains it; if the universe is expanding, maybe at some point these photons were close enough to each other to equilibrate, but this can't be the answer either because it did not have enough time to equilibrate (more on this later). So the problem remains: the universe is uniform, implying the CMB photons were once in equilibrium with each other but the accelerating universe

makes that impossible now and at early times, it wouldn't have had enough time. This problem will later be solved by **inflation theory**.

1.3 CP Violation And Baryogenesis

In the 1920s, efforts to reconcile relativity and quantum mechanics began. It was an arduous journey with many, many dead ends and failures. Paul Dirac tried his hand at making a relativistic modification of the Schrodinger equation, and the equations of motion admitted particles with negative energy states. What he had discovered was **antimatter**. Experimental verifications of antimatter were made a few years later.

Here's the interesting thing: there's nothing written in the cosmological stone that says that the universe prefers matter over anti-matter. The laws of physics treat them almost exactly the same, and the symmetry that swaps particles with antiparticles (combined with a mirror reflection) is called CP (charge conjugation and parity). But by brief observation of the fact that we are not dead, there clearly is more matter than anti-matter in the universe. This means that at some point in the early universe, something tipped the balance, and that requires CP to be violated. The Standard Model does violate CP a little bit, but nowhere near enough. Why? We don't know, but the theory of the matter anti-matter asymmetry is called **baryogenesis**.

1.4 Curvature Problem

The universe, as we know it, is flat. What does that mean? We live in 3D space, how can 3D space be flat? Here is some visual intuition. Suppose you take points in space; this could be distant stars. If you were to connect 3 stars by a straight line, you would get a triangle. If the internal angles of the triangle sum to 180, then the geometry of our universe is flat. If the sum of internal angles is NOT 180, then the universe globally has nonzero curvature.

Observational data has shown that the universe is flat. There's nothing crazy about this, but similar to the CP violation, there is nothing written that says that the universe must be flat; it just is. But is it possible that we are wrong? Yes. Perhaps our measurements are not precise enough; perhaps we are not looking on scales large enough.

Obviously, there are many many more problems relating to cosmology; these are the ones that are the most popular to discuss. As we go along, we will develop tools and theories to solve these problems, but there will be some problems that are left unsolved. Our goal is not to explain everything, but to develop the tools to eventually answer these questions.

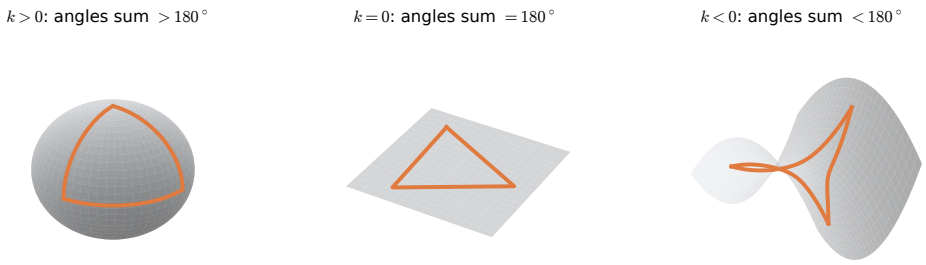


Figure 1.3: Triangles made of straight lines (geodesics) on a positively curved, flat, and negatively curved surface. Only on the flat surface do the internal angles add up to 180°.

1.5 Problems

1. Newton's Shell Theorem

Newton's Shell Theorem states that

(1) a spherically symmetric body affects external objects gravitationally as though all of its mass were concentrated at a point at its center, and
(2) if the body is a spherically symmetric shell (i.e., a hollow ball), no net gravitational force is exerted by the shell on any object inside, regardless of the object's location within the shell.

a). Let us first start by using the universal law of gravitation. For a continuous mass distribution, the force on a test mass m_s at position $\boldsymbol{r}'$ is

$$\boldsymbol{F}_g = -Gm_s \int_V \rho(\boldsymbol{r}) \frac{\boldsymbol{r}' - \boldsymbol{r}}{|\boldsymbol{r}' - \boldsymbol{r}|^3} \, d^3r$$

where $\rho(\boldsymbol{r})$ is the mass density and V is the volume of the mass that we are integrating over. Assuming that the mass density of a thin spherical shell is uniform, show that the force inside the shell is 0 for every $\boldsymbol{r}'$ inside the sphere.

b). Suppose that the density is not constant but still only depends on the radius, $\rho(r)$. Does this result still hold true?

c). Now take $\boldsymbol{r}'$ to be outside the sphere. Show that the gravitational force reduces down to that of 2 point masses separated by some distance d, where d is the distance between the centers of mass of the 2 objects.

d). Using Gauss's law for gravity

$$\oint_S \boldsymbol{g} \cdot d\boldsymbol{A} = -4\pi G M_{\text{enc}}$$

where $\boldsymbol{g}$ is the gravitational field and M_{enc} is the mass enclosed by the surface S, show that you retrieve the exact same results with almost no work.

2. Dark Matter In Galaxies

In the dark matter section, we argued that if the mass distribution dies off at large distances from the galactic center, then the dominating variable is the $1/r$ in (1.3). Now consider a more realistic distribution of the form

$$\rho(r) = \frac{\rho_0}{(1 + r/r_0)^\alpha}$$

a). Derive the rotation curve $v(r)$ for a value of α which gives a flat rotation curve at $r \gg r_0$ as indicated by measurements.

b). At $r = 10^5$ lightyears the measurement yields $v_{\text{meas}} = 225$ km/s, while the visible matter alone predicts $v_{\text{calc}} = 15$ km/s. Calculate the visible as well as the true galaxy mass enclosed within that radius. What is the percentage of dark matter in the galaxy? What is the average dark matter mass density? (Use $G = 6.67 \times 10^{-11}$ m^3 kg^{-1} s^{-2}.)

3. Hubble's Law

a). Show that if the universe expands uniformly, so that every distance is multiplied by the same factor $a(t)$, then every observer sees the other galaxies receding with a speed proportional to their distance, $v = Hd$ with $H = \dot{a}/a$. Why does this mean that being at the "center" of the expansion is meaningless?

b). With $H_0 = 70$ km s^{-1} Mpc^{-1}, compute the **Hubble time** $1/H_0$ in years and the **Hubble distance** c/H_0 in Mpc and in lightyears. These set the rough age and size of the observable universe.

2

General Relativity

"For your homework, you are going to generalize general relativity. Call it 'generaler relativity' "

– Andrew Dotson

One of the hardest parts of writing these notes is figuring out where to begin and where to go. The backbone of cosmology is general relativity, and if we assume that the reader has no background in relativity, then we would be screwed because to responsibly talk about GR, we would need an entire undergraduate education in math and physics just to begin. Unfortunately, I am not interested in making these notes 1000 pages long. That means that I will have to cut corners here and there and our discussion on relativity will not be complete. Perhaps one day in the future I will make my own notes on GR, but until then, we are just going to have to derive only what we will need.

So here is the plan. We are going to follow the route that Sean Carroll takes in *Spacetime and Geometry* [1], just a lot faster. We start in flat spacetime with special relativity, because that is where the language of four-vectors and tensors is the least scary. Then we let the geometry curve, which forces us to invent the metric, the covariant derivative, and the curvature tensors. Only then do we turn on gravity, through the equivalence principle and Einstein's equation, and the very last thing we do is apply all of it to the universe as a whole. If you already know GR, you can skim this chapter and jump to the section on cosmology. If you don't, I would recommend reading it with a pencil, because every single object we build here will show up again in the perturbation theory chapters.

2.1 Special Relativity And Flat Spacetime

Before we can curve anything, we need to be comfortable with flat spacetime. In special relativity, space and time are glued together into a single four-dimensional object called **spacetime**, and a point in it (an **event**) is labeled by four coordinates

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z). \tag{2.1}$$

Greek indices like μ, ν, σ run over all four values $0, 1, 2, 3$, and Latin indices like i, j, k only run over the spatial ones $1, 2, 3$. We will also set $c = 1$ for the rest of the book, so time and distance are measured in the same units (a year and a lightyear are the same thing). The upper index on x^μ is not an exponent; it's a label, and the reason it's upstairs will become clear in the next section.

The central idea of special relativity is that different inertial observers disagree about time intervals and about distances, but they all agree on one combination of the two. For two events separated by $(\Delta t, \Delta x, \Delta y, \Delta z)$, this combination is the **spacetime interval**

$$\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2. \tag{2.2}$$

As much as it absolutely PAINS me to write the negative sign on the time part, this is the convention that relativists (and Carroll) use, so we have to be consistent. We can package the interval using a matrix called the **Minkowski metric**,

$$\Delta s^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu, \qquad \eta_{\mu\nu} = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \tag{2.3}$$

where we have used the **Einstein summation convention**: any index that appears once upstairs and once downstairs is summed over, so $\eta_{\mu\nu}\Delta x^\mu \Delta x^\nu$ secretly has two sums in it. The sign of Δs^2 tells us about causality. If $\Delta s^2 < 0$ the separation is **timelike**, and a massive particle can travel between the two events. If $\Delta s^2 = 0$ it is **null**, and only light can connect them. If $\Delta s^2 > 0$ it is **spacelike**, and nothing can get from one to the other, so they can't influence each other. The set of null directions from an event forms the **light cone**, and it's the light cone, not some absolute notion of simultaneity, that organizes cause and effect.

2.1.1 Lorentz Transformations

The transformations between inertial frames are exactly the ones that leave the interval alone. If we write a linear change of coordinates as $x^{\mu'} = \Lambda^{\mu'}{}_{\nu} x^{\nu}$, then demanding $\eta_{\mu'\nu'} \Delta x^{\mu'} \Delta x^{\nu'} = \eta_{\rho\sigma} \Delta x^{\rho} \Delta x^{\sigma}$ for every separation means

$$\eta_{\rho\sigma} = \Lambda^{\mu'}{}_{\rho} \Lambda^{\nu'}{}_{\sigma} \, \eta_{\mu'\nu'}, \tag{2.4}$$

or in matrix form $\eta = \Lambda^T \eta \Lambda$. Matrices satisfying this are **Lorentz transformations**. They include ordinary rotations of the spatial axes, and they also include **boosts**, which mix time and space. A boost along x with velocity v is

$$\Lambda^{\mu'}{}_{\nu} = \begin{bmatrix} \gamma & -\gamma v & & \\ -\gamma v & \gamma & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2}}. \tag{2.5}$$

A nice way to see that this is a "rotation" in the t-x plane is to define the **rapidity** ϕ by $v = \tanh\phi$, so that $\gamma = \cosh\phi$ and $\gamma v = \sinh\phi$. The boost then has $\cosh\phi$ on the diagonal and $-\sinh\phi$ off the diagonal, which is a rotation with hyperbolic functions instead of circular ones, and the minus sign in η is exactly why the hyperbolic functions show up. Rapidities also add, $\phi_{\text{total}} = \phi_1 + \phi_2$, which is a much cleaner rule than the relativistic velocity addition formula.

2.1.2 Proper Time And Four-Velocity

For a massive particle moving along a path $x^{\mu}(\lambda)$ through spacetime (its **worldline**), the interval between neighbouring points is timelike. The time measured by a clock carried along with the particle is called the **proper time** τ, and it's defined by

$$d\tau^2 = -\eta_{\mu\nu} dx^{\mu} dx^{\nu} = dt^2 \left(1 - v^2\right) \quad \Longrightarrow \quad d\tau = \frac{dt}{\gamma}. \tag{2.6}$$

Since everyone agrees on the interval, everyone agrees on the proper time, which makes it a good parameter for the worldline. The **four-velocity** is the tangent vector to the worldline,

$$U^{\mu} = \frac{dx^{\mu}}{d\tau} = \gamma(1, v^x, v^y, v^z), \tag{2.7}$$

and plugging into the definition of proper time shows that it always has the same "length"

$$\eta_{\mu\nu} U^{\mu} U^{\nu} = -1. \tag{2.8}$$

Multiplying by the rest mass gives the **four-momentum** $p^\mu = mU^\mu = (E, \boldsymbol{p})$, whose time component is the energy $E = \gamma m$, and the normalization becomes the famous $E^2 = \boldsymbol{p}^2 + m^2$. Massless particles don't have a proper time (their interval is null), so we parametrize their worldlines with some other parameter, but they still have a four-momentum with $p^\mu p_\mu = 0$, meaning $E = |\boldsymbol{p}|$.

Problem 1.

For a bougie vacation get-away, you decide to board a rocket ship to God knows where. You instruct the pilot to accelerate so that you constantly feel $g = 9.8$ m/s^2, so you can experience the comfort of Earth's gravity on the rocket. The pilot does this for 10^8 seconds of your (proper) time and then decelerates at the same rate for another 10^8 seconds, so that you come to rest.

a). What is your speed relative to Earth as a function of your proper time?

b). How much time passed on Earth during your trip?

Solution: a). It's tempting to just use $dv/dt = g$, but that can't be right, because for a long enough time it would push us past the speed of light. The trick is that "you feel g" means that in your *instantaneous rest frame*, after a small proper time $d\tau$ your speed has increased by $g\,d\tau$. In terms of rapidity this is easy, because rapidities simply add, so

$$\frac{d\phi}{d\tau} = g \quad \Longrightarrow \quad \phi(\tau) = g\tau \quad \Longrightarrow \quad \boxed{v(\tau) = c\tanh\left(\frac{g\tau}{c}\right)}$$

where we have put c back so that we can plug in numbers. (If you prefer, you can get the same thing by expanding the velocity addition formula $\frac{v+g\,d\tau}{1+vg\,d\tau/c^2}$ to first order, which gives $dv = (1 - v^2/c^2)g\,d\tau$, and integrating.) Your speed approaches c but never reaches it. During the deceleration phase the argument just runs backwards, $v = c\tanh\left(g(2T - \tau)/c\right)$ for $T \le \tau \le 2T$ with $T = 10^8$ s.

b). Earth time and your proper time are related by $dt = \gamma\,d\tau$, and $\gamma = \cosh\phi = \cosh(g\tau/c)$. By symmetry the two halves of the trip contribute the same amount, so

$$\Delta t_{\text{Earth}} = 2\int_0^T \cosh\left(\frac{g\tau}{c}\right) d\tau = \frac{2c}{g}\sinh\left(\frac{gT}{c}\right).$$

With $gT/c = (9.8)(10^8)/(3\times 10^8) \approx 3.27$ and $c/g \approx 3.06\times 10^7$ s, we get $\sinh(3.27) \approx 13.1$ and

$$\boxed{\Delta t_{\text{Earth}} \approx 8.0\times 10^8 \text{ s} \approx 25 \text{ years}}$$

while you only aged 2×10^8 s, or about 6.3 years. Your twin back home is almost 19 years older than you, and you would also have travelled about $2(c^2/g)(\cosh 3.27 - 1) \approx 2.3 \times 10^{17}$ m, or about 24 lightyears.

2.2 Vectors, Dual Vectors, And Tensors

The main players in GR are tensors. We already met two of them in disguise, the four-velocity U^μ and the Minkowski metric $\eta_{\mu\nu}$. When I was younger, I thought that tensors were these mystical objects that constantly evaded my understanding. As it turns out, like most things in physics, they're not mystical; I was just too dumb to understand them. One way that we can discuss them is through the generalization of a vector, and how a vector transforms.

2.2.1 Covariant and Contravariant Vectors

What is a vector? In elementary physics, one would say that a vector is a collection of numbers. This...Is incorrect. Take for example the collection of numbers $[2K, 3m/s, -7m/s, 4N]$. Even though this is a collection of numbers, we would not call this a vector just for the simple fact that not every single component transforms the same way. In the first component, we have temperature which is a scalar quantity (has no direction). In the next 2, we have what looks like to be velocity, which would have magnitude and direction. Likewise with the last component being force. If we were to rotate our coordinate system (turn our heads), every component transforms differently (we'll make this clear in a second). It is how this object transforms under coordinate transformations that makes a vector a vector and a tensor a tensor.

Every single vector can be decomposed as follows

$$\boldsymbol{f} = f^x\ \hat{\boldsymbol{e}}_x + f^y\ \hat{\boldsymbol{e}}_y + f^z\ \hat{\boldsymbol{e}}_z$$

Where the f^i are the vector components and the $\hat{\boldsymbol{e}}_i$ are the basis vectors. My claim is that the basis vectors and the vector components transform differently to each other under coordinate transformations. Let's first look at the vector components. Consider the the following vector

$$d\boldsymbol{r} = dx\ \hat{\boldsymbol{x}} + dy\ \hat{\boldsymbol{y}} + dz\ \hat{\boldsymbol{z}}$$

And then we perform a coordinate change so that we get the vector $d\boldsymbol{r}$ expressed in terms

of a new coordinate system

$$d\boldsymbol{r'} = dx'\hat{\boldsymbol{x}}' + dy'\hat{\boldsymbol{y}}' + dz'\hat{\boldsymbol{z}}'$$

How can we relate $d\boldsymbol{r}'$ in terms of the coordinates of $d\boldsymbol{r}$? Let us just look at the dx' term in $d\boldsymbol{r}'$. From multivariable calculus, we know that a change of variables

$$dx' = \frac{\partial x'}{\partial x}dx + \frac{\partial x'}{\partial y}dy + \frac{\partial x'}{\partial z}dz$$

where it is clear that $x'(x, y, z)$ is a function of the original basis elements x, y and z. We compact the above and say that a single basis element changes as

$$dx' = \frac{\partial x'}{\partial r^\mu}dr^\mu \tag{2.9}$$

and therefore the components of $d\boldsymbol{r}'$ transform as

$$dr'^\mu = \frac{\partial r'^\mu}{\partial r^\nu}dr^\nu \tag{2.10}$$

Any object that that transforms like

$$f'^\mu = \frac{\partial r'^\mu}{\partial r^\nu}f^\nu \tag{2.11}$$

is called a **contravariant vector**. It is denoted with indices superscripted instead of subscripted.

Problem 2.

Convert the vector $\boldsymbol{v} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}}$ from the cartesian basis $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$ into the polar coordinates system (r, ϕ). What is the $\frac{\partial r'^\mu}{\partial r^\nu}$ matrix and write the new coordinates in terms of the old using equation (2.11).

Solution: Suppose the coordinates (x, y). We can parametrize this point in terms of the cartesian basis vectors as $\boldsymbol{v} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}}$ and relate (x, y) using the following transformation

$$x = r\cos(\phi) \tag{2.12}$$

$$y = r\sin(\phi) \tag{2.13}$$

The inverse parametrization would then be

$$r = \sqrt{x^2 + y^2} \tag{2.14}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \tag{2.15}$$

Taking partial derivatives of the inverse parametrization with respect to x and y gives us that the transformation matrix is

$$\boxed{J = \begin{bmatrix} \frac{x}{\sqrt{x^2+y^2}} & \frac{y}{\sqrt{x^2+y^2}} \\ \frac{-y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{bmatrix}}$$

For the initiated, they might remember this to be the **Jacobian matrix**. Once we have the Jacobian, we can relate the vector $\boldsymbol{v}'$ in the polar basis in terms of the old cartesian basis

$$\boldsymbol{v}' = \left(\frac{x}{\sqrt{x^2+y^2}}x + \frac{y}{\sqrt{x^2+y^2}}y\right)\hat{\boldsymbol{r}} + \left(\frac{-y}{x^2+y^2}x + \frac{x}{x^2+y^2}y\right)\hat{\boldsymbol{\phi}}$$

$$\boxed{= \sqrt{x^2+y^2}\ \hat{\boldsymbol{r}}}$$

And this makes sense because the vector $\boldsymbol{v} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}}$ is the radial vector in cartesian coordinates!

Contravariant transformations are usually the transformations that we think of first when we think about coordinate transformations, but they are not the only ones! We can also transform the basis vectors themselves. A very simple example would be if we instead of using the basis vectors $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$, we use $2\hat{\boldsymbol{x}}$ and $-4\hat{\boldsymbol{y}}$. The second example and less obvious example of this would the gradient of a scalar function. The gradient of a scalar function $f(r)$ should not change depending on the coordinate system that we are using. Let us deduce how they must transform.

Suppose we have a gradient in one coordinate system

$$[\nabla f(r)]_\mu \equiv \frac{\partial f(r)}{\partial r^\mu}$$

A scalar function doesn't care what coordinates we use, its value at a point is just its value at that point. So in the new coordinates, the components of the gradient are just the derivatives of the same function with respect to the new coordinates, $\partial f/\partial r'^\mu$. Using the

chain rule, we find that

$$\frac{\partial f(r)}{\partial r'^{\mu}} = \frac{\partial f(r)}{\partial r^{\nu}} \frac{\partial r^{\nu}}{\partial r'^{\mu}} = [\nabla f(r)]_{\nu} \frac{\partial r^{\nu}}{\partial r'^{\mu}} \tag{2.16}$$

This transformation relies on

$$\frac{\partial r^{\nu}}{\partial r'^{\mu}} \tag{2.17}$$

which is the inverse of the transformation that we saw for the contravariant transformation (by the chain rule, $\frac{\partial r'^{\mu}}{\partial r^{\nu}} \frac{\partial r^{\nu}}{\partial r'^{\sigma}} = \delta^{\mu}_{\sigma}$). Such a transformation is called **covariant**, and objects that transform this way are also called **dual vectors** or **one-forms**.

Problem 3.

Find the cartesian gradient $\nabla \equiv \langle \partial_x, \partial_y \rangle$ in terms of polar coordinates.
Solution: We should be a little careful. We want to find the gradient in cartesian coordinates in terms of polar coordinates. Thus, when we use (2.16), we note that we want ∂_x in terms of ∂_r and ∂_ϕ, and thus the r coordinates are actually polar coordinates and the r' are cartesian. This means that the $\frac{\partial r^{\nu}}{\partial r'^{\mu}}$ matrix is the inverse of the one we found in problem 2 (written in polar coordinates).

$$J = \begin{bmatrix} \cos(\phi) & -\frac{\sin(\phi)}{r} \\ \sin(\phi) & \frac{\cos(\phi)}{r} \end{bmatrix}$$

Then using a wee bit of algebra, we find that

$$\partial_x = \cos(\phi)\partial_r - \frac{\sin(\phi)}{r}\partial_\phi \tag{2.18}$$

$$\partial_y = \sin(\phi)\partial_r + \frac{\cos(\phi)}{r}\partial_\phi \tag{2.19}$$

And if we really wanted to, we could write

$$\nabla = \langle \cos(\phi)\partial_r - \frac{\sin(\phi)}{r}\partial_\phi, \sin(\phi)\partial_r + \frac{\cos(\phi)}{r}\partial_\phi \rangle$$

Now that we have an understanding of contravariant and covariant transformations, we make the claim as follows: a **vector** is an object composed with a contravariant component and a covariant component such that under coordinate transformations, it is left unchanged.

To see this, consider once again

$$\boldsymbol{f} = f^x\ \hat{\boldsymbol{e}}_x + f^y\ \hat{\boldsymbol{e}}_y + f^z\ \hat{\boldsymbol{e}}_z$$

We can rewrite this as

$$\boldsymbol{f} = f^\mu \hat{\boldsymbol{e}}_\mu$$

Under a coordinate transformation

$$\boldsymbol{f}' = f'^\mu \hat{\boldsymbol{e}}'_\mu = \frac{\partial r'^\mu}{\partial r^\nu} f^\nu \frac{\partial r^\sigma}{\partial r'^\mu} \hat{\boldsymbol{e}}_\sigma = \delta^\sigma_\nu f^\nu \hat{\boldsymbol{e}}_\sigma = f^\sigma \hat{\boldsymbol{e}}_\sigma = \boldsymbol{f} \tag{2.20}$$

Where we have used (2.11), (2.16), and the inverse property of the transformations in the second and third equality. Intuitively we can think about it like this: imagine you have a pencil in space with the tip facing the door of your room. If you change your coordinate system (say, rotate your head), relative to your eyes, the pencil might be in a new location, but it doesn't change the fact that it is still a pencil in space with the tip pointing at the door.

Problem 4.

One can think of the full gradient operator (not just its components) as an inner product of a vector and its dual

$$\nabla = \frac{\partial}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial}{\partial z}\hat{\boldsymbol{z}} = \langle \partial_x, \partial_y, \partial_z \rangle \cdot \langle \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}} \rangle$$

If this is the case, find the gradient operator in the cylindrical basis.
Solution: It might be a strange thing to refer to $\langle \partial_x, \partial_y, \partial_z \rangle$ as a vector and $\langle \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}} \rangle$ as its dual, but it's not such a crazy notion. We will elaborate on this more in a second. We now want to find the gradient operator above written in the cylindrical basis. To do this, we will need figure out how the dual basis transforms AS WELL AS how the basis vectors transform. Using the result of Problem 3, we can write

$$\langle \partial_x, \partial_y, \partial_z \rangle = \langle \cos(\phi)\partial_r - \frac{\sin(\phi)}{r}\partial_\phi, \sin(\phi)\partial_r + \frac{\cos(\phi)}{r}\partial_\phi, \partial_z \rangle$$

Now what we need is to find how $\langle \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}} \rangle$ transforms. Since this is a contravariant vector, we must use the (2.11) rule. In doing so, we get

$$\hat{\boldsymbol{x}} = \cos(\phi)\hat{\boldsymbol{r}} - \sin(\phi)\hat{\boldsymbol{\phi}}$$

$$\hat{\boldsymbol{y}} = \sin(\phi)\hat{\boldsymbol{r}} + \cos(\phi)\hat{\boldsymbol{\phi}}$$
$$\hat{\boldsymbol{z}} = \hat{\boldsymbol{z}}$$

I should note that any r dependencies drop out because we are working with unit vectors. Technically, if your basis did not have unit norm, you would need to include the r dependence. Now that we have all the ingredients, we can just take the inner product

$$\nabla = \left(\cos(\phi)\partial_r - \frac{\sin(\phi)}{r}\partial_\phi\right)\left(\cos(\phi)\hat{\boldsymbol{r}} - \sin(\phi)\hat{\boldsymbol{\phi}}\right) + \left(\sin(\phi)\partial_r + \frac{\cos(\phi)}{r}\partial_\phi\right)\left(\sin(\phi)\hat{\boldsymbol{r}} + \cos(\phi)\hat{\boldsymbol{\phi}}\right) + \partial_z\hat{\boldsymbol{z}}$$

Fully simplifying, we should get

$$\boxed{\nabla = \partial_r\hat{\boldsymbol{r}} + \frac{1}{r}\partial_\phi\hat{\boldsymbol{\phi}} + \partial_z\hat{\boldsymbol{z}}} \tag{2.21}$$

Which is indeed the gradient in cylindrical coordinates.

2.2.2 From Vectors To Tensors

In the simplest explanation, a tensor is just a multi-index generalization of a vector. Let us explore a 2 dimensional generalization, because from there it's actually pretty obvious what a general tensor would be. A rank 2 tensor is an object an object comprised of covariant and contravariant components such that under a coordinate transformation remains invariant. For example,

$$\boldsymbol{g} = g^{\mu\nu}\hat{\boldsymbol{e}}_\mu \otimes \hat{\boldsymbol{e}}_\nu \tag{2.22}$$

Here we see that the $g^{\mu\nu}$ component is contravariant and the $\hat{\boldsymbol{e}}_\mu$ and $\hat{\boldsymbol{e}}_\nu$ are covariant. The notation might be a bit confusing; basis vectors are not always covariant and components are not always contravariant. Take the case of the gradient where the ∂_μ are the covariant and the basis vectors $\langle\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}\rangle$ are contravariant. How, then, should we denote when a vector is contravariant and when it is covariant? The first is the position of the indices. As the saying goes; "co goes down." The second would be to use ∂_μ to denote a covariant vector and dr^μ to denote a contravariant vector. Like I said, it might be weird to denote vectors this way, but it is actually very intuitive, and it comes from the notion of a differential. We

know from multivariable calculus that the total derivative of a function can be written as

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

Another way to think about this is to rewrite this as an inner product

$$df = \langle \partial_x, \partial_y, \partial_z \rangle \cdot \langle dx, dy, dz \rangle$$

In this way, we see that the total derivative is nothing more than the dot product of the gradient with the differential. We know from the beginning of the section that the differentials dr^μ transform contravariantly and we know from the part afterwards that the gradient transforms covariantly. It is appropriate, then, to say that the notation for a covariant vector and a contravariant vector should be ∂_μ and dr^μ respectively. When we do this, a 2 tensor can be written as some combination of ∂_μ and dr^μ together with the tensor components g. For example, we might have this

$$\boldsymbol{g} = g_\mu^{\ \nu}\, dr^\mu \otimes \partial_\nu$$

How does something like this transform under coordinate transformations? Exactly how each component would change!

$$\boldsymbol{g'} = g'_\mu{}^\nu\, dr'^\mu \otimes \partial'_\nu = g_\mu^{\ \nu} \frac{\partial r'^\mu}{\partial r^\lambda} dr^\lambda \frac{\partial r^\sigma}{\partial r'^\nu} \partial_\sigma$$

And thus we see that the components of this 2 tensor transform as

$$g'_\mu{}^\nu = g^\sigma_\lambda \frac{\partial r'^\nu}{\partial r^\sigma} \frac{\partial r^\lambda}{\partial r'^\mu} \tag{2.23}$$

This is exactly what makes a tensor a tensor. So when physicists say that a tensor is an object that transform like a tensor, they mean exactly the transformation rule above. Obviously, this is for 1 case of a (1,1) tensor, but you just have to remember the transformation rule for the covariant and the contravariant part. Now that we understand how a 2 tensor is composed and how it transforms under coordinate transformations, a general N tensor is just

$$T = T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} \frac{\partial}{\partial x^{\mu_1}} \dots \frac{\partial}{\partial r^{\mu_r}} dr^{\nu_1} \dots dr^{\nu_s} \tag{2.24}$$

and under coordinate transformations, it transforms as

$$T'^{\mu_1 \dots \mu_n}{}_{\nu_1 \dots \nu_s} = \frac{\partial r'^{\mu_1}}{\partial r^{\rho_1}} \dots \frac{\partial r'^{\mu_n}}{\partial r^{\rho_n}} \frac{\partial r^{\sigma_1}}{\partial r'^{\nu_1}} \dots \frac{\partial r^{\sigma_s}}{\partial r'^{\nu_s}} T^{\rho_1 \dots \rho_n}{}_{\sigma_1 \dots \sigma_s} \tag{2.25}$$

Problem 5.

Using (2.25), argue that the Levi-Civita symbol ϵ^{ijk} is NOT a tensor (even though we might call it the Levi-Civita tensor)

Solution: If the Levi-Civita symbol was truly a tensor, then under a coordinate transformation, we would expect

$$\bar{\epsilon}^{\,ijk} = \frac{\partial r'^i}{\partial r^\alpha}\frac{\partial r'^j}{\partial r^\beta}\frac{\partial r'^k}{\partial r^\gamma}\epsilon^{\alpha\beta\gamma}$$

And this is true for all coordinate transformations. The right hand side is something we can actually evaluate. Contracting any matrix M with $\epsilon^{\alpha\beta\gamma}$ like this antisymmetrizes it, and the result is

$$\frac{\partial r'^i}{\partial r^\alpha}\frac{\partial r'^j}{\partial r^\beta}\frac{\partial r'^k}{\partial r^\gamma}\epsilon^{\alpha\beta\gamma} = \det\left(\frac{\partial r'}{\partial r}\right)\epsilon^{ijk},$$

which is just the definition of the determinant written with indices. Let us, then, consider a reflection about some plane, for example $x \to -x$. Its Jacobian has determinant -1, so if ϵ were a tensor we would need

$$\bar{\epsilon}^{\,ijk} = (-1)\epsilon^{ijk}.$$

However, the Levi-Civita symbol is *defined* to have the same values ($+1$, -1 or 0) in every coordinate system, so $\bar{\epsilon}^{\,ijk} = \epsilon^{ijk}$. The two can't both be true, so the Levi-Civita symbol is NOT a tensor. It picks up an extra factor of the Jacobian determinant, and objects like this are called **tensor densities** (or pseudo-tensors). For a general (non-orthogonal) transformation the determinant doesn't even have to be ± 1, which is why in curved space one defines the honest tensor $\sqrt{|g|}\,\epsilon_{\mu\nu\rho\sigma}$ instead.

2.2.3 The Metric Tensor

Earlier I said that the main players in general relativity are tensors, and there's no tensor more important than the metric tensor. What is the point of the metric tensor? What even is it? The metric is an object that encodes how to find lengths and angles on the geometry that you are working with. Remember all the way back in multivariable calculus, we started with flat geometry and said that we can encode the lengths and angles of vectors on this flat surface using the dot product. You might even have defined the dot product to be

$$\boldsymbol{v}\cdot\boldsymbol{u} = v_\mu u^\mu$$

As it turns out, this is a very specific case, and is only true if the geometry is flat. We could get the same result if wedged the identity in the above

$$\boldsymbol{v} \cdot \boldsymbol{u} = \delta_{\mu\nu} v^\nu u^\mu$$

How might we generalize this for an arbitrary geometry? Well, we still want to find the lengths and angles of the vectors $\boldsymbol{u}$ and $\boldsymbol{v}$, so we can't really change that. The only thing that we can really change in the above this the $\delta_{\mu\nu}$, and that's exactly what we are going to do. We can define the inner product between 2 vectors on a manifold to be

$$\boldsymbol{v} \cdot \boldsymbol{u} = g_{\mu\nu} v^\nu u^\mu$$

We call $g_{\mu\nu}$ the **metric** and our claim is that all the information of the geometry is stored in the metric. Because general relativity is a theory relating the geometry of spacetime to the matter and energy of the universe, we would expect that the metric is the object that stores the information about spacetime itself. For example, suppose we are working in flat Euclidean space. We can find that the line differential line element is given by

$$ds^2 = g_{ij} dr^i dr^j = dx^2 + dy^2 + dz^2 \tag{2.26}$$

and the spacetime line element of special relativity is the infinitesimal version of the interval, $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = -dt^2 + dx^2 + dy^2 + dz^2$, so in flat spacetime and Cartesian coordinates the metric is just the Minkowski metric (2.3).

2.2.3.0.1 Properties of the metric. We've been talking about the metric $g_{\mu\nu}$, but not much about the **metric tensor**. What exactly is the tensor? Well, the tensor is

$$\boldsymbol{g} = g_{\mu\nu} dx^\mu \otimes dx^\nu \tag{2.27}$$

Sometimes people use $g_{\mu\nu}$ and $\boldsymbol{g}$ interchangeably, similar to how we might call u_ν a vector. What properties can we derive from this? Well, immediately we see that the metric tensor must be symmetric. Why? Well, the definition of a symmetric tensor is if you swap the indices, the object remains unchanged: $g_{\mu\nu} = g_{\nu\mu}$. Well, in this case, if we swap the indices

$$g_{\nu\mu} dx^\mu \otimes dx^\nu$$

The tensor product commutes, and therefore we still get the same object $\boldsymbol{g}$ even after permutations of the indices. Thus, the metric tensor is a symmetric tensor. Why might this matter to us? Well, it matters because symmetries puts restrictions on the number of

degrees of freedom. Later on, we will see that the Einstein's equations are tensor differential equations, and it might help us a great deal if we could reduce the number of independent terms.

Problem 6.

Show that a symmetric 2 tensor in n dimensions must necessarily have $\frac{1}{2}(n)(n+1)$ independent entries.

Solution: Let $g_{\mu\nu}$ be a symmetric tensor in n dimensions. The definition of a symmetric tensor imposes that $g_{\mu\nu} = g_{\nu\mu}$. This is just a basic counting problem. How many ways can we choose μ and ν (with replacement) such that order doesn't matter? This one is a little bit tricky just because of the replacement, but no matter, we can always break this up into 2 pieces: the first piece is when μ and ν are not the same and the second is when μ and ν are the same (the second part is due to us replacing). Let us answer the first: how many ways can we choose μ and ν such that they are not the same? Using the multiplication rule, there are n ways to choose the first index, and $n-1$ ways to choose the second. However; we double counted (because $\mu = 1$ and $\nu = 3$ is the same as $\mu = 3$ and $\nu = 1$). So we have to divide by 2. The first piece is $\frac{1}{2}(n)(n-1)$. As for the second piece: how many ways can we have the index be the same? Well that's just n (corresponding to the diagonals). When we add these up, we get

$$\boxed{\frac{n(n-1)}{2} + n = \frac{n(n-1)+2n}{2} = \frac{n(n+1)}{2}}$$

Problem 6 tells us if we live in 4 dimensions, 3 spatial and 1 temporal (which we do), then the metric must necessarily produce at most 10 independent terms.

Another important property of the metric is that it acts to raise and lower an index. For example, if we had a vector v^μ, we can lower its index to v_μ using the metric. How? Suppose we have a vector v^μ, if we act on it with the metric, we get

$$u_\nu = g_{\mu\nu} v^\mu$$

We get a vector u_ν with an index ν on the bottom (because the μ index is summed over, and so the only free index is the covariant ν). This new object is a dual vector, and it is so naturally attached to v^μ that we just give it the same name, $v_\nu \equiv g_{\mu\nu} v^\mu$. It's built so that contracting it with v^ν gives the dot product of $\boldsymbol{v}$ with itself, $v_\nu v^\nu = g_{\mu\nu} v^\mu v^\nu$. In flat space with Cartesian coordinates the metric is the identity (or η), so lowering an index does

nothing (or flips the sign of the time component), which is why in elementary physics we never had to care about the difference. What this means is that we can transform a vector from being contravariant to covariant and back by slapping it with the metric. This raising and lowering indices also brings about another property of the metric. Suppose we lower the index of a vector v^μ

$$v_\nu = g_{\mu\nu} v^\mu \tag{2.28}$$

We can raise it back up again with the metric with all upper indices by the same argument

$$g^{\mu\nu} v_\nu = v^\mu \tag{2.29}$$

Combining the two, we get

$$v^\mu = g^{\mu\sigma} g_{\sigma\rho} v^\rho \implies g^{\mu\sigma} g_{\sigma\rho} = \delta^\mu_\rho$$

Which means that the metric with all upper indices is the inverse of the metric with all lower indices! This property of the metric will become *extremely* useful in the future.

Problem 7.

Let

$$\boldsymbol{X}^{\mu\nu} = \begin{bmatrix} 2 & 0 & 1 & 0 \\ -1 & 0 & 3 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad g_{\mu\nu} = \begin{bmatrix} -1 & & & \\ & a^2 & & \\ & & a^2 & \\ & & & a^2 \end{bmatrix}$$

Find $\boldsymbol{X}_\mu^{\ \mu}$

Solution: One might be tempted to say that $\boldsymbol{X}_\mu^{\ \mu}$ is the trace $\boldsymbol{X}$ and say that the answer is 2, but that's not entirely correct. Remember, all these operations that we are doing is with respect to the metric $g_{\mu\nu}$, and thus if we have a tensor $\boldsymbol{X}^{\mu\nu}$, we need to lower the ν index. To do this, we slap it with the metric with all lowered indices

$$\boldsymbol{X}_\mu^{\ \mu} = \boldsymbol{X}^{\mu\nu} g_{\mu\nu}$$

Do not be mistaken; the above is NOT matrix multiplication! It is element-wise multiplication, and then summed up. Since the only nonzero entries of the metric are on the diagonals, we will only look at the diagonals for both tensors. However, for $\boldsymbol{X}$, the only nonzero entry on the diagonal is the first entry, so we just multiply the first

entry of $\boldsymbol{X}$ and the first entry of the metric and find that

$$\boxed{\boldsymbol{X}_{\mu}^{\ \mu} = -2}$$

All these properties of the metric, and yet we have neglected the most important property of all: coordinate invariance. Because the metric tensor is a tensor, it doesn't matter what coordinate system we are using; it will still describe the same thing. For example, if we had the flat Minkowski metric

$$g_{\mu\nu} = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

in cartesian coordinates (dx^2, dy^2, dz^2), we would get

$$g_{\mu\nu} dx^{\mu} dx^{\nu} = -dt^2 + d\boldsymbol{x}^2$$

Obviously. But if we were to do a coordinate transformation into spherical coordinates

$$x = r\cos(\phi)\sin(\theta) \tag{2.30}$$

$$y = r\sin(\phi)\sin(\theta) \tag{2.31}$$

$$z = r\cos(\theta) \tag{2.32}$$

The metric isn't just $ds^2 = -dt^2 + dr^2 + d\theta^2 + d\phi^2$, there are more factors that are encoded in the metric. If we find the differentials of dx, dy, and dz above, squaring them and then adding them together, we would find that

$$-dt^2 + dx^2 + dy^2 + dz^2 = -dt^2 + dr^2 + r^2 d\theta^2 + r^2\sin^2(\theta) d\phi^2$$

You can double check this using Mathematica, or by hand, since all the cross terms cancel. Anyways, we now see the metric under this coordinate change is

$$g_{\mu\nu} = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & r^2 & \\ & & & r^2\sin^2(\theta) \end{bmatrix} \tag{2.33}$$

Although 2 observers using 2 different coordinate systems might not agree on the same points, they will always agree on ds^2, and that is what we mean when we say that the metric

tensor is coordinate invariant.

Problem 8.

What is the difference between the metric of flat space in spherical coordinates vs. the metric of an actual 2-sphere ($\mathbb{S}^2$)?
Solution: Suppose we have an actual sphere embedded in 3 space $\mathbb{R}^3$. We know that the parametrization of the surface is

$$\begin{aligned} x &= R_0 \cos(\phi)\sin(\theta) \\ y &= R_0 \sin(\phi)\sin(\theta) \\ z &= R_0 \cos(\theta) \end{aligned}$$

Then when we take differentials, square them, and add them up, we find that

$$\boxed{ds^2 = R_0^2\, d\theta^2 + R_0^2 \sin^2(\theta) d\phi^2} \tag{2.34}$$

What is the difference? For flat space in spherical coordinates, there is a dr^2 component, and the space is still flat; we just chose curvy coordinates on it. For the 2-sphere there is no dr^2 component, because the radius of the sphere does not change, and the leftover two-dimensional space is honestly curved. No change of coordinates can turn (2.34) into $dx^2 + dy^2$, even over a small patch (try drawing a map of the Earth without distorting anything). Telling these two situations apart, curvy coordinates versus curvy space, is exactly what the curvature tensors will do for us later in this chapter.

2.2.4 The Energy-Momentum Tensor

There is one more tensor that we need from flat spacetime, and it will turn out to be the thing that tells spacetime how to bend. But what is the energy momentum tensor? There are 2 ways to answer this question. The first and most straightforward is: it is the conserved Noether current for a field that is (spacetime) translationally invariant. In much the same way that momentum and energy are conserved for a particle Lagrangian that is space and time translationally invariant, spacetime translation invariance of a field Lagrangian gives rise to the conservation of the energy momentum tensor. This description is deeply unsatisifying. The other way to say it is that the energy momentum tensor is a symmetric rank $(2,0)$ (or, lowering indices, $(0,2)$) tensor that encodes the density and flux of energy and momentum at a given point. Basically it just tells you how much energy and

momentum stuff has and what is doing at any given point. It is given by

$$T_{\mu\nu} \tag{2.35}$$

The T_{00} component is the energy density as measured by an observer, T_{0i} are the momentum densities in the spatial directions, and the T_{ij} are the stress and shears. Some basic properties is that it is typically symmetric and obeys the continuity equation

$$\nabla_\mu T^{\mu\nu} = 0 \tag{2.36}$$

In flat spacetime the covariant derivative ∇_μ is just ∂_μ (we will define ∇_μ properly in a few sections), and the $\nu = 0$ component is the tensorial analogue of the continuity equation that we are familiar with in electrodynamics

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \boldsymbol{J} \tag{2.37}$$

The question that we are interested in is: what kind of energies are allowed in our universe? We will see later that, with the constraints of isotropy and homogeneity, the only fluids that are allowed are called **perfect fluids**. A perfect fluid is a fluid that has no viscosity, heat conductivity, nor shear stresses (otherwise it would mean there is a preferred direction and that's not possible in our super-symmetric universe). In the rest frame of the fluid, we should expect that the fluid's 4 velocity is

$$U^\mu = \langle 1, 0, 0, 0 \rangle$$

in which case the energy momentum tensor will just reduce down to (remembering that there are no preferred directions)

$$T_{\mu\nu} = \begin{bmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{bmatrix}$$

In this frame, in order to get $T_{\mu\nu}$, we would need to construct

$$T_{\mu\nu} = (\rho + P) U_\mu U_\nu + P \eta_{\mu\nu}$$

It's fairly easy to double-check that this gives the correct result. In the fluid's rest frame, the only nonzero contribution from the 4 velocities is in the $T_0 0$ component. To get the pressure along the diagonal, we just add it with the Minkowski metric. Once we allow curved

spacetime later on, the same expression holds with $\eta_{\mu\nu}$ replaced by a general metric (this is the principle of general covariance, which we will meet properly when we talk about gravity)

$$T_{\mu\nu} = (\rho + P)U_\mu U_\nu + P g_{\mu\nu} \tag{2.38}$$

Oftentimes for convenience, we will use the energy momentum tensor with 1 up 1 down indices

$$T^\mu{}_\nu = g^{\mu\lambda} T_{\lambda\nu} = \begin{bmatrix} -\rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{bmatrix} \tag{2.39}$$

2.3 Curved Spacetime

Everything that we have done so far is really with respect to a flat metric, even when we used curvy coordinates. But as we know, space is not always flat. The surface of the Earth is a curved two-dimensional space, and the presence of gravity, as we will see, demands that spacetime itself must curve. The mathematical setting for a curved space is a **manifold**. I won't give the formal definition (Carroll spends a whole chapter on it), but the idea is simple: a manifold is a space that might be curved and twisted on large scales, but that looks like ordinary flat $\mathbb{R}^n$ if you zoom in far enough. The Earth looks flat from your backyard, which is why maps of your neighbourhood work fine even though a map of the whole globe can't.

On a manifold, we can still put down coordinates x^μ (at least on patches), and everything we said about vectors, dual vectors and tensors goes through unchanged, as long as we only ever talk about objects at a single point. The transformation rules (2.11), (2.16) and (2.25) hold with $\partial x'^\mu / \partial x^\nu$ now being the Jacobian of any smooth change of coordinates, not just a Lorentz transformation. The geometry is stored in a metric $g_{\mu\nu}(x)$ that can now depend on position,

$$ds^2 = g_{\mu\nu}(x)\, dx^\mu dx^\nu, \tag{2.40}$$

which we take to be symmetric, invertible, and with one negative and three positive eigenvalues (so it has the same signature as η).

The "looks flat if you zoom in" statement can be made precise. At any point p, we can always find coordinates (called **locally inertial** or **Riemann normal coordinates**) in which

$$g_{\mu\nu}(p) = \eta_{\mu\nu}, \qquad \partial_\sigma g_{\mu\nu}(p) = 0. \tag{2.41}$$

The metric and its first derivatives can be made to look exactly like special relativity at a

point. What we can't do, in general, is also make the *second* derivatives vanish. The leftover second derivatives are what curvature is, and the rest of this section is about building a tensor that captures them. Counting tells you why this has to be true. The Jacobian $\partial x'^{\mu}/\partial x^{\nu}$ has 16 numbers to play with, which is more than enough to set the 10 components of $g_{\mu\nu}$ to $\eta_{\mu\nu}$. Its first derivatives $\partial^2 x'^{\mu}/\partial x^{\nu}\partial x^{\sigma}$ have $4 \times 10 = 40$ components, which is exactly enough to kill the $10 \times 4 = 40$ first derivatives of the metric. But the second derivatives of the Jacobian have $4 \times 20 = 80$ components, while the second derivatives of the metric have $10 \times 10 = 100$. The 20 that are left over can't be removed by any choice of coordinates, and we'll see that the Riemann tensor has exactly 20 independent components in four dimensions.

2.3.1 The Covariant Derivative

In flat space with Cartesian coordinates, taking derivatives of vectors is easy, you just take the derivative of each component. On a manifold (or even in flat space with curvy coordinates) this goes wrong, because derivatives of tensors don't transform like tensors. This is apparent in the line that looks something like

$$\frac{\partial}{\partial x^{\mu}}\left(v^{\nu}\right) = \frac{\partial x'^{\alpha}}{\partial x^{\mu}}\frac{\partial}{\partial x'^{\alpha}}\left(\frac{\partial x^{\nu}}{\partial x'^{\beta}}v'^{\beta}\right) = \frac{\partial x'^{\alpha}}{\partial x^{\mu}}\frac{\partial x^{\nu}}{\partial x'^{\beta}}\partial'_{\alpha}v'^{\beta} + \frac{\partial x'^{\alpha}}{\partial x^{\mu}}\frac{\partial^2 x^{\nu}}{\partial x'^{\alpha}\partial x'^{\beta}}v'^{\beta}$$

Where if we had a vector v^{μ} as function of x, we transform the vector to be in terms of a new coordinate system x', then we get the above. As we can see from the last equality, the first term is what we would expect from our discussion of how tensors transform. However, the chain rule also admitted an extra term on the right. The presence of this term is what makes the regular derivative of a tensor not a tensor because, simply put, it just doesn't transform like one.

The extra term came from the fact that the basis vectors themselves change from point to point, and the ordinary derivative doesn't know about that. So we want a new derivative operator, the **covariant derivative** ∇_{μ}, which does transform like a tensor. The natural guess is to take the ordinary derivative and correct it with a term that is linear in the vector,

$$\nabla_{\nu}V^{\mu} = \partial_{\nu}V^{\mu} + \Gamma^{\mu}_{\nu\lambda}V^{\lambda}, \tag{2.42}$$

where the correction coefficients $\Gamma^{\mu}_{\nu\lambda}$ are called the **connection coefficients**. They are *not* the components of a tensor. In fact, they have to transform in a non-tensorial way precisely so that their bad term cancels the bad term in $\partial_{\nu}V^{\mu}$ above. There are many connections that do this job, and to pick out a unique one we make two very natural demands:

(1) **No torsion**: $\Gamma^{\mu}_{\nu\lambda} = \Gamma^{\mu}_{\lambda\nu}$ (symmetric in the lower indices).
(2) **Metric compatibility**: $\nabla_\lambda g_{\mu\nu} = 0$, which means that lengths and angles don't change when we move vectors around. Raising and lowering indices then commutes with taking derivatives.

Writing out the second condition using the rule for dual vectors below, we find

$$\partial_\lambda g_{\mu\nu} = \Gamma^{\rho}_{\lambda\mu} g_{\rho\nu} + \Gamma^{\rho}_{\lambda\nu} g_{\mu\rho}. \tag{2.43}$$

Now we play a little game. Write this equation three times with the indices cyclically permuted, add the first two, and subtract the third. Using the symmetry of Γ in its lower indices, almost everything cancels,

$$\begin{aligned}\partial_\lambda g_{\mu\nu} + \partial_\mu g_{\lambda\nu} - \partial_\nu g_{\mu\lambda} &= \left(\Gamma^{\rho}_{\lambda\mu} g_{\rho\nu} + \Gamma^{\rho}_{\lambda\nu} g_{\rho\mu}\right) + \left(\Gamma^{\rho}_{\mu\lambda} g_{\rho\nu} + \Gamma^{\rho}_{\mu\nu} g_{\rho\lambda}\right) - \left(\Gamma^{\rho}_{\nu\mu} g_{\rho\lambda} + \Gamma^{\rho}_{\nu\lambda} g_{\rho\mu}\right) \\ &= 2\Gamma^{\rho}_{\lambda\mu} g_{\rho\nu}. \end{aligned} \tag{2.44}$$

Multiplying by the inverse metric gives the unique torsion-free, metric-compatible connection, called the **Christoffel symbols**,

$$\boxed{\Gamma^{\rho}_{\alpha\beta} = \frac{1}{2} g^{\rho\sigma} \left(\partial_\beta g_{\alpha\sigma} + \partial_\alpha g_{\beta\sigma} - \partial_\sigma g_{\alpha\beta}\right)} \tag{2.45}$$

So the connection is completely determined by the metric and its first derivatives. In locally inertial coordinates (2.41) the first derivatives of the metric vanish, so $\Gamma = 0$ at that point and $\nabla_\mu = \partial_\mu$ there. That's the precise sense in which curved space looks flat if you zoom in. We have been a little out of order, because we used the rule for dual vectors to write down (2.43). Demanding that ∇ obeys the product rule and reduces to ∂ on scalars ($\nabla_\nu(\omega_\mu V^\mu) = \partial_\nu(\omega_\mu V^\mu)$) forces the covariant derivative acting on co-vectors to take the form

$$\nabla_\nu \omega_\mu = \partial_\nu \omega_\mu - \Gamma^{\lambda}_{\nu\mu} \omega_\lambda \tag{2.46}$$

For the full derivative of a general tensor $T^{\mu_1\mu_2\cdots}_{\nu_1\nu_2\cdots}$, we see that

$$\nabla_\lambda T^{\mu_1\mu_2\cdots}_{\nu_1\nu_2\cdots} = \partial_\lambda T^{\mu_1\mu_2\cdots}_{\nu_1\nu_2\cdots} + \sum_i \Gamma^{\mu_i}_{\lambda\sigma} T^{\mu_1\ldots\sigma\cdots}_{\nu_1\nu_2\cdots} - \sum_j \Gamma^{\sigma}_{\lambda\nu_j} T^{\mu_1\mu_2\cdots}_{\nu_1\cdots\sigma\ldots} \tag{2.47}$$

The pattern is: one $+\Gamma$ for every upper index and one $-\Gamma$ for every lower index. As Baumann says, "we have to sacrifice beauty for speed," and a proper general relativity course will cover this in more detail!

Problem 10.

Let us do a basic calculation. Suppose we have a vector

$$V^\mu = \langle 0, t^2, 0, 0 \rangle$$

with the metric being the flat FRW metric $ds^2 = -dt^2 + a^2(t)(dx^2 + dy^2 + dz^2)$ that we will meet at the end of this chapter. Find the covariant derivative $\nabla_t V^x$. (You may use the result of Problem 9 below for the Christoffel symbols.)

Solution: First, since we have a vector, we should be using the rule for vectors, which becomes

$$\nabla_t V^\mu = \partial_t V^\mu + \Gamma^\mu_{t\lambda} V^\lambda$$

but the only nonzero component of V^μ is V^x and so the above can just be written as

$$\nabla_t V^x = \partial_t V^x + \Gamma^x_{tx} V^x$$

Since the background metric is the FRW metric, we can just use the result from problem 9 to get

$$\Gamma^i_{tj} = H(t)\delta^i_j$$

$$\begin{aligned} \implies \nabla_t V^x &= \partial_t V^x + H(t)\delta^x_x V^x \\ &= \boxed{2t + t^2 H(t)} \end{aligned}$$

I should note that although the vector only has an x component, the full covariant derivative $\nabla_\nu V^\mu$ has more than one nonzero component. For example, $\nabla_x V^t = \Gamma^t_{xx} V^x = a\dot{a}\, t^2$, because moving in the x direction in an expanding universe tilts the vector into the time direction. Doing these derivatives are a bit annoying, so we can just use EinsteinPy to get the full covariant derivative if we wanted to (I paste the code below)

```
import sympy as sp
from einsteinpy.symbolic import MetricTensor, ChristoffelSymbols

# Define symbols and coordinates.
syms = sp.symbols('t x y z')
t, x, y, z = syms
coords = syms
```

```
8
9  # Define the scale factor a(t) as a symbolic function.
10 a = sp.Function('a')(t)
11 # Define the FRW metric (spatially flat in Cartesian coordinates).
12 # ds^2 = -dt^2 + a(t)^2*(dx^2 + dy^2 + dz^2)
13 g = sp.Matrix([
14     [-1,       0,       0,       0],
15     [ 0, a**2,       0,       0],
16     [ 0,       0, a**2,       0],
17     [ 0,       0,       0, a**2]
18 ]).tolist()
19
20 # Create a MetricTensor object.
21 metric = MetricTensor(g, syms)
22 # Define our vector field V^μ = (0, t^2, 0, 0).
23 V = sp.Matrix([0, t**2, 0, 0]).tolist()
24 chi = ChristoffelSymbols.from_metric(metric).tensor()
25 # Set up an array to hold the components of the covariant derivative.
26 # The covariant derivative has two indices:
27 # mu labels the component of the vector and nu labels the direction of differentiation.
28 n = 4  # number of dimensions
29 cov_deriv = sp.MutableDenseNDimArray(sp.zeros(n, n), (n, n))
30
31 # Compute the covariant derivative:
32 for mu in range(n):
33     for nu in range(n):
34         # Partial derivative with respect to coordinate nu.
35         expr = sp.diff(V[mu][0], coords[nu])
36         # Add the connection piece: sum_lam Gamma^mu_{nu lam} * V^lam.
37         for lam in range(n):
38             expr += chi[mu, nu, lam] * V[lam][0]
39         cov_deriv[mu, nu] = sp.simplify(expr)
40
41 cov_deriv
```

The output (rows labelled by μ, columns by ν) will be

$$\begin{bmatrix} 0 & t^2 a(t) \frac{d}{dt} a(t) & 0 & 0 \\ \frac{t^2 \frac{d}{dt} a(t)}{a(t)} + 2t & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Problem 9.

Although these calculations are ugly, it's good to get your hands a little dirty from time to time. Calculate the Γ^i_{0j} component of the Christoffel symbol for the FRW metric

$$ds^2 = -dt^2 + a^2(t)\gamma_{ij}dx^i dx^j$$

where γ_{ij} is a 3×3 matrix which depends only on the spatial coordinates.
Solution: We will use (2.45) with $\rho = i$, $\alpha = 0$ and $\beta = j$,

$$\Gamma^i_{0j} = \frac{1}{2} g^{i\sigma} \left(\partial_j g_{0\sigma} + \partial_0 g_{j\sigma} - \partial_\sigma g_{0j} \right).$$

The metric has no time-space cross terms, so $g^{i\sigma}$ is only nonzero when σ is a spatial index k, and $g_{0k} = 0$. That kills the first and third terms in the parentheses. The time derivative in the second term will only act on the $a^2(t)$, which gives

$$\frac{d}{dt} a^2(t) = 2a\dot{a} = 2a^2 H$$

Where $H(t)$ is the Hubble parameter and is defined as $H \equiv \dot{a}/a$. This leaves us with

$$\begin{aligned} \Gamma^i_{0j} &= \frac{1}{2} g^{ik}\, \partial_0 g_{jk} \\ &= \frac{1}{2} g^{ik} \left(2a^2 H \right) \gamma_{jk}. \end{aligned}$$

All that is left is to find g^{ik}, which is just the inverse of the spatial part of the metric, $g^{ik} = a^{-2}\gamma^{ik}$, where γ^{ik} is the matrix inverse of γ_{ik}. Subbing this back in, $\gamma^{ik}\gamma_{jk} = \delta^i_j$ and

$$\boxed{\Gamma^i_{0j} = H(t)\delta^i_j}$$

The same kind of calculation gives the only other nonzero components involving time, $\Gamma^0_{ij} = a^2 H \gamma_{ij} = a\dot{a}\gamma_{ij}$, while $\Gamma^0_{00} = \Gamma^0_{0i} = \Gamma^i_{00} = 0$. The purely spatial ones Γ^i_{jk} are the Christoffel symbols of γ_{ij} alone, and they vanish for flat space in Cartesian coordinates.

2.3.2 Parallel Transport And Geodesics

In flat space, moving a vector from one point to another without changing it is trivial: keep its components the same. In curved space, "the same" has to be defined, and the covariant derivative is what defines it. A vector V^μ is **parallel transported** along a curve $x^\mu(\lambda)$ if

its covariant derivative along the curve vanishes,

$$\frac{dx^\nu}{d\lambda}\nabla_\nu V^\mu = \frac{dV^\mu}{d\lambda} + \Gamma^\mu_{\nu\sigma}\frac{dx^\nu}{d\lambda}V^\sigma = 0. \tag{2.48}$$

On a sphere this has a surprising consequence. Take a vector at the north pole pointing along a line of longitude, carry it down to the equator, along the equator a quarter of the way around, and back up to the pole, keeping it "as parallel as possible" the whole time. When it comes back, it has rotated by 90°. In flat space this never happens. The fact that parallel transport around a closed loop can change a vector is the cleanest signal that a space is curved, and we will use it to define curvature in a second.

A **geodesic** is the curved-space version of a straight line: it's a curve whose tangent vector is parallel transported along itself, so it never turns. With the tangent vector $U^\mu = dx^\mu/d\tau$ (the four-velocity), the condition is $U^\nu\nabla_\nu U^\mu = 0$, and we can write it out. If we take the covariant derivative

$$\nabla_\nu U^\mu = \partial_\nu U^\mu + \Gamma^\mu_{\nu\lambda}U^\lambda$$

and contract it with U^ν, the first term becomes a derivative along the curve by the chain rule, $U^\nu\partial_\nu U^\mu = \frac{dx^\nu}{d\tau}\frac{\partial U^\mu}{\partial x^\nu} = \frac{dU^\mu}{d\tau}$. So

$$U^\nu\nabla_\nu U^\mu = \frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{d\tau}\frac{dx^\lambda}{d\tau}.$$

Finally, we can write the **geodesic equation** as

$$\boxed{\frac{d^2x^\rho}{d\tau^2} + \Gamma^\rho_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0 \quad \Longleftrightarrow \quad U^\nu\nabla_\nu U^\mu = 0} \tag{2.49}$$

In flat space with Cartesian coordinates, $\Gamma = 0$ and this is just $d^2x^\mu/d\tau^2 = 0$, a straight line at constant velocity. One can also show (Carroll does this carefully) that geodesics are the curves that extremize the proper time between two events, so a timelike geodesic is the path of *longest* proper time, which is the geometric version of the statement that the travelling twin is the younger one.

2.3.3 The Riemann Tensor

We saw that parallel transport around a closed loop can rotate a vector, and only in a curved space. The way to turn this into a formula is to look at an infinitesimally small loop, and the commutator of covariant derivatives does exactly that: $\nabla_\mu\nabla_\nu V^\rho$ moves the vector a little bit in the ν direction then the μ direction, and the commutator subtracts the other order. In flat space the order doesn't matter (partial derivatives commute), so whatever is

left over is curvature. Mathematically, we can write it as

$$[\nabla_\mu, \nabla_\nu] V^\rho = \nabla_\mu \nabla_\nu V^\rho - \nabla_\nu \nabla_\mu V^\rho \equiv R^\rho_{\sigma\mu\nu} V^\sigma \tag{2.50}$$

This tensor $R^\rho_{\sigma\mu\nu}$ that generates the twisting due to spacetime is called the **Riemann** tensor. It's role in differential geometry is to encode the information of the **intrinsic curvature** of the manifold. That is, it describes the innate curvature of the manifold without knowledge of an embedding in a higher dimensional space. Another way to see this is to use the fact that the covariant derivative is a function of the metric and derivatives (from the Christoffel symbols)

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma} \tag{2.51}$$

This, unfortunately, is disgusting. But it is a tensor, and since it's built from second derivatives of the metric that can't be removed by a change of coordinates, it is zero in every coordinate system if and only if the space is flat. With all indices down, it has the symmetries

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} = -R_{\rho\sigma\nu\mu} = R_{\mu\nu\rho\sigma}, \qquad R_{\rho[\sigma\mu\nu]} = 0, \tag{2.52}$$

which cut the naive $4^4 = 256$ components down to $\frac{1}{12}n^2(n^2-1) = 20$ in four dimensions, exactly the number of second derivatives of the metric that we couldn't get rid of. It also isn't even the object that shows up in the field equations. We can define an object by contracting 2 of the indices of the Riemann tensor

$$R_{\mu\nu} = R^\lambda_{\mu\lambda\nu} \tag{2.53}$$

Intuitively, the contraction of 2 of the indices is almost like an "average." Where the Riemann tensor tells us all the curvature details, the **Ricci tensor** tells us how volume elements deviate from that of flat space. Or in other words, it measures the net convergence or divergence of geodesics (the free-fall trajectories) in a way that is independent of the more subtle "shearing" effects. The Ricci tensor is symmetric, and like the Riemann tensor it can be written in terms of the metric through the Christoffel symbols

$$R_{\mu\nu} \equiv \partial_\lambda \Gamma^\lambda_{\mu\nu} - \partial_\nu \Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\lambda\rho}\Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\rho} \tag{2.54}$$

It's not much better in terms of ugliness but it's still something! Oftentimes, we will use software to compute these tensors from the metric (have fun doing it by hand), but it's still good to get your hands a bit dirty and to appreciate the efforts of those that came before without computers!

Problem 11.

Calculate R_{00} for the flat FRW universe $ds^2 = -dt^2 + a^2(t)\delta_{ij}dx^i dx^j$, using the Christoffel symbols from Problem 9.

Solution: Using (2.54) and subbing in the indices gives us

$$R_{00} = \partial_\lambda \Gamma^\lambda_{00} - \partial_0 \Gamma^\lambda_{0\lambda} + \Gamma^\lambda_{\lambda\rho}\Gamma^\rho_{00} - \Gamma^\rho_{0\lambda}\Gamma^\lambda_{0\rho}$$

We can make some simplifications. From Problem 9, $\Gamma^\lambda_{00} = 0$ for every λ, so the first and third terms vanish. Now the Ricci tensor simplifies to

$$R_{00} = -\partial_0 \Gamma^\lambda_{0\lambda} - \Gamma^\rho_{0\lambda}\Gamma^\lambda_{0\rho}$$

Remember that $\Gamma^\lambda_{0\lambda} = \Gamma^0_{00} + \Gamma^i_{0i} = 3H(t)$, so

$$\partial_0(3H(t)) = 3\left(\frac{\ddot{a}a - \dot{a}^2}{a^2}\right)$$

What about the term $\Gamma^\rho_{0\lambda}\Gamma^\lambda_{0\rho}$? We can use our result that $\Gamma^i_{0j} = H(t)\delta^i_j$

$$\Gamma^\rho_{0\lambda}\Gamma^\lambda_{0\rho} = \Gamma^i_{0j}\Gamma^j_{0i} = H^2(t)\delta^i_j\delta^j_i = 3H^2(t)$$

(remember that $\delta^i_i = 3$). Then we can finally calculate the Ricci tensor component to being

$$R_{00} = -3\left(\frac{\ddot{a}a - \dot{a}^2}{a^2}\right) - 3\frac{\dot{a}^2}{a^2}$$

And I guess if you wanted to fully simplify

$$\boxed{R_{00} = -3\left(\frac{\ddot{a}}{a}\right)}$$

All this just for 1 component. If you were a student of mine, I would make you calculate the entire $R_{\mu\nu}$ by hand because I love causing pain. But hypothetically, if you needed to compute the Ricci tensor, nowadays you wouldn't have to do it by hand. We can use EinsteinPy

```
import sympy as sp
from einsteinpy.symbolic import MetricTensor, RicciTensor, RicciScalar

# Define symbols and coordinates.
```

```
syms = sp.symbols('t x y z')
t, x, y, z = syms

# Define the scale factor a(t) as a symbolic function.
a = sp.Function('a')(t)

# Define the FRW metric (spatially flat in Cartesian coordinates).
# ds^2 = -dt^2 + a(t)^2*(dx^2 + dy^2 + dz^2)
g = sp.Matrix([
    [-1,       0,       0,       0],
    [ 0, a**2,       0,       0],
    [ 0,       0, a**2,       0],
    [ 0,       0,       0, a**2]
]).tolist()

# Create a MetricTensor object.
metric = MetricTensor(g, syms)
# Create the Ricci tensor and print it
Ric = RicciTensor.from_metric(metric)
Ric.tensor()
```

$$R_{\mu\nu} = \begin{bmatrix} -3\ddot{a}/a & 0 & 0 & 0 \\ 0 & a\ddot{a} + 2\dot{a}^2 & 0 & 0 \\ 0 & 0 & a\ddot{a} + 2\dot{a}^2 & 0 \\ 0 & 0 & 0 & a\ddot{a} + 2\dot{a}^2 \end{bmatrix}$$

Once we have the Ricci tensor, we can go one step further and contract the indices of the Ricci tensor with the metric to get a trace. This trace is called **the Ricci scalar**, or sometimes just the scalar curvature. If the Ricci tensor is an averaging out over the directions of the Riemann tensor, the Ricci scalar is a scalar that tells you the average curvature at that point. If you had a ball of fluid in free fall (following the geodesic), then the Ricci scalar tells you whether or not on average the volume of fluid will increase (positive), decrease (negative), or stay the same (0). Mathematically, it is given by

$$R = R^{\mu}_{\mu} = g^{\mu\nu} R_{\mu\nu} \tag{2.55}$$

Problem 12.

Later in this chapter, when we derive the metric for the FRW universe, we will force

the condition that in order to satisfy isotropy and homogeneity, the spatial curvature must be a constant. Using the full Ricci tensor above, compute the four-dimensional Ricci scalar of the flat FRW metric. Is it a constant?

Solution: In the case of the flat FRW metric that we used to calculate $R_{\mu\nu}$, the flat metric is just $g^{\mu\nu} = \text{diag}(-1, a^{-2}, a^{-2}, a^{-2})$. Then the trace becomes

$$R = g^{00}R_{00} + g^{ij}R_{ij} = \frac{3\ddot{a}}{a} + \frac{3}{a^2}\left(a\ddot{a} + 2\dot{a}^2\right)$$

Or if we wanted to simplify

$$\boxed{R = \frac{6\left(a\ddot{a} + \dot{a}^2\right)}{a^2}} \tag{2.56}$$

Wait a second... this isn't a constant! Does this mean that our isotropy and homogeneity assumptions are invalid? No! Isotropy and homogeneity only constrain the *spatial* slices: they say the curvature can't depend on the spatial coordinates, and indeed, it does not. The symmetries say nothing about time dependence, and the spacetime curvature changes in time because the universe expands. (The spatial slices themselves are flat here, with zero curvature.) If you wanted the four-dimensional curvature to be constant (say, 0), then you would need to solve the above differential equation for $a(t)$.

We've been working with the flat FRW metric. If we introduced nonzero spatial curvature, then the metric would become (we will derive this shortly)

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2\right]$$

If we write the metric in this way, what becomes of the Ricci scalar? Back to EinsteinPy we go!

```
import sympy as sp
from einsteinpy.symbolic import MetricTensor, RicciTensor, RicciScalar

# Define symbols and coordinates.
syms = sp.symbols('t r th ph')
t, r, th, ph = syms

# Define the scale factor a(t) as a symbolic function.
a = sp.Function('a')(t)
k = sp.Symbol('k')
R = sp.Symbol('R')
```

```
12
13 # Define the curved FRW metric in spherical coordinates.
14 g = sp.Matrix([
15     [-1,       0,       0,       0],
16     [ 0, (a**2)/(1-(k*r**2)),       0,       0],
17     [ 0,       0, a**2*r**2,       0],
18     [ 0,       0,       0, a**2*r**2*sp.sin(th)**2]
19 ]).tolist()
20
21 # Create a MetricTensor object.
22 metric = MetricTensor(g, syms)
23 # Create the Ricci tensor and print it
24 Ric = RicciTensor.from_metric(metric)
25 Ric.tensor()
26
27 # Calculate the Ricci scalar and print it
28 R = RicciScalar.from_riccitensor(Ric)
29 R.simplify()
30 R.expr
```

$$R = 6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2}\right) \tag{2.57}$$

And as we can see, if we take $k = 0$, then we just retrieve back the result of Problem 12. Also note that the curvature still does not depend on any of the spatial coordinates.

2.3.4 The Bianchi Identity And The Einstein Tensor

There is one more property of the Riemann tensor that we need, and it's a differential identity. In locally inertial coordinates, where $\Gamma = 0$ at a point, the Riemann tensor is just $\partial\Gamma - \partial\Gamma$, and a short calculation shows that its derivatives satisfy

$$\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu} = 0. \tag{2.58}$$

This is a tensor equation, so if it's true in one coordinate system it's true in all of them. It's called the **Bianchi identity**. Contracting it twice with the metric gives $\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu R$, which we can write as

$$\nabla^\mu G_{\mu\nu} = 0, \qquad G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu}. \tag{2.59}$$

The symmetric tensor $G_{\mu\nu}$ is called the **Einstein tensor**. The fact that its divergence vanishes *identically*, for any metric whatsoever, is going to be the key to writing down the field equation, because the energy-momentum tensor also has vanishing divergence.

2.4 Gravitation

Now that we have all of the geometry, we can finally go about showing that gravity is the geometry of spacetime. In my opinion, the most elegant way is through extremizing an action (the Einstein-Hilbert action). The issue with this is that it requires Lagrangian field theory, and it's a bit of a departure from the story we've been telling, which is how gravity emerges from geometry. So, like Carroll, we will instead go through the principle of equivalence and general covariance, and then guess the field equation.

2.4.1 The Equivalence Principle

Einstein's most famous thought experiment was the accelerating rocket-ship. Galileo had already noticed that all objects fall the same way in a gravitational field, regardless of their mass or composition (the inertial mass and the gravitational mass are the same). Einstein's conclusion was that, in a small enough region, gravity is indistinguishable from acceleration. A person in a closed box can't tell whether the box is sitting on Earth or accelerating at g through empty space, and a person in a freely falling box feels no gravity at all. This is the **equivalence principle**: in a small enough region of spacetime, the laws of physics reduce to those of special relativity in a freely falling frame. Compare this with (2.41). Freely falling frames are exactly the locally inertial coordinates of a curved spacetime, which is the first hint that gravity is curvature. This is a beautiful thought experiment, but it extends to more than just beauty because we can make logical predictions from it. Without doing any math, we can deduce things like gravitational time dilation and gravitational redshift. For example, when I was taking general relativity, we discussed the movie *Interstellar*. Part of the plot included characters experiencing time passing differently than others simply due to the fact that they are orbiting a super-massive black hole. The claim (which is known to be true because satellites need to adjust their clocks) is that gravitational wells (like a black hole, or just heavy objects like the sun or Earth) cause time to be dilated. We can see this using the twin paradox in special relativity (Problem 1). If you have a twin on a rocket going at some huge speed to a nearby star and back, the twin on Earth will have aged more than the twin on the rocket. The paradox comes from the fact that in special relativity, there are no preferred reference frames, so it's equally valid to say that relative to the twin on the rocket, the twin on Earth is the one that is moving. So if neither twin can agree on who is moving, then why does one of them age more than the other? The solution to the paradox is a cheat in my opinion, which is that the twin on the rocket ship has to accelerate and then decelerate in order to turn around, so the two twins are not equivalent. The Earth twin follows a straight worldline (a geodesic), and straight worldlines have the longest proper time.

As we see, it is the acceleration that causes 2 observers to experience a change in time. If we apply the equivalence principle, then we can make the conclusion that gravity must cause time differences between observers. Here's a quick way to see it with the equivalence principle. A photon sent up from the floor to the ceiling of a rocket accelerating at g arrives a time h later (height h, $c = 1$), by which point the ceiling is moving away from where the floor was at emission with extra speed gh, so the photon is Doppler shifted to the red by $\Delta\omega/\omega \approx -gh$. By the equivalence principle, a photon climbing a height h in a gravitational field must be redshifted by the same amount, $\Delta\omega/\omega = -gh = -\Delta\Phi$, where Φ is the Newtonian potential. Since the number of wave crests is fixed, clocks higher up in the potential must tick faster than clocks lower down. GPS satellites have to correct for this every day.

2.4.2 General Covariance

In the previous section, we talked about the effects of gravity in a constant background gravitational potential Φ (constant acceleration). Now, if we have a gravitational field that varies in space and time, then the obvious way to write the effects of gravity is just for every point in space, go into a locally freely falling frame (where there is no gravity), write the equations of motion of special relativity, and then transform back. Obviously this can't be efficient. So what do we do? The prescription of **general covariance** is to begin in the absence of gravity, then replace $\eta_{\mu\nu}$ by $g_{\mu\nu}$ and partial derivatives by covariant derivatives ("comma goes to semicolon"), so that the equations of motion are tensor equations, valid in any coordinate system. This will guarantee to satisfy the equivalence principle because if they are form invariant, then they will also hold true in a locally "free-falling" frame. There are some subtleties about general covariance, and I implore you to check them out. * For example, $\partial_\mu T^{\mu\nu} = 0$ in special relativity becomes $\nabla_\mu T^{\mu\nu} = 0$ in curved spacetime. To see how this works for the equations of motion for a particle, and to see that it really does reproduce what free fall should look like, consider a particle in the absence of gravity

$$\frac{d^2x}{d\tau^2} = 0$$

where τ is the proper time of the particle. There is another subtlety here and it is the fact that for massless particles like light, their spacetime trajectories are on the lightcone and so therefore their proper time is always 0. It's better to use some other affine parameter λ to parametrize the trajectory of our particle. Let us start by choosing our free falling

* It has to do with the fact that we could always transform our coordinate systems from one to another, this is called a **diffeomorphism** and all it means is that physics looks the same no matter your coordinate system. The nontrivial part of general covariance is that these coordinate changes are physical and are due to the effects of gravity.

coordinates to be $\xi^\mu(x)$ where x are our lab frame coordinates. If we are starting in the free falling frame, then that means that there must exist a transformation that takes the lab frame metric $g_{\mu\nu}$ to our free falling frame with the Minkowski metric $\eta_{\mu\nu}$

$$\eta_{\mu\nu} = g_{\alpha\beta}\frac{\partial x^\alpha}{\partial \xi^\mu}\frac{\partial x^\beta}{\partial \xi^\nu}, \quad g_{\alpha\beta} = \eta_{\mu\nu}\frac{\partial \xi^\mu}{\partial x^\alpha}\frac{\partial \xi^\nu}{\partial x^\beta}$$

In the free falling frame, we should have that $\frac{d^2\xi^\mu}{d\tau^2} = 0$ (and using λ if the particle is massless) so we can incorporate the coordinate changes via the chain rule

$$\begin{aligned}\frac{d^2\xi^\mu}{d\tau^2} &= \frac{d}{d\tau}\left(\frac{\partial \xi^\mu}{\partial x^\alpha}\frac{dx^\alpha}{d\tau}\right) \\ &= \frac{\partial^2 \xi^\mu}{\partial x^\alpha \partial x^\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} + \frac{\partial \xi^\mu}{\partial x^\alpha}\frac{d^2x^\alpha}{d\tau^2} \\ &= 0\end{aligned}$$

In the second line of the above, we can use the fact that $\frac{\partial x^\rho}{\partial \xi^\mu}\frac{\partial \xi^\mu}{\partial x^\alpha} = \delta^\rho_\alpha$ and multiply both sides by $\frac{\partial x^\rho}{\partial \xi^\mu}$

$$\left(\frac{\partial x^\rho}{\partial \xi^\mu}\right)\frac{\partial^2 \xi^\mu}{\partial x^\alpha \partial x^\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} + \left(\frac{\partial x^\rho}{\partial \xi^\mu}\right)\frac{\partial \xi^\mu}{\partial x^\alpha}\frac{d^2x^\alpha}{d\tau^2} = \left(\frac{\partial x^\rho}{\partial \xi^\mu}\right)\frac{\partial^2 \xi^\mu}{\partial x^\alpha \partial x^\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} + \frac{d^2x^\rho}{d\tau^2} = 0$$

If we define the object

$$\Gamma^\rho_{\alpha\beta} \equiv \frac{\partial x^\rho}{\partial \xi^\mu}\frac{\partial^2 \xi^\mu}{\partial x^\alpha \partial x^\beta} \tag{2.60}$$

then we can rewrite the free falling equation in the lab coordinates as

$$\frac{d^2x^\rho}{d\tau^2} + \Gamma^\rho_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0 \tag{2.61}$$

This is exactly the form of the geodesic equation (2.49). Are these Γ's the same as the Christoffel symbols? We can check by starting with

$$\frac{\partial}{\partial x^\lambda}g_{\alpha\beta} = \eta_{\mu\nu}\frac{\partial}{\partial x^\lambda}\left(\frac{\partial \xi^\mu}{\partial x^\alpha}\frac{\partial \xi^\nu}{\partial x^\beta}\right)$$

Then taking the product rule

$$= \eta_{\mu\nu}\left(\frac{\partial^2 \xi^\mu}{\partial x^\lambda \partial x^\alpha}\frac{\partial \xi^\nu}{\partial x^\beta} + \frac{\partial^2 \xi^\nu}{\partial x^\lambda \partial x^\beta}\frac{\partial \xi^\mu}{\partial x^\alpha}\right)$$

Then recognizing that we can just sub in (2.60) and get

$$= \eta_{\mu\nu}\left(\Gamma^{\rho}_{\lambda\alpha}\frac{\partial\xi^{\mu}}{\partial x^{\rho}}\frac{\partial\xi^{\nu}}{\partial x^{\beta}} + \Gamma^{\rho}_{\lambda\beta}\frac{\partial\xi^{\mu}}{\partial x^{\alpha}}\frac{\partial\xi^{\nu}}{\partial x^{\rho}}\right)$$

If we distribute the Minkowski metric $\eta_{\mu\nu}$ through the parentheses, then we get

$$\frac{\partial}{\partial x^{\lambda}}g_{\alpha\beta} = \Gamma^{\rho}_{\lambda\alpha}g_{\rho\beta} + \Gamma^{\rho}_{\lambda\beta}g_{\alpha\rho} \tag{2.62}$$

This is exactly the metric compatibility condition (2.43), and (2.60) is obviously symmetric in α and β. So the coefficients that show up in free fall are the Christoffel symbols (2.45)! This is the payoff: freely falling particles move on geodesics of the spacetime metric, and the Christoffel symbols are what encode the "force" of gravity in a general coordinate system. The things we used to call gravitational forces are just what geodesics look like in coordinates that aren't freely falling.

2.4.3 The Newtonian Limit

If gravity is geometry, we had better get Newton back in the appropriate limit. Newtonian gravity applies to particles moving slowly ($dx^i/d\tau \ll dt/d\tau$) in a gravitational field that is weak and static. Weak means the metric is close to Minkowski, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$, and static means nothing depends on t. For a slow particle, the geodesic equation (2.49) is dominated by the time-time term,

$$\frac{d^2x^{\mu}}{d\tau^2} + \Gamma^{\mu}_{00}\left(\frac{dt}{d\tau}\right)^2 = 0. \tag{2.63}$$

For a static metric, (2.45) gives to first order in h

$$\Gamma^{\mu}_{00} = \frac{1}{2}\eta^{\mu\sigma}\left(\partial_0 h_{0\sigma} + \partial_0 h_{0\sigma} - \partial_\sigma h_{00}\right) = -\frac{1}{2}\eta^{\mu i}\partial_i h_{00}. \tag{2.64}$$

So $\Gamma^0_{00} = 0$, which means $dt/d\tau$ is constant, and the spatial components become

$$\frac{d^2x^i}{dt^2} = \frac{1}{2}\partial_i h_{00}. \tag{2.65}$$

Compare this with Newton's $\ddot{\boldsymbol{x}} = -\nabla\Phi$. They agree if $h_{00} = -2\Phi$, that is

$$\boxed{g_{00} = -(1 + 2\Phi)} \tag{2.66}$$

The Newtonian potential lives in the time-time component of the metric. Notice that this is the gravitational time dilation from before: a clock at rest at potential Φ measures

$d\tau = \sqrt{-g_{00}}\, dt \approx (1+\Phi)dt$, so clocks deeper in the potential well ($\Phi < 0$) tick slower. We will meet this exact structure again in Chapter 6, where the metric of a perturbed universe has $g_{00} = -a^2(1+2\Psi)$ and Ψ plays the role of the Newtonian potential.

2.4.4 Einstein's Equation

We've spent a great deal of time talking about the motion of particles in a gravitational field, but we have yet to talk about the dynamics of the field itself. In Newtonian gravity, the field equation is Poisson's equation,

$$\nabla^2\Phi = 4\pi G\rho. \tag{2.67}$$

We want a tensor generalization of this. On the right hand side, the energy density ρ is the 00 component of the energy-momentum tensor, so the right hand side should be proportional to $T_{\mu\nu}$. On the left hand side, by (2.66) the potential is part of the metric, so we want a symmetric tensor built out of second derivatives of the metric. The natural candidates are $R_{\mu\nu}$ and $Rg_{\mu\nu}$. But the right hand side satisfies $\nabla^\mu T_{\mu\nu} = 0$, so the left hand side had better satisfy the same thing identically. By (2.59), the combination that does this is the Einstein tensor, and so we arrive at **Einstein's equation**,

$$\boxed{R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G\, T_{\mu\nu}} \tag{2.68}$$

The constant $8\pi G$ is fixed by demanding that we get Poisson's equation back. Taking the trace of (2.68) gives $-R = 8\pi G T$, so we can rewrite it as $R_{\mu\nu} = 8\pi G\left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}\right)$. For dust at rest ($T_{00} = \rho$, $T = -\rho$) the 00 component is $R_{00} = 4\pi G\rho$, and for a weak static field one can check that $R_{00} \approx \partial_i\Gamma^i_{00} = -\frac{1}{2}\nabla^2 h_{00} = \nabla^2\Phi$, which is exactly Poisson's equation. Even though for all of the text we've set $c = 1$, here it is in all of its glory,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}, \tag{2.69}$$

so that we can gawk at it. This truly is one of humanity's greatest intellectual achievements. It's ten coupled, nonlinear partial differential equations for the metric, and as John Wheeler put it, spacetime tells matter how to move, and matter tells spacetime how to curve. As beautiful as it is, behind the curtains of each tensor is a really really ugly calculation. But maybe that speaks to a greater universal truth: all roses have thorns.

There is one thing that we are allowed to add. Since $\nabla^\mu g_{\mu\nu} = 0$, we could add a term $\Lambda g_{\mu\nu}$

to the left hand side without breaking anything,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{2.70}$$

Λ is called the **cosmological constant**. Moving it to the right hand side, it looks like a perfect fluid with $\rho_\Lambda = \Lambda/8\pi G$ and $P_\Lambda = -\rho_\Lambda$, an energy density of empty space itself. Einstein added it to get a static universe, called it his biggest blunder when the universe turned out to be expanding, and it came back in 1998 as the best explanation we have for the accelerating expansion. We will treat it as one of the fluids in the universe.

2.5 Cosmology

Now we can finally apply all of this to the universe. Solving Einstein's equation in general is hopeless, so we need to make our lives a bit easier and ask: what spacetimes are we allowed to have? The set of spacetime geometries that we could have is technically infinite. However, if we can impose symmetry, then the set becomes a smaller infinity, and for the universe as a whole there is a lot of symmetry.

2.5.1 The Cosmological Principle

We are going to make an assumption about the universe that will allow us to simplify our calculations. We will relax these assumptions in the future when we talk about perturbation theory.

The Cosmological Principle: The universe is homogeneous (translationally invariant) and isotropic (rotationally invariant) on the largest of scales.

Basically what this means is that if you rotate your head and move around, the universe looks the same everywhere. This might seem like a crazy idea because if you were to actually rotate your head and move forwards and backwards, the universe does NOT look the same everywhere (unless you're dead). But on scales larger than about 100 Mpc, the distribution of galaxies really does look the same everywhere, and when you look at the CMB (figure 1.2), its temperature is the same in every direction to one part in 10^5. Isotropy about every point implies homogeneity, and together they say that space at a fixed time is **maximally symmetric**: it has as many symmetries as a space can possibly have.

2.5.2 Maximally Symmetric Spaces

The most general form that a metric can take on is

$$ds^2 = g_{ij}(\boldsymbol{x}, t)dx^i dx^j \tag{2.71}$$

We will talk about time dependence on the metric shortly, but for now, let us just focus on the spatial part (hence using Latin indices). Isotropy means that elements of the metric can only depend on the magnitude of $\boldsymbol{x}$, which is just r. We should then be using spherical polar coordinates

$$ds^2 = f(r)dr^2 + r^2 d\Omega^2$$

Where $d\Omega^2$ is the solid angle element (basically all the angular dependencies). The argument for why cross terms like $d\phi d\theta$ vanish is because this violates the isotropy condition because it would introduce directional dependence on ϕ and θ and that clearly cannot happen under isotropy. We are going to use symmetry arguments to figure out what $f(r)$ must be. You can theoretically skip this subsection if you wanted to, but I am including it for completeness (and also for myself).

I am now going to make a slight change in the factor multiplying dr^2, and this is purely for simplicity. There is nothing stopping us from parametrizing the metric this way, and so we are going to rewrite the above metric as

$$ds^2 = e^{2\alpha(r)}dr^2 + r^2 d\Omega^2 \tag{2.72}$$

Homogeneity forces us to conclude that if the geometry of space is the same everywhere, then the curvature must be the same everywhere. How do we compute curvature from the metric? We now know: we compute the Ricci scalar of the three-dimensional metric

$$R = g^{ij}R_{ij}$$

where R is the Ricci curvature scalar, and R_{ij} is the Ricci curvature tensor. If I showed you what the full calculation of the Ricci tensor contracted with the metric is, it would be 5 pages long. And also, I doubt it would be very meaningful at the current moment, and so what I will do is write and paste code to do the calculation in python.

```python
import sympy
from sympy import symbols, Function, sin, exp, simplify
from einsteinpy.symbolic import MetricTensor, RicciTensor, RicciScalar

# Define spherical coordinates
```

```
syms = sympy.symbols("r theta phi")
r, th, ph = syms

# Define alpha(r) as a symbolic function
alpha = sympy.Function("alpha")(r)

# Define the 3D metric components:
g = sympy.diag(exp(2*alpha), r**2, r**2 * sin(th)**2).tolist()

# Construct the Metric object for a 3-dimensional Riemannian space.
# Signature "+++" indicates a Riemannian (all positive) metric.
metric_3d = MetricTensor(g, syms, config="ll")
Ric = RicciTensor.from_metric(metric_3d)

# Compute the Ricci Scalar
ricci_scalar = RicciScalar.from_riccitensor(Ric)
R_expr = simplify(ricci_scalar.expr)
R_expr
```

The answer that you get, after full-simplifying, is

$$R = \frac{2}{r^2}\left[1 - \frac{d}{dr}\left(re^{-2\alpha(r)}\right)\right] \tag{2.73}$$

If the curvature is the same everywhere, $R = C$ for some constant C, then we can write the above as

$$\frac{C}{2}r^2 = 1 - \frac{d}{dr}\left(re^{-2\alpha(r)}\right) \implies \frac{d}{dr}\left(re^{-2\alpha(r)}\right) = 1 - \frac{C}{2}r^2$$

We can solve for $e^{2\alpha(r)}$ by integrating and then doing a bit of algebra

$$re^{-2\alpha(r)} = r - \frac{C}{6}r^3 + C_2 \implies e^{2\alpha(r)} = \frac{1}{1 - \frac{C}{6}r^2 + C_2 r^{-1}}$$

where C_2 is an integration constant. You can check that the C_2 term doesn't contribute to R at all (its contribution to $re^{-2\alpha}$ is a constant, which the derivative kills). But it does do two bad things. The metric component g_{rr} blows up or goes negative near $r = 0$, so the space isn't smooth there, and the origin $r = 0$ becomes a special place, which violates homogeneity (the full Riemann tensor would not be the same everywhere, even though its trace is). So C_2 must vanish. Calling $k \equiv C/6$, the metric for a fully homogeneous and isotropic 3-space is

$$ds^2 = \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \tag{2.74}$$

and the constant $k = R/6$ determines the curvature. We can always rescale r to set k to $+1$, 0, or -1 (and move the overall size into the scale factor that we will add in a moment). Let

us look at what types of geometries are allowed in our universe.

k=0:

If we set k = 0, then we get

$$ds^2 = dr^2 + r^2 d\Omega^2$$

Which is is just the flat space metric. This obviously makes sense because if the curvature is 0 everywhere, then the geometry must be flat.

k > 0:

If the curvature is positive, then the metric (setting k=1 without loss of generality) becomes

$$ds^2 = \frac{dr^2}{1 - r^2} + r^2 d\Omega^2$$

We can deduce what the geometry is by doing a coordinate transformation

$$r = \sin(\chi)$$

$$\implies dr = \cos(\chi) d\chi$$

and when we substitute this back, we get

$$ds^2 = \frac{\cos^2(\chi) d\chi^2}{1 - \sin^2(\chi)} + \sin(\chi)^2 d\Omega^2$$

$$= d\chi^2 + \sin^2(\chi) d\Omega^2 \tag{2.75}$$

If we look back to equation (2.34), we realize that this is similar looking to the metric of a 2-sphere, with one more angle. As it turns out, this is actually the metric of a 3-sphere, the surface $X_0^2 + X_1^2 + X_2^2 + X_3^2 = 1$ in four-dimensional Euclidean space (exercise in the problems). What we have found is that if the of the universe is positive, then the resulting geometry is spherical.

k < 0:

Once again, we set $k = -1$ without loss of generality, and when we do this, we find that

$$ds^2 = \frac{dr^2}{1 + r^2} + r^2 d\Omega^2$$

We can't use the same parametrization as the previous case because it won't cancel out the

$1 + r^2$ in the denominator. Let us make this parametrization instead

$$r = \sinh(\chi)$$

$$\implies dr = \cosh(\chi)d\chi$$

And then when we sub this back into the metric, we find

$$\begin{aligned} ds^2 &= \frac{\cosh^2(\chi)d\chi^2}{1 + \sinh^2(\chi)} + \sinh^2(\chi)d\Omega^2 \\ &= d\chi^2 + \sinh^2(\chi)d\Omega^2 \end{aligned} \tag{2.76}$$

Problem 13.

What is the geometry described by the metric in (2.76)? You can deduce this by starting with the parametrization of the sphere and modifying it such that we get the metric in (2.76).

Solution: In the problems, you are asked to find the metric for the 3-sphere, and you'll find that the parametrization is

$$\begin{aligned} X_0 &= \cos\chi \\ X_1 &= \sin\chi\cos\theta \\ X_2 &= \sin\chi\sin\theta\cos\phi \\ X_3 &= \sin\chi\sin\theta\sin\phi \end{aligned}$$

And when we find the metric, we get

$$ds^2 = d\chi^2 + \sin^2(\chi)d\Omega^2$$

If the metric in (2.76) is the same as that of the above except with $\sinh^2(\chi)$ instead of $\sin^2(\chi)$, then we should replace all χ dependent functions with their hyperbolic counterparts. Doing so gives us the following parametrization

$$\begin{aligned} X_0 &= \cosh(\chi) \\ X_1 &= \sinh(\chi)\cos(\theta) \\ X_2 &= \sinh(\chi)\sin(\theta)\cos(\phi) \\ X_3 &= \sinh(\chi)\sin(\theta)\sin(\phi) \end{aligned} \tag{2.77}$$

But we know exactly what this is! This is the parametrization of a 3-hyperboloid.

Using $\cosh^2 - \sinh^2 = 1$, the coordinates satisfy

$$-X_0^2 + X_1^2 + X_2^2 + X_3^2 = -1,$$

and if we compute the line element using the *Minkowski* metric $-dX_0^2 + dX_1^2 + dX_2^2 + dX_3^2$ of the embedding space (not the Euclidean one), we get exactly (2.76). So a space of constant negative curvature is a hyperboloid sitting inside a flat Minkowski space, the same way the sphere sits inside flat Euclidean space. (You can write Mathematica code to double-check.)

This concludes that the only geometries allowed in a homogeneous and isotropic universe are flat, spherical, or hyperbolic. The parameter that determines the geometry is called the curvature k.

2.5.3 The Expanding Universe

We've so far created a static, symmetric universe. However, as we know, the universe is not static; it is expanding. How can we include the universe expanding into the metric? Because of homogeneity, the expansion of the universe must happen uniformly, and so therefore the only thing that we need to add is an **expansion parameter** to the spatial component of the metric

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right) \tag{2.78}$$

This is called the **Friedmann-Robertson-Walker (FRW) metric.** We normalize $a(t_0) = 1$ today. We can stare at this for a bit and figure out what this is saying. When we say that the universe is expanding, we mean that space *itself* is expanding (figure 2.1): the coordinate grid is fixed, but the physical distance between grid lines grows like $a(t)$. There are 2 coordinate systems we will occupy ourselves with. The first is when we are expanding WITH the universe (like in figure 2.1), in which case, the coordinates of an object relative to a chosen origin don't change, even though the physical distances might. This coordinate system is called **co-moving**. Because we are co-moving with it. Duh. The second coordinate system of interest is the **physical** coordinate system, in which case we will literally see the objects move further away. It is given by

$$\boldsymbol{r}'_{\text{physical}} = a(t)\boldsymbol{r}_{\text{co-moving}} \tag{2.79}$$

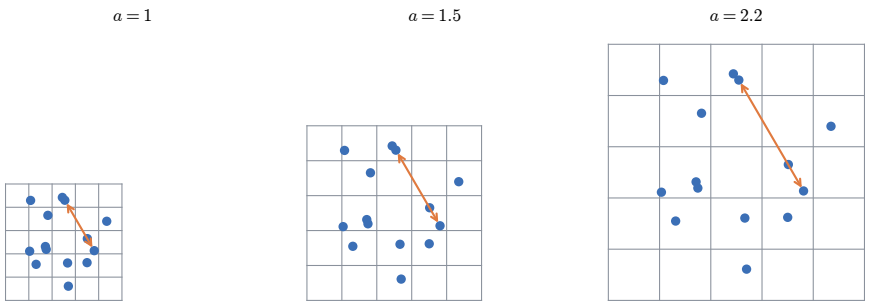


Figure 2.1: The same comoving grid and galaxies at three different times. The comoving coordinates of every galaxy stay fixed, but every physical distance (like the orange arrow) is stretched by the same factor $a(t)$.

How fast are objects truly moving away from us? We can find out by taking the derivative to find the velocity

$$\boldsymbol{v} = \frac{d\boldsymbol{r}'}{dt} = \frac{da(t)}{dt}\boldsymbol{r} + a(t)\frac{d\boldsymbol{r}}{dt} \tag{2.80}$$

We can define a new parameter

$$H \equiv \frac{\dot{a}}{a} \tag{2.81}$$

which we call the **Hubble parameter**. Using $\dot{a} = Ha$, the physical velocity will become

$$\boldsymbol{v}_{\text{physical}} = H\boldsymbol{r}' + \boldsymbol{v}_{\text{peculiar}} \tag{2.82}$$

where $\boldsymbol{r}' = a\boldsymbol{r}$ is the physical position and $\boldsymbol{v}_{\text{peculiar}} = a\,d\boldsymbol{r}/dt$ is the velocity of the object relative to the comoving grid (that is, if the object is truly moving instead of moving because space is expanding). For galaxies that are far enough away, the peculiar velocities are small compared to the first term, and we get **Hubble's law** $v = H_0 d$.

2.5.3.0.1 Redshift. The expansion also stretches light. Consider a light ray travelling radially towards us in the flat FRW metric. Light travels on null paths, $ds^2 = 0$, so $dt = a\,dr$ and the comoving distance travelled between emission at t_e and observation at t_0 is $r = \int_{t_e}^{t_0} dt/a(t)$. Two successive wave crests, emitted δt_e apart, have to cover the same comoving distance, so $\delta t_e/a(t_e) = \delta t_0/a(t_0)$. The period, and hence the wavelength, is stretched by the expansion,

$$1 + z \equiv \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{a(t_0)}{a(t_e)} = \frac{1}{a(t_e)}, \tag{2.83}$$

which defines the **redshift** z. Redshift is just a measure of how much the universe has expanded since the light was emitted, which is why cosmologists use z (or a) as a clock. A

photon's momentum falls like $p \propto 1/a$, and as we'll see the same is true of the momentum of any freely moving particle.

2.5.4 The Contents Of The Universe

By isotropy, the energy-momentum tensor of whatever fills the universe can't pick out a direction, so in the rest frame of the comoving observers it must be the perfect fluid (2.38) with $\rho(t)$ and $P(t)$ depending only on time. Its conservation, $\nabla_\mu T^\mu{}_\nu = 0$, is our first real calculation in an expanding universe.

Problem 14.

Using the form of the energy momentum tensor given in (2.39), find the condition in which $T^\mu{}_\nu$ satisfies the conservation equation

$$\nabla_\mu T^\mu{}_\nu = 0$$

for the FRW metric.

Solution: We first begin by expanding our previous definition of the covariant derivative on tensors (in the specific case of 1 up 1 down)

$$\nabla_\mu T^\mu{}_\nu = \partial_\mu T^\mu{}_\nu + \Gamma^\mu_{\mu\lambda} T^\lambda{}_\nu - \Gamma^\lambda_{\mu\nu} T^\mu{}_\lambda = 0$$

Notice that the μ indices are being contracted, so this just corresponds to 4 continuity equations for each component of ν. If we choose the $\nu = 0$ component, then we are working with the energy density.

$$\nabla_\mu T^\mu{}_0 = \partial_\mu T^\mu{}_0 + \Gamma^\mu_{\mu\lambda} T^\lambda{}_0 - \Gamma^\lambda_{\mu 0} T^\mu{}_\lambda = 0$$

Let us look at the first term. $T^\mu{}_\nu$ is diagonal and only depends on time, so the only surviving piece is $\partial_0 T^0{}_0 = -\dot{\rho}$. For the second term, diagonality forces $\lambda = 0$, so we need $\Gamma^\mu_{\mu 0} = \Gamma^0_{00} + \Gamma^i_{i0} = 0 + 3H$, giving $3HT^0{}_0 = -3H\rho$. For the last term, diagonality forces $\lambda = \mu$, so we need $\Gamma^\mu_{\mu 0} T^\mu{}_\mu$ summed over μ, and only the spatial ones are nonzero, $\Gamma^i_{i0} T^i{}_i = H \cdot P$ for each of the 3 spatial directions, giving $3HP$. Putting it all together, $-\dot{\rho} - 3H\rho - 3HP = 0$, and the continuity equation is

$$\boxed{\dot{\rho} + 3H(\rho + P) = 0} \tag{2.84}$$

This is just the first law of thermodynamics, $dE = -P\,dV$, for a comoving volume $V \propto a^3$ with energy $E = \rho V$.

Most texts will define the **equation of state parameter** $w \equiv P/\rho$, in which case equation (2.84) can be rewritten as

$$\begin{aligned}\dot{\rho} &= -3\rho H(t)\left(1 + \frac{P}{\rho}\right) \\ &= -3\rho H(t)(1+w)\end{aligned}$$

If we use our definition of $H = \frac{\dot{a}}{a}$, then we have

$$\frac{\dot{\rho}}{\rho} = -3\frac{\dot{a}}{a}(1+w) \tag{2.85}$$

And for constant w we can solve this differential equation by separation of variables

$$\begin{aligned}\int \frac{d\rho}{\rho} &= -3(1+w)\int \frac{da}{a} \\ \implies \rho(a) &\propto a^{-3(1+w)}\end{aligned} \tag{2.86}$$

Our interest now is to calculate the energy densities for the "stuff" in our universe. The 2 simplest types of fluids that exist are matter and radiation, and both have different values of w. Let us focus on the matter content first.

For a perfect fluid, matter is a fluid in which the pressure is tiny compared to its energy density $P \ll \rho$ (the particles are moving slowly, so their kinetic energy is negligible compared to their rest mass). In this case, $w \approx 0$, and using (2.86) we find that

$$\rho_{\text{matter}} \propto a^{-3} \tag{2.87}$$

Which makes sense because as the volume of the universe expands, the energy density should die with the volume (since $\rho = E/V$ and $V \propto a^3$). What constitutes matter in our universe? Baryons (the protons and neutrons that make up ordinary matter, plus their electrons) and dark matter.

Now let us look at radiation. How do we define radiation? We can derive the relationship between radiation pressure and energy density from kinetic theory, but I think the simplest explanation is this: For a perfect radiation fluid, the distribution of the momentum is isotropic. The average contribution from each component will be 1/3. As such, we should

expect that the pressure (relating to the momentum) is 1/3 that of the energy density. Thus

$$P_{\text{rad}} = \frac{1}{3}\rho_{\text{rad}}$$

This means that $w = 1/3$ and the energy density falls as

$$\rho_{\text{rad}} \propto a^{-3(1+1/3)} = a^{-4} \tag{2.88}$$

And this makes sense because not only does the number of photons per volume fall with the volume, the energy of each photon also redshifts like a^{-1} by (2.83), hence the a^{-4}. What constitutes radiation in our universe? Light (obviously), neutrinos while they are relativistic (not so obvious), and in the early universe, every particle whose mass is much smaller than the temperature.

The last thing that we should consider is dark energy. We saw that a cosmological constant behaves like a fluid with $P_\Lambda = -\rho_\Lambda$, so $w = -1$. If that is the case, then

$$\rho_\Lambda \propto a^{-3(1-1)} = a^0 \tag{2.89}$$

The energy density of a cosmological constant doesn't dilute at all; it's the energy of empty space, and as space grows there is just more of it. We should note that the term "dark energy" is an umbrella term to describe whatever is causing the accelerated expansion, and the cosmological constant is the simplest (and so far perfectly consistent) candidate. I know that I blasted through this discussion while other texts might have a more lengthy exposition on each type of matter. The reason that I am not putting much of an emphasis is because...there isn't much that you *need* to know beyond the value of w. Chapter 3 will justify $w = 1/3$ for radiation properly with statistical mechanics.

2.5.5 The Friedmann Equations

We now have all the ingredients to put the FRW metric into Einstein's equation (2.68). We've come all this way for what? Ultimately, we are here to find an equation to solve for the evolution of our universe; more specifically, an equation for the scale factor $a(t)$. This involves solving the Einstein equation using the FRW metric and the perfect fluid energy momentum tensor. Let us begin deriving the Friedmann equations, 1 component at a time. Starting with the time-time component,

$$R_{00} - \frac{1}{2}g_{00}R = 8\pi G\, T_{00}$$

We already solved for R_{00} in problem 11, it's just $-\frac{3\ddot{a}}{a}$ (and it's the same for $k \neq 0$). g_{00} is just -1, R is given in equation (2.57), and T_{00} is just the energy density ρ. Combining them all gives

$$-\frac{3\ddot{a}}{a} + 3\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{k}{a^2}\right) = 8\pi G\rho$$

The $\ddot{a}$ terms cancel, and we are left with

$$\boxed{H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}} \tag{2.90}$$

This is called the **first Friedmann equation**. The term "first" is foreshadowing that there will exist a second Friedmann equation and indeed there is! It comes from the spatial parts of the Einstein equation, but there is a shortcut, which is to use the continuity equation (2.84) instead. We can take time derivatives of the first Friedmann equation and get

$$2H\dot{H} = \frac{8\pi G}{3}\dot{\rho} + \frac{2kH}{a^2}$$

And then subbing the continuity equation for $\dot{\rho}$ into the above

$$2H\dot{H} = \frac{8\pi G}{3}\left(-3H\left(\rho + P\right)\right) + \frac{2kH}{a^2}$$

and this becomes

$$\dot{H} = -4\pi G\left(\rho + P\right) + \frac{k}{a^2}$$

but this isn't the form of the equation that most people are familiar with. We can eliminate the curvature term by using Friedmann 1

$$\frac{k}{a^2} = \frac{8\pi G}{3}\rho - H^2 \implies \dot{H} = -4\pi G\left(\rho + P\right) + \frac{8\pi G}{3}\rho - H^2$$

We can write everything in terms of a and its derivatives as

$$\dot{H} + H^2 = \frac{\ddot{a}a - \dot{a}^2}{a^2} + \frac{\dot{a}^2}{a^2} = \frac{\ddot{a}}{a}$$

and putting the two together gives

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + 3P\right)} \tag{2.91}$$

And this is the **second Friedmann equation**. There are a few things to note. The first is that while you have 2 differential equations, they aren't independent. Any two of Friedmann

1, Friedmann 2 and the continuity equation imply the third (this is the Bianchi identity at work). The second thing to note is that if there are multiple fluids existing in our universe, then the energy density and pressure would just be the sums over all fluids. The third is that ordinary matter and radiation have $\rho + 3P > 0$, so by (2.91) they always *decelerate* the expansion, which is what gravity being attractive looks like in cosmology. Accelerated expansion needs $P < -\rho/3$, which is what dark energy (and later, inflation) provides. The last bit is to observe that in Friedmann 1, the curvature term $-k/a^2$ acts like an extra fluid with $\rho_k \propto a^{-2}$. If the universe has nonzero spatial curvature, then the curvature contributes to how the scale factor evolves. Today, the curvature's contribution is measured to be tiny. Why it's so tiny is a puzzle that we will come back to in the chapter on inflation.

We can continue to make further simplifications! Let us look at Friedmann 1 in a flat universe; we can define the **critical density** today to be the density that would make the universe exactly flat

$$H_0^2 = \frac{8\pi G}{3}\rho_{0,\text{crit}} \implies \rho_{0,\text{crit}} = \frac{3H_0^2}{8\pi G}$$

which has a numerical value of

$$\begin{aligned}\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G} &= 1.9 \times 10^{-29} h^2 \text{ g cm}^{-3} \\ &= 2.8 \times 10^{11} h^2 M_\odot \text{Mpc}^{-3} \\ &= 1.1 \times 10^{-5} h^2 \text{ protons cm}\end{aligned}$$

where we wrote the Hubble constant as $H_0 = 100\,h$ km s^{-1} Mpc^{-1}, so $h \approx 0.67$–0.73 is a dimensionless number (not Planck's constant!). The critical density is only about five protons per cubic meter. The reason we do this is because now we can define the dimensionless density parameters

$$\Omega_i = \frac{\rho_{i,0}}{\rho_{0,\text{crit}}}, \qquad \Omega_k \equiv -\frac{k}{H_0^2}, \tag{2.92}$$

where $\rho_{i,0}$ is the density of fluid i today. Using $\rho_r \propto a^{-4}$, $\rho_m \propto a^{-3}$ and $\rho_\Lambda \propto a^0$, Friedmann 1 becomes

$$\frac{H^2}{H_0^2} = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda \tag{2.93}$$

With the constraint (from evaluating it today, where $a = 1$ and $H = H_0$) that

$$\Omega_r + \Omega_m + \Omega_\Lambda + \Omega_k = 1 \tag{2.94}$$

This is purely notational, and if you were ever to read a paper on cosmology, these are the typical notations that you will see. The measured values (from the CMB, supernovae and galaxy surveys) are roughly $\Omega_m \approx 0.31$ (of which only about 0.05 is baryons), $\Omega_\Lambda \approx 0.69$,

$\Omega_r \approx 9 \times 10^{-5}$ (photons and neutrinos) and $|\Omega_k| \lesssim 0.01$.

2.5.6 Simple Solutions And Our Universe

As you can see from Friedmann 1, the differential equations are typically very nonlinear and exact solutions will be difficult to come by. We can make some simplifications to make our lives a bit easier and illustrate the different eras of the universe. First, the easy case: a universe dominated by a single fluid with constant w. Then $\rho \propto a^{-3(1+w)}$, and in a flat universe Friedmann 1 is $\dot{a}/a \propto a^{-3(1+w)/2}$, or $a^{3(1+w)/2-1}\, da \propto dt$. Integrating gives

$$a(t) \propto \begin{cases} t^{2/3(1+w)} & w \neq -1 \\ t^{1/2} & w = 1/3 \quad \text{radiation domination (RD)} \\ t^{2/3} & w = 0 \quad \text{matter domination (MD)} \\ e^{Ht} & w = -1 \quad \Lambda \text{ domination } (\Lambda\text{D}) \end{cases} \tag{2.95}$$

with $H = H_0\sqrt{\Omega_\Lambda}$ constant in the last case. This tells us how the scale factor evolved during the different eras of the universe (figure 2.2). Because the different fluids dilute at different rates, the universe goes through a sequence of eras: radiation (falling like a^{-4}) dominates early on, then matter (a^{-3}), and finally the cosmological constant. The moment when the energy densities of 2 different fluids are equal is called **equality** (figure 2.3). Matter-radiation equality happened at $a_{\text{eq}} = \Omega_r/\Omega_m \approx 1/3400$, and matter-$\Lambda$ equality at $a_{m\Lambda} = (\Omega_m/\Omega_\Lambda)^{1/3} \approx 0.77$, only about 4 billion years ago.

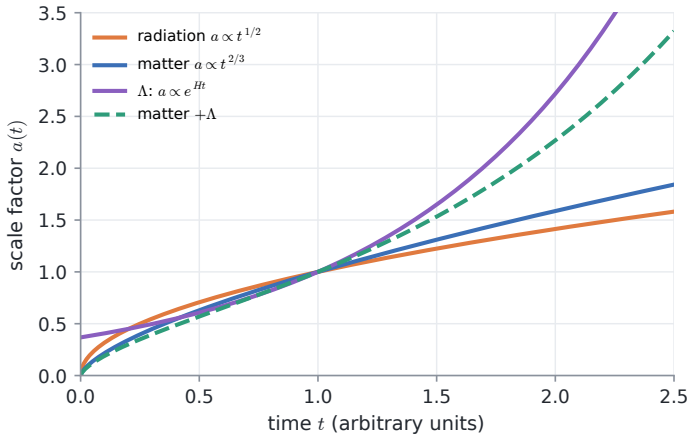


Figure 2.2: The scale factor in single-component universes, all normalized so that $a = 1$ at $t = 1$, together with a universe containing matter and a cosmological constant, which decelerates at first and then accelerates.

Before we do a two component universe, we will introduce a neat trick. Because the

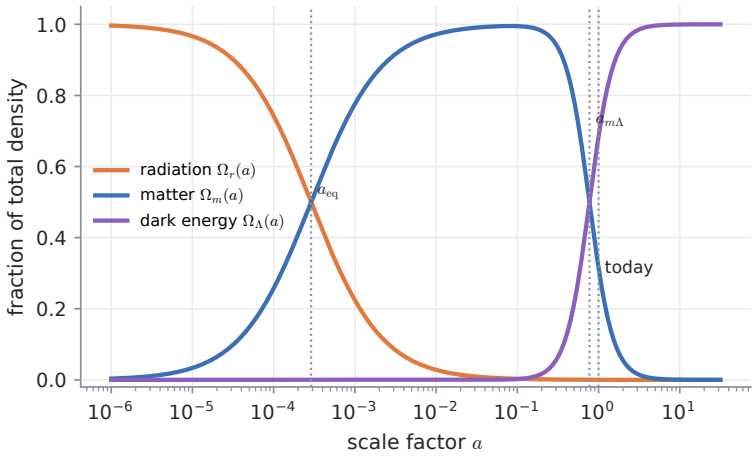


Figure 2.3: The fraction of the total energy density in radiation, matter and the cosmological constant as a function of the scale factor, for $\Omega_m = 0.315$ and $\Omega_r = 9.1 \times 10^{-5}$. The crossings are called equality.

tensors are inherently invariant under coordinate transformations, we can use whatever time coordinate we want. Remember that in our derivation of the FRW universe, we reparametrized space in terms of a variable χ

$$ds^2 = -dt^2 + a^2(t)\left[d\chi^2 + S_k^2(\chi)d\Omega^2\right]$$

where $S_k(\chi)$ is a function governed by the curvature

$$S_k(\chi) \equiv \begin{cases} \sinh(\chi) & k = -1 \\ \chi & k = 0 \\ \sin(\chi) & k = +1 \end{cases}$$

The fact that the scale factor only acts on the spatial part makes it a bit ugly because there is an asymmetry between the space and time coordinates. If we define **conformal time**

$$d\eta \equiv \frac{dt}{a(t)} \tag{2.96}$$

then the metric becomes

$$ds^2 = a^2(\eta)\left[-d\eta^2 + d\chi^2 + S_k^2(\chi)d\Omega^2\right] \tag{2.97}$$

The whole metric is now just an overall factor times a static metric. Such a rescaling is called a **conformal** transformation, because it preserves angles (but not lengths), and

in particular it preserves light cones: radial light rays move on $d\chi = \pm d\eta$, exactly as in Minkowski space. This makes conformal time very convenient for studying how light and perturbations propagate, which is why we'll use it constantly in the perturbation chapters. A comoving distance η is also the maximum distance light could have travelled since the Big Bang, so η is called the **comoving horizon**. Derivatives with respect to η are denoted by primes, and since $dt/d\eta = a$, we have

$$\mathcal{H} \equiv \frac{a'}{a} = \frac{1}{a}\frac{da}{dt}\frac{dt}{d\eta} = aH,$$

which is called the **conformal Hubble parameter**.

But maybe it's not enough to have a single component universe. Maybe we care about 2 component universes. The important example for our considerations will be the flat universe with matter and radiation, which describes our universe up until dark energy shows up. What will the first and second derivatives of a become in conformal time? Remember: $\frac{da}{d\eta} = \frac{da}{dt}\frac{dt}{d\eta} = \dot{a}a$. This means $\dot{a} = \frac{1}{a}a'$ where the prime denotes derivatives with respect to conformal time. The second derivative is

$$\frac{d^2a}{dt^2} = \frac{d}{dt}\dot{a} = \frac{d\eta}{dt}\frac{d}{d\eta}\frac{a'}{a} = \frac{1}{a}\left(\frac{a''a - (a')^2}{a^2}\right)$$

Subbing these back into the Friedmann equations (with $k = 0$)

$$\left(\frac{a'}{a^2}\right)^2 = \frac{8\pi G}{3}\rho$$

$$\implies (a')^2 = \frac{8\pi G}{3}\rho a^4$$

And then the second Friedmann becomes

$$\frac{1}{a^2}\left[\left(\frac{a''a - (a')^2}{a^2}\right)\right] = -\frac{4\pi G}{3}(\rho + 3P)$$

And using $(a')^2 = \frac{8\pi G}{3}\rho a^4$ from Friedmann 1, we get

$$\frac{a''a - (\frac{8\pi G}{3}\rho a^4)}{a^4} = -\frac{4\pi G}{3}(\rho + 3P)$$

$$\implies a'' = \frac{4\pi G}{3}(\rho - 3P)\,a^3 \tag{2.98}$$

We will note here that the scale factor is now a function of the conformal time $a(\eta)$. How

exactly does this simplify our calculations? Well, we can write the total energy density as

$$\rho = \rho_m + \rho_r = \frac{\rho_{\rm eq}}{2}\left[\left(\frac{a_{\rm eq}}{a}\right)^3 + \left(\frac{a_{\rm eq}}{a}\right)^4\right]$$

Where a_{eq} is the scale factor at matter-radiation equality. This intuitively makes sense because at equality, we expect the densities to be the same so they will contribute equal amounts. Away from equality, they will scale like how we expect them to scale. This now means that for Friedmann 2

$$a'' = \frac{4\pi G}{3}\left(\rho_m - 3P_m + \rho_r - 3P_r\right)a^3$$

We note that for radiation, $\rho_r - 3P_r = 0$ by our definition of radiation and that $P_m = 0$ also by definition. This leaves

$$a'' = \frac{4\pi G}{3}\left(\rho_m\right)a^3 = \frac{4\pi G}{3}\frac{\rho_{eq}a_{eq}^3}{2}$$

The right hand side is a constant, so this is nice because then we can just integrate twice!

$$a(\eta) = \frac{\pi G}{3}\rho_{\rm eq}a_{\rm eq}^3\eta^2 + C\eta + D \tag{2.99}$$

The Big Bang is at $a(\eta = 0) = 0$, so $D = 0$. To fix C, look at very early times, when radiation dominates. Friedmann 1 says $(a')^2 = \frac{8\pi G}{3}\rho a^4 \to \frac{8\pi G}{3}\frac{\rho_{\rm eq}}{2}a_{\rm eq}^4$ as $a \to 0$, so $C = a'(0) = \left(\frac{4\pi G}{3}\rho_{\rm eq}\right)^{1/2}a_{\rm eq}^2$. Putting it together, we have

$$a(\eta) = a_{\rm eq}\left[\left(\frac{\eta}{\eta_*}\right)^2 + 2\left(\frac{\eta}{\eta_*}\right)\right]$$

with

$$\eta_* \equiv \left(\frac{\pi G}{3}\rho_{\rm eq}a_{\rm eq}^2\right)^{-1/2}$$

At early times ($\eta \ll \eta_*$) the linear term wins and $a \propto \eta$, which is radiation domination ($a \propto t^{1/2}$ is the same as $a \propto \eta$, since $dt = a\,d\eta$). At late times the quadratic term wins and $a \propto \eta^2$, which is matter domination. Equality happens at $a = a_{\rm eq}$, which is $\eta_{\rm eq} = (\sqrt{2}-1)\eta_*$. We will use this solution a lot in the perturbation theory chapters. There are other 2 component universes you can explore (matter plus Λ is Problem 2 below), but I don't really think it's necessary to spend that much time on it. The big take away is that we now have an equation for the scale factor in the form of the Friedmann equations, and from there we can deduce the past, present, and future of the universe! As fun as this chapter is, this is

not really the main focus of cosmology. We know how spacetime bends, we know how fluids behave, and we know how the universe expands. But the story of the universe is more than just the geometry of spacetime; we will need to discuss the thermal history of the universe.

2.6 Problems

1. The Big Bang

Starting with Friedmann 2,

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P)$$

Argue that the **strong energy condition** ($\rho + 3P \geq 0$) means that $a(t)$ is concave, and therefore that if we run the expansion backwards, a must reach zero at some finite time in the past, at most $1/H_0$ ago. Call that time $t = 0$. This is called the **the Big Bang**! Does the scale factor vanishing necessarily mean that space has 0 volume?

2. Matter And Dark Energy

A flat universe, containing matter and a cosmological constant, expands according to the Friedmann equation

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\left(\rho_\Lambda + \frac{\rho_0}{a^3}\right)$$

Solve this to find

$$a(t) = \beta(\sinh \alpha t)^{2/3}$$

with α and β constant. [Hint: you may find the substitution $b = a^{3/2}$ helpful.] Verify the expected asymptotic behaviour for early and late times.

Our universe today is flat, with $\Omega_m \approx 0.3$ and $\Omega_\Lambda \approx 0.7$ and $H_0^{-1} \approx 14$Gyr. Determine:

- The age of the universe.

- The age at which the expansion of the universe first started to accelerate.

- The age at which the energy density in matter and the cosmological constant were equal.

A cosmologically ignorant civilization measures $H_0^{-1} \approx 14$ Gyr but refuses to countenance the possibility of a cosmological constant. How old would they believe their universe to be if they assumed it was flat with $\Omega_m = 1$?

3. Einstein's Biggest Blunder

Einstein introduced the cosmological constant into his field equations to avoid the conclusion

that the universe is expanding. In this problem, you will see that this was misguided.

a). Show that for a perfect fluid with positive density and pressure there is no static solution to the Einstein equations.

b). Consider now a universe with pressureless matter and a cosmological constant. Show that it is possible to obtain a static solution-called the Einstein static universe-if

$$\Lambda = 4\pi G\rho_m$$

What is the spatial curvature of this solution?

c). Show that the Einstein static universe is unstable to small perturbations. Hint: Consider a small perturbation around the static solution

$$\rho_m(t) = \rho_{m,0}[1 + \delta(t)]$$
$$a(t) = 1 + \epsilon(t)$$

with $|\delta| \ll 1$ and $|\epsilon| \ll 1$. Show that the perturbations δ and ϵ are related to each other and that they grow exponentially with time.

4. Dark Stars

With the arrival of the Einstein field equations, people were eager to begin solving it to find whatever solutions that they could find. We will consider the most symmetric and static object that we can solve for. Consider a spherically symmetric, static object with the metric

$$ds^2 = -g_{00}(r)dt^2 + g_{rr}(r)dr^2 + r^2 d\Omega^2$$

Using the Einstein equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0$$

a). Solve for the functions $g_{00}(r)$ and $g_{rr}(r)$ (you may use EinsteinPy if you wish. Or you can do it by hand).

b). The solution for your metric will admit 2 singularities. Using the scalar curvature R, determine which is a true singularity and which is an artifact of our coordinate system

c). At the coordinate singularity, determine what happens to measurements of time and measurements space. What happens to these measurements below the "event horizon"?

d). Once you have passed the event horizon, show that in order for you to leave this object,

your velocity would need to be greater than the speed of light. It should be fairly obvious by now, but do you know what object you have just discovered?

5. The 3-Sphere

Show that the embedding $X_0 = \cos\chi$, $X_1 = \sin\chi\cos\theta$, $X_2 = \sin\chi\sin\theta\cos\phi$, $X_3 = \sin\chi\sin\theta\sin\phi$ describes the unit 3-sphere $\sum_i X_i^2 = 1$, and that the induced metric $\sum_i dX_i^2$ is $d\chi^2 + \sin^2\chi\, d\Omega^2$. What is the total volume of the 3-sphere of radius a? (This would be the volume of a closed universe with $k = +1$.)

3

The Thermal History Of The Universe

"Our whole universe was in a hot dense state..."

– Barenaked Ladies

In the last chapter, we saw space and time interconnected with each other and how the universe expands. However, the universe is more than just the stage in which the actors perform; it also involves the actors themselves. What we are really interested in this chapter is the thermal history of the universe. Run the expansion backwards and everything gets denser and hotter, since radiation redshifts like a^{-1}, and so the temperature of the universe goes like $T \propto 1/a$. Today the universe is a chilly 2.7 K. When it was a thousand times smaller, it was a few thousand kelvin, hot enough to ionize hydrogen. When it was a billion times smaller, it was hot enough to break apart atomic nuclei, and even earlier it was hot enough to create every particle in the Standard Model out of the thermal bath. This chapter tells that story in order, starting from the hottest moments that we understand and following the universe as it cools: matter beating antimatter, dark matter freezing out, neutrinos decoupling, the first nuclei forming, and finally atoms forming and releasing the light that we see today as the cosmic microwave background.

The reason we can tell this story at all is that, for most of its early history, the universe was in (or very close to) thermal equilibrium, so we can describe it with statistical mechanics. The interesting events, the ones that left fossils that we can still measure, all happened when some particle species *fell out* of equilibrium. So our plan is to first build up the equilibrium statistical mechanics that we need, then learn how to describe departures from

equilibrium with the Boltzmann equation, and then turn the crank on the timeline. The quantum mechanical initial conditions of the universe will get their own chapter (inflation) after this one.

Before we get started, I would like to define some useful constants. In this chapter we set $\hbar = c = k_B = 1$, so temperature has units of energy and we can measure everything in electron volts (1 eV $\approx$ 11,600 K). Starting off, we have the reduced Planck mass

$$M_{\rm Pl} \equiv \sqrt{\frac{\hbar c}{8\pi G}} = 2.4 \times 10^{18}{\rm GeV} \tag{3.1}$$

In flat space, Friedmann 1 becomes

$$H^2 = \frac{\rho}{3M_{\rm Pl}^2}. \tag{3.2}$$

We will also use the proton mass $m_p \approx 0.938$ GeV and the electron mass $m_e = 0.511$ MeV as rough landmarks. These are purely notational and to avoid clutter (because famously statistical mechanics is cluttered with constants).

3.1 The Tools: Equilibrium Statistical Mechanics

I believe that one of the biggest successes in modern physics is statistical mechanics. A lot of people believe that the birth of quantum mechanics came from the double slit experiment, photoelectric effect, Stern Gerlach. While I believe that these are important experiments and people need to know them, I don't believe that they were the first or the most consequential to the development of quantum mechanics. I believe that honor goes to the ultraviolet catastrophe, which was solved by none other than Max Planck, and fittingly, the blackbody spectrum he found is exactly what we will see in the CMB at the end of this chapter. As much as I would love to go into the nitty gritty details, that is not the focus of these notes. We will derive the main results of statistical mechanics, and use them to describe the things that we need.

3.1.1 The Partition Function

Because the systems that we are dealing with are in equilibrium (we will argue why later in the chapter), it would be good to have an equilibrium distribution for the thermodynamic quantities that we want to solve. However, for calculation purposes, it's not so much the distributions that are interested in but instead their averages. There is a way to retrieve all of the relevant thermodynamic averages from a single function, and this function is called the **partition function.** It is the normalization factor for the probability distribution of a

system in thermal equilibrium with a heat bath at fixed temperature T. The fact that we can we can retrieve all quantities of interest makes the partition function a very important object in statistical mechanics, and even in quantum field theory as it is the object that gives us the correlation functions. Since derivatives of $\ln Z$ give averages, the partition function plays the role of a **moment generating function**, and we will derive it from first principles. The characteristic property of equilibrium is the maximization of entropy, subject to whatever constraints we impose. There are a lot of ways to formalize the entropy, but we will use the Gibbs entropy as it makes a deep connection between information theory and statistical mechanics. We define the Gibbs entropy as

$$S = -k_B \sum_{i=1}^{n} p_i \ln(p_i) \tag{3.3}$$

I have written out the entropy with the Boltzmann constant k_B but moving forward we will suppress all fundamental constants. The p_i is the probability of a being in a certain microstate with energy E_i. The system is in contact with the reservoir, so its energy can fluctuate. We can define the average fluctuation to be

$$\langle E \rangle = \sum_i E_i p_i \tag{3.4}$$

Since we are dealing with probabilities, we will also need to fix a normalization constraint

$$\sum_i p_i = 1 \tag{3.5}$$

If we maximize the entropy given these constraints, then we hope that it will give us the equilibrium distribution in the canonical ensemble. How do we maximize the constrained entropy? By using Lagrange multipliers

$$S = -\sum_i p_i \ln(p_i) - \alpha \left(\sum_i p_i - 1 \right) - \beta \left(\sum_i E_i p_i - E \right)$$

If we take derivatives of the entropy with respect to the probabilities, we will get

$$\frac{\partial S}{\partial p_i} = -\ln(p_i) - 1 - \alpha - \beta \left(E_i \right) = 0$$

If we rearrange and then solve for the p_i, we find

$$\ln(p_i) = -1 - \alpha - \beta E_i$$

If we redefine $\tilde{\alpha} = 1 + \alpha$, then we have

$$p_i = e^{-\tilde{\alpha}} e^{-\beta E_i}$$

Now imposing normalization to unity, we get

$$\sum_i p_i = \sum_i e^{-\tilde{\alpha}} e^{-\beta E_i} = 1$$

We find that the normalization constant is $e^{-\tilde{\alpha}} = 1/Z$ and we can define the canonical partition function to be

$$Z = \sum_i e^{-\beta E_i} \tag{3.6}$$

For those that are well versed in statistical mechanics, you might identify the Lagrange multiplier β as the inverse temperature $1/T$. This speaks to a deeper truth that the Lagrange multipliers for the Gibbs entropy are thermodynamic observables. Now that we have the partition function...now what? My claim is that once you have the partition function, then all desired quantities are stored within it!

Problem 1.

Show that average energy can be extracted from the partition function. Then with the internal energy obtained, show that the Helmholtz free energy defined as

$$F = E - TS$$

can be obtained also from the partition function.

Solution:

First we need to calculate the average energy. We know that the average energy is defined as

$$\langle E \rangle = \sum_i E_i p_i$$

We notice that from the definition of the partition function, we can just bring down a factor of E_i by taking a derivative

$$\frac{\partial}{\partial \beta} Z = \frac{\partial}{\partial \beta} \sum_i e^{-\beta E_i} = -\sum_i E_i e^{-\beta E_i}$$

But we know that $p_i = e^{-\beta E_i}/Z$ and so if we sub that in and divide out Z, we get

$$\boxed{-\frac{1}{Z}\frac{\partial Z}{\partial \beta} = \sum_i E_i p_i = \langle E \rangle}$$

Or sometimes as you will see it

$$\boxed{-\frac{\partial \ln Z}{\partial \beta} = \langle E \rangle} \tag{3.7}$$

As for the Helmholtz free energy, We just need to subtract the above by TS, but this requires us to know the entropy as a function of the partition function. This isn't horrible, all we need to do is use

$$S = -\sum_i p_i \ln(p_i)$$

We know that $p_i = e^{-\beta E_i}/Z$ and therefore $\ln(p_i) = -\beta E_i - \ln(Z)$

$$S = \sum_i \frac{e^{-\beta E_i}}{Z}\left(\beta E_i + \ln(Z)\right) = \frac{\beta}{Z}\sum_i e^{-\beta E_i} E_i + \frac{\ln(Z)}{Z}\sum_i e^{-\beta E_i}$$

The second term just evaluates to $\ln(Z)$ by definition of Z. The first term is nothing more than just the average energy. Thus we have

$$S = \beta \langle E \rangle + \ln(Z)$$

For now on, I am going to drop the brackets around E because it is understood that it is the average energy. We also recognize that β is the inverse temperature, so when we multiply the entropy by temperature, that term will cancel out. Thus the free energy is

$$\begin{aligned} F = E - TS &= E - T\left(\beta E + \ln(Z)\right) \\ &= E - E - T\ln(Z) \end{aligned}$$

The first and second terms cancel because $T\beta = 1$. Thus the free energy can be written as

$$\boxed{F = -T\ln Z} \tag{3.8}$$

As you can see, the partition function is a power tool. In trying to find the free energy, we inadvertently found the entropy as a function of the partition function. Standard problems

in statistical mechanics will ask you to find the partition function for some system and ask you to find the average energy and heat capacity and blah blah blah. All that you really need to know is that once you have the partition function, you have it all.

The derivation of the canonical partition function hinged on the fact that we were in thermal equilibrium with a heat bath. We can go one step further and relax the condition that the number of particles is fixed. We will allow the number of particles to fluctuate, and the thermodynamic quantity that allows that to happen is called the **chemical potential** μ. It is very confusing because μ is the chemical potential and it is also a spacetime index. But I am sure you are smart enough to understand context. If we allow exchange of both energy and particles with the heat bath, what becomes of our partition function? Well, first and foremost it graduates to being the **grand canonical partition function** with the fancy dash through it Ξ. Get it? Because it's *grand*. Anyways, to get our partition function, we do exactly what we did before because are still in equilibrium, we write the entropy

$$S = -\sum_{N,i} p_{N,i} \ln p_{N,i} - \alpha \left(\sum_{N,i} p_{N,i} - 1 \right) - \beta \left[\sum_{N,i} p_{N,i} \left(E_{i,N} - \mu N \right) - \langle E - \mu N \rangle \right]$$

We maximize the entropy subject to those constraints but we won't do it here because it's literally the same procedure as last time. The resulting grand partition function is

$$\Xi = \sum_{N=0}^{\infty} \sum_{i} \exp\left[-\beta \left(E_{i,N} - \mu N \right) \right] \tag{3.9}$$

This grand partition function has the same properties as our canonical partition function.

3.1.2 The Particle Distribution Functions

With the possession of the grand partition function, we can now derive the distribution functions for the 2 types of particles in our universe (the reason why there are only 2 types of particles, fermions and bosons, has to do with quantum field theory, through the spin-statistics theorem). The fact that we are allowing particle number to fluctuate means that the "distribution function" we are solving for are really just the average number of particles at some energy state $\langle n \rangle$. Let us start with fermionic particles. Fermions are half integer spin particles that obey the Pauli exclusion principle, which just states that no 2 fermions can occupy the same quantum state. Thus, without loss of generality (explained in a bit why), we can just consider single quantum state system. Let us suppose that the energy of this quantum state is E. The grand partition function becomes (after imposing

the Pauli exclusion principle)

$$\begin{aligned}\Xi &= \sum_{N=0}^{1} \exp\left[-\beta\left(E_N - \mu N\right)\right] \\ &= e^{-\beta * 0 * (E-\mu)} + e^{-\beta * 1 * (E-\mu)} \\ &= 1 + e^{-\beta(E-\mu)}\end{aligned}$$

For a single particle state, there are only 2 possible configurations for fermions to obey the exclusion principle: either 1 particle is in it or 0 particles are in it. Once we have our partition function, we can now find the distribution for the occupation number, which is defined as

$$\langle n \rangle = \frac{1}{\Xi} \sum_{N=0}^{1} N\, e^{-\beta N(E-\mu)}$$

Again, using the fact that there can only be either 0 particles or a single particle, we have

$$\langle n \rangle = \frac{1}{1 + e^{-\beta(E-\mu)}} e^{-\beta(E-\mu)}$$

or sometimes as it's known

$$f_{FD}(E) = \frac{1}{1 + e^{\beta(E-\mu)}} \tag{3.10}$$

which has the name **the Fermi-Dirac distribution.** Now, you might ask: we derived this using a single particle state. What if there were multiple particle states? How would we include that? Well, the answer is that since fermions obey the exclusion principle, each particle state can only be occupied by a single fermion at most. What that means is that *each energy state will have its own FD distribution.* In general, if you had many particle states (of non interacting fermions), then the partition function would just be

$$\Xi = \prod_{i=1}^{N} \Xi_i = \prod_{i=1}^{N} \left(1 + e^{-\beta(E_i - \mu)}\right)$$

in which case, the Fermi Dirac distribution just becomes

$$f_{FD}(E_i) = \frac{1}{1 + e^{\beta(E_i - \mu)}} \tag{3.11}$$

and we can find the total particle number N by

$$N = \sum_i \langle n_i \rangle$$

Oftentimes, as you will see later, we will replace this sum with an integral.

Problem 2.

Take the limit as the temperature becomes very low $\beta \gg 1$ and show what happens to the Fermi Dirac distribution. If you allow the temperature to warm up a bit, what do you expect to see happen to the FD distribution?

Solution:

The FD distribution is given as

$$f_{FD}(E) = \frac{1}{1 + e^{\beta(E-\mu)}}$$

Now, in the limit that $\beta \to \infty$, if $E > \mu$the denominator becomes very very large, and the function is equal to 0. Now, if $E < \mu$, then we have essentially

$$\lim_{\beta \to \infty} \frac{1}{1 + e^{-\beta|E-\mu|}} = 1.$$

Thus, if we plot the FD distribution as a function of the energy, then we would actually see a step function with the step happening at $E = \mu$. The energy in which the $E = \mu$ is called the **Fermi energy**, and so oftentimes you might see that μ gives the Fermi energy. Physically what this is telling us that when the temperatures get very cold (relative to the Fermi energy), all of the fermions will try to occupy the lowest energy state that it can. However, because fermions obey the exclusion principle, they can't all occupy the same lowest energy so the fermions will just fill the next available energy state until all the fermions are "stacked" on top of each other. The energy of the highest energy state occupied by the last fermion is the Fermi energy. We then expect that for very low nonzero temperatures, it is the fermions near the top that get kicked up first, and so we should see the FD "smear" out from being a step function over a width of a few T. A plot is shown in figure 3.1 to visualize this.

This covers the fermions, what about the other half of the universe? Well, they are called bosons. The force carriers of the standard model are the gauge bosons. They are classified by their integer spins, and most importantly, they do NOT obey the Pauli exclusion principle. This means that if we were to try to calculate the partition function for bosons, we would need to sum over all possible number of particles that can occupy that state. Let us consider our 1 particle state toy model and see if we can squeeze out the partition function. We would now need to use the grand partition function in its full glory

$$\Xi = \sum_{N=0}^{\infty} \exp\left[-\beta\left(E_N - \mu N\right)\right]$$

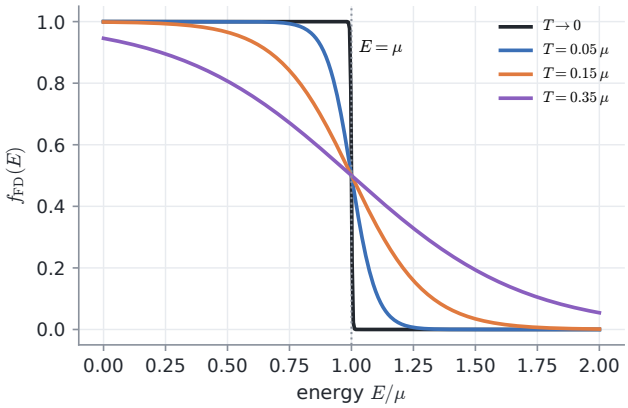


Figure 3.1: The Fermi-Dirac distribution at several temperatures. At $T \to 0$ it is a step at $E = \mu$, and as the temperature rises the step smears out over a width of a few T.

This can seem daunting at first, but remember that this is a single state system, so we can just ignore the sum over all energies. That single energy state will have N particles allowed in it, so we can just write the

$$\Xi = \sum_{N=0}^{\infty} \exp\left[-\beta N(E-\mu)\right]$$

This seems daunting at first, but realize that this is nothing more than just the geometric series. The base is e and the only variable being summed is N. Every other term are just constants. Thus, we know how to evaluate geometric series they are just

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \tag{3.12}$$

As such, we should expect that the grand partition function is just

$$\Xi = \frac{1}{1-e^{-\beta(E-\mu)}} \tag{3.13}$$

and you would be correct! Now to calculate the average particle occupation number, we just need to find

$$\langle n \rangle = \frac{1}{\Xi} \sum_{N=0}^{\infty} N e^{-\beta N(E-\mu)} \tag{3.14}$$

Now surely *this* is difficult to solve? Sorta. It depends. There are 2 ways to go about doing

this. The first is to just look up the answer. The second is to use a trick. The trick is that

$$\sum_n x^n = \frac{1}{1-x} \implies \frac{\partial}{\partial x}\sum_n x^n = \sum_n nx^{n-1} = \frac{1}{(1-x)^2}$$
$$= \sum_n nx^n = \frac{x}{(1-x)^2}$$

where in the second line we multiplied both sides by x. If we use this with $x = e^{-\beta(E-\mu)}$, we find that

$$\langle n \rangle = \frac{1}{\Xi}\frac{e^{-\beta(E-\mu)}}{\left(1-e^{-\beta(E-\mu)}\right)^2}$$

With having the partition just being the term in the denominator, we have that

$$\langle n \rangle = \frac{e^{-\beta(E-\mu)}}{1-e^{-\beta(E-\mu)}}$$

Which simplifies down to

$$f_{BE}(E) = \frac{1}{e^{\beta(E-\mu)}-1} \tag{3.15}$$

and the name of this distribution is called the **Bose-Einstein distribution**. We note here that the product property of the partition function for many states works all the same for the Bose-Einstein distribution. Note that the BE distribution doesn't have a nice limit at low temperatures like the FD, and the reason for this is that the BE distribution diverges as $E \to \mu$ (which is why we need μ to be less than the lowest energy). What does this mean physically? Since bosons don't have to obey the Pauli exclusion principle, at low temperatures, they can just pile up on top of each other at the lowest energy state. There is a critical temperature in which this condensation happens, and the phase that is exhibited is called **Bose Einstein condensation.**

3.1.3 The Density Of States

Before we can start making any reasonable calculations using the particle distributions, we need to know at any given energy level, how many states are available for the particle to occupy? This quantity is known as **the density of states** and it is typically denoted as $g(E)dE$. Sometimes you will hear the definition of the density of states being the number of available quantum states within $E + dE$ and E. If we think about a particle confined into a box of length L, the volume is L^3. In phase space, we can parametrize the momentum state as

$$\langle p_x, p_y, p_z \rangle = \langle \frac{2\pi\hbar}{L}n_x, \frac{2\pi\hbar}{L}n_y, \frac{2\pi\hbar}{L}n_z \rangle \tag{3.16}$$

Then, as we can see, in momentum space, the number of states with momentum p_n and the next available momentum p_{n+1} is

$$\Delta p_x \Delta p_y \Delta p_z = \left(\frac{2\pi\hbar}{L}\right)^3 \Delta n = \frac{(2\pi\hbar)^3}{V} \Delta n_x \Delta n_y \Delta n_z$$

In this example, each state takes up one cell of momentum space with $\Delta n_x = \Delta n_y = \Delta n_z = 1$, so the number of states per unit volume of momentum space is

$$\frac{\Delta N}{\Delta p_x \Delta p_y \Delta p_z} = \frac{V}{(2\pi\hbar)^3}.$$

This derivation is only considering 3D space but you can see how easily this translates to arbitrary dimensions. From here, we can use dimensional analysis. The quantity that we just derived is the number of states available per some unit momentum. To get the number of states with some momentum in some infinitesimally small element d^3p, multiply by d^3p

$$dN = \frac{V}{(2\pi\hbar)^3} d^3p$$

Assuming a rotationally invariant box, this just becomes

$$dN = \frac{V}{(2\pi\hbar)^3} 4\pi p^2 dp$$

The above equation is an equation of the momentum. We were originally interested in the density of states for some energy. No matter! We can use the dispersion relation. For a non relativisic particle, the dispersion relation is

$$E = \frac{p^2}{2m} \implies p = \sqrt{2mE} \tag{3.17}$$

$$dp = \frac{m}{\sqrt{2mE}} dE \tag{3.18}$$

and thus, the density of states is (remembering to divide through by the volume)

$$g(E)dE = \frac{1}{(2\pi\hbar)^3} 4\pi \left(2mE\right) \left(\frac{m}{\sqrt{2mE}} dE\right) = \frac{1}{(2\pi\hbar)^3} 4\pi\sqrt{2} m^{3/2} \sqrt{E} dE \tag{3.19}$$

Problem 3.

Derive the density of (energy) states (DOS) $g(E)dE$ for the 2-dimensional case. Assume that the particles are "free" inside this box so that you may use the particle-in-a-box quantization (an example of a non-free particle DOS problem will be worked out later in a future problem)

Solution: Perhaps in this problem we will go a bit slower to justify why the density of states is the way that it is. Suppose you have a particle that is free but confined in some box. From quantum mechanics, we know that the solutions to the Schrodinger equation are

$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

with the A term representing positive momentum states and the B term negative ones. In the confinement of a box, the boundary conditions (you can think of it as periodic if you wish) force the k values to take on

$$k = \frac{2n\pi}{L}$$

Now, there are 2 ways to obtain the momentum quantization. The first way is to remember from basic physics that $p = \hbar k$. The second is to use the fact that the momentum operator in position space is $i\hbar\frac{\partial}{\partial x}$ and this will bring down the coefficients in the exponent. In either case, you get the same result which is that

$$p = \frac{2\pi n\hbar}{L}$$

and if we were in 2 dimensions, the p_x and p_y are independent of each other so we have

$$\langle p_x, p_y\rangle = \langle \frac{2\pi\hbar}{L}n_x, \frac{2\pi\hbar}{L}n_y\rangle$$

The number of states between p_n and the next p_{n+1} is

$$\Delta p_x \Delta p_y = \left(\frac{2\pi\hbar}{L}\right)^2 \Delta n_x \Delta n_y$$

Which, so far, isn't different from the 3D case except for the power. Each state takes up an area $(2\pi\hbar/L)^2$ of momentum space, so the number of states in a small area d^2p is

$$dN = \left(\frac{L}{2\pi\hbar}\right)^2 d^2p$$

Since we are in 2D and the particle is free (and no preferred momentum direction), $d^2p = 2\pi p dp$

$$\implies g(p)dp = \frac{1}{L^2}\left(\frac{L}{2\pi\hbar}\right)^2 2\pi p\, dp$$

Remembering our nonrelativisic dispersion relation, we can find the density of energy states as

$$\begin{aligned} g(E)dE &= \left(\frac{1}{2\pi\hbar}\right)^2 2\pi\sqrt{2mE}\frac{mdE}{\sqrt{2mE}} \\ &= \boxed{\frac{m}{2\pi\hbar^2}dE} \end{aligned}$$

In general, then, we find that the density of states $g(E)$ is

$$g(E) = \frac{1}{V}\frac{dN}{dE} \tag{3.20}$$

The reason we need this is because if we have the particle number distribution, we would like to use it to find the average energy and average pressure. For that to happen, we need to know the number of allowed states for some given energy. Hence, the necessity for the density of states. This allows us to write the number density of particles as (dropping $\hbar$, and with $g(E)$ already a density per unit volume)

$$n(T) = \int dE\ f(E)g(E) \tag{3.21}$$

And if we wanted to calculate the energy density

$$\rho(T) = \int dE\ f(E)g(E)\ E \tag{3.22}$$

and the pressure

$$P(T) = \int dE\ f(E)g(E)\frac{p^2(E)}{3E} \tag{3.23}$$

with

$$f(E) = \frac{1}{1 \pm e^{\beta(E-\mu)}}$$

The pressure formula comes from kinetic theory: a particle with momentum p and velocity $v = p/E$ bouncing off a wall transfers momentum at a rate that averages to $pv/3$ per particle over all directions. We note that all of these are densities because $g(E)$ is the number of states per unit volume. Let us see how we can use these distributions for calculations that

we might be interested in.

Problem 4.

Here is a famous statistical mechanics calculation that I believe really solidifies the power and importance of these results. Suppose we have a neutron star of mass M. A neutron star is the corpse of a star that was not large enough to collapse into a black hole. Neutrons, as we know, are spin 1/2 particles, so they are fermions.
a). Using the Fermi-Dirac distribution, show that the Fermi temperature (defined to be the temperature of a Fermi gas where the average kinetic energy is equal to the Fermi energy) is much, much greater than the typical temperature of a neutron star (approximately 10 million Kelvin) and therefore the star can be approximated as a Fermi gas at very low temperatures.
b). Now that we know what the distribution should look like approximately, find the pressure inside the star.
Solution:
a). The first thing that we need to do is to find the Fermi energy. At very low temperatures the FD distribution is a step function, $f = \Theta(\mu - E)$, and neutrons have spin 1/2, so there are $g = 2$ spin states per momentum state. The number density is then (setting $\hbar = 1$)

$$n = 2\int_0^\mu \frac{4\pi\sqrt{2}m^{3/2}}{(2\pi)^3}E^{1/2}\,dE = \frac{\sqrt{2}m^{3/2}}{\pi^2}\cdot\frac{2}{3}\mu^{3/2} \implies \boxed{\mu = \frac{1}{2m}\left(3\pi^2 n\right)^{2/3}}$$

which is the same as saying that the Fermi momentum is $p_F = (3\pi^2 n)^{1/3}$. For a neutron star of mass $M = 1.4M_\odot$ and radius $R = 10$ km, the number density is $n = M/(m_n \cdot \frac{4}{3}\pi R^3) \approx 4\times 10^{44}$ m^{-3}. Putting $\hbar$ back, $p_F = \hbar(3\pi^2 n)^{1/3}$, and the Fermi energy is

$$\mu = \frac{p_F^2}{2m_n} \approx 1.7\times 10^{-11}\text{ J} \approx 107\text{ MeV},$$

which corresponds to a Fermi temperature $T_F = \mu/k_B \approx 1.2\times 10^{12}$ K. That is about 10^5 times hotter than the actual temperature of a neutron star, so we can approximate the neutron star as a free Fermi gas at absolute zero. (The Fermi velocity comes out to about half the speed of light, so a relativistic treatment would change the numbers somewhat, but not the conclusion.)
b). We can use the pressure formula, with $p^2/(3E) \to p^2/(3m) = 2E/3$ for non-

relativistic particles,

$$P = \int_0^\mu \frac{\sqrt{2}m^{3/2}}{\pi^2} E^{1/2} \cdot \frac{2E}{3}\, dE = \frac{2\sqrt{2}m^{3/2}}{3\pi^2} \cdot \frac{2}{5}\mu^{5/2} \implies \boxed{P = \frac{2}{5}n\mu}$$

Plugging in, $P \approx 0.4 \times (4 \times 10^{44}) \times (1.7 \times 10^{-11}) \approx 3 \times 10^{33}$ Pa. This is the **degeneracy pressure**, which has nothing to do with temperature. It's the Pauli exclusion principle pushing back, and it's what holds the neutron star up against gravity.

3.1.4 Adding Relativity

The calculations that we have done so far assume that the particles are non-relativistic. It's obvious then what the next step should be: to include the relativistic dispersion relations. For now, we will set the chemical potential μ to be 0. The justification for this is two-fold. The first is that it will simplify our calculations, and the second is that in the early universe, particles are either created and destroyed freely in fast reactions (like photons, whose number isn't conserved, forcing $\mu_\gamma = 0$) or have very small μ relative to the temperature (like electrons). For the latter, this is because particles and antiparticles were almost equally abundant (there is a slight asymmetry, about one part in a billion, but relative to the temperature, it is tiny). So for all intents and purposes, we can set the chemical potential to 0 and not lose much. Our calculations for the number density becomes

It is easiest to integrate over momentum instead of energy, with the relativistic energy being $E = \sqrt{p^2 + m^2}$. Including g internal degrees of freedom (spin states, colors, and so on), the number density is

$$n(T) = g \int_0^\infty \frac{dp}{2\pi^2} \frac{p^2}{\exp\left[\beta\sqrt{p^2 + m^2}\right] \pm 1} \tag{3.24}$$

And similarly for the energy density

$$\rho(T) = g \int_0^\infty \frac{dp}{2\pi^2} \frac{p^2\sqrt{p^2 + m^2}}{\exp\left[\beta\sqrt{p^2 + m^2}\right] \pm 1} \tag{3.25}$$

We will hold off on the calculation for pressure in a second because it follows quite nicely from what we are going to do. First, we have to deal with the above integrals. We can do a bit of variable substitution to make it more palatable. Let $\xi \equiv p/T$ and $x = m/T$. If we

remember that $\beta = 1/T$, then the number and energy densities just become

$$n(T) = \frac{gT^3}{2\pi^2}\int_0^\infty d\xi\ \frac{\xi^2}{\exp\left[\sqrt{x^2+\xi^2}\right] \pm 1} \tag{3.26}$$

$$\rho(T) = \frac{gT^4}{2\pi^2}\int_0^\infty d\xi\ \frac{\xi^2\sqrt{\xi^2+x^2}}{\exp\left[\sqrt{\xi^2+x^2}\right] \pm 1} \tag{3.27}$$

What's left to do is to define the integrals

$$I_\pm(x) = \int_0^\infty d\xi\ \frac{\xi^2}{\exp\left[\sqrt{x^2+\xi^2}\right] \pm 1} \tag{3.28}$$

$$J_\pm(x) = \int_0^\infty d\xi \frac{\xi^2\sqrt{\xi^2+x^2}}{\exp\left[\sqrt{\xi^2+x^2}\right] \pm 1} \tag{3.29}$$

Oftentimes these integrals will have to be evaluated numerically. We are still given a bit of scrap to work with, and that is in the extreme limits of these integrals. For example, at temperatures *much* larger than the mass of the particles, we can say that $x \to 0$ and in this limit, the integrals just become

$$I_\pm(0) = \int_0^\infty d\xi\ \frac{\xi^2}{e^\xi \pm 1}$$

$$J_\pm(0) = \int_0^\infty d\xi \frac{\xi^3}{e^\xi \pm 1}$$

Cosmology is not the study of how to perform integrals, so the wise thing to do in this case is to throw it into Mathematica!

```
FermiI = Integrate[xi^2/(Exp[xi] + 1), {xi, 0, Infinity}]
BoseI = Integrate[xi^2/(Exp[xi] - 1), {xi, 0, Infinity}]
FermiJ = Integrate[xi^3/(Exp[xi] + 1), {xi, 0, Infinity}]
BoseJ = Integrate[xi^3/(Exp[xi] - 1), {xi, 0, Infinity}]

output: (3 Zeta[3])/2
output: 2 Zeta[3]
output: (7 \[Pi]^4)/120
output: \[Pi]^4/15
```

This allows us to conclude that

$$n = \frac{\zeta(3)}{\pi^2} g T^3 \begin{cases} 1 & \text{bosons} \\ \frac{3}{4} & \text{fermions} \end{cases} \tag{3.30}$$

and

$$\rho = \frac{\pi^2}{30} g T^4 \begin{cases} 1 & \text{bosons} \\ \frac{7}{8} & \text{fermions} \end{cases} \tag{3.31}$$

The last thing that we need to do is solve for the pressure. In the ultra relativistic limit, $p \gg m$ and so $E = p$. From this, we get

$$P = g T^4 \int_0^\infty \frac{d\xi}{2\pi^2} \frac{1}{e^\xi \pm 1} \frac{\xi^3}{3}$$

But wait! The above integral is the exact same as that of energy density except for a factor 1/3. Thus we justify our earlier claim that

$$P = \frac{1}{3}\rho$$

as expected. The factor $\zeta(3) \approx 1.202$ is the Riemann zeta function, and these two formulas, with their 3/4 and 7/8 for fermions, are the most used results in this chapter.

3.1.5 Non-Relativistic Particles

Particles during the early universe were mostly relativistic and this was due to the fact that the temperature of the universe was very hot. Sorta a non-statement. The real question is: what happens to particles when the temperature drops below their mass? Do they still contribute to the equilibrium distribution?

Problem 5.

In non-relativistic limit, what happens to number density?

Solution:

In the non-relativistic limit, $x = m/T \gg 1$, the exponential in the denominator dominates over the ± 1. Thus what we have is

$$n(T) = \frac{g T^3}{2\pi^2} \int_0^\infty d\xi \frac{\xi^2}{e^{\sqrt{x^2+\xi^2}}}$$

If the mass of the particle is now significantly larger than its momentum, then we can expand the square root in the denominator to first order as

$$\sqrt{x^2+\xi^2} \approx x + \frac{\xi^2}{2x}$$

$$\implies n(x) = \frac{gT^3}{2\pi^2}\int_0^\infty d\xi\ \xi^2 e^{-x-\xi^2/2x}$$

The integral above is just the second moment of the Gaussian. Thus what we are really calculating is

$$\begin{aligned} n(x) &= \frac{gT^3}{2\pi^2}e^{-x}\int_0^\infty d\xi\ \xi^2 e^{-\xi^2/2x} \\ &= \frac{gT^3}{2\pi^2}e^{-x}\left(\sqrt{\frac{\pi}{2}}x^{3/2}\right) \end{aligned}$$

In the second line we used Mathematica to compute the integral. This gives us the number density to be

$$n = g\left(\frac{mT}{2\pi}\right)^{3/2} e^{-m/T} \tag{3.32}$$

(If the particle has a chemical potential, $e^{-m/T}$ becomes $e^{-(m-\mu)/T}$.) Which means that as the universe cools down, particles whose mass is larger than the temperature have an exponential suppression factor in their number density. Physically, the thermal bath no longer has enough energy to create them in pairs, but they can still annihilate, so in equilibrium they disappear. The same exponential suppression shows up in ρ and P, and since $P \approx nT \ll \rho \approx mn$, they behave like pressureless matter.

3.2 The Hot Big Bang In Equilibrium

3.2.1 Counting Degrees Of Freedom

Problem 5 tells us that as the temperature of the early universe cools down, a species goes from having $n \propto T^3$ (relativistic) to being exponentially suppressed, and this transition happens when the temperature drops to around the particle's mass. So at any temperature, the energy budget of the universe is dominated by the species that are still relativistic, each contributing $\frac{\pi^2}{30}g_i T_i^4$ (times 7/8 for fermions). Table 3.1 lists the particles of the Standard Model with their masses and internal degrees of freedom. In general, we can have a collection of different particles with different temperatures (a species that has stopped interacting with

the rest can cool at its own rate), so the total energy density is

$$\rho_{\rm tot} = \sum_{i=\text{species}} \frac{g_i}{2\pi^2} T_i^4 J_\pm(x_i)$$

Table 3.1: The particles of the Standard Model, with approximate masses and internal degrees of freedom g. For example, a quark has 2 spin states, 3 colors and an antiparticle, so $g = 12$, while the photon has 2 polarizations.

	Particle	Mass	Spin	g
Quarks	$t, \bar{t}$	173 GeV	1/2	12
	$b, \bar{b}$	4.2 GeV	1/2	12
	$c, \bar{c}$	1.3 GeV	1/2	12
	$s, \bar{s}$	95 MeV	1/2	12
	$d, \bar{d}$	4.7 MeV	1/2	12
	$u, \bar{u}$	2.2 MeV	1/2	12
Gluons	g	0	1	16
Leptons	$\tau^\pm$	1.78 GeV	1/2	4
	$\mu^\pm$	106 MeV	1/2	4
	$e^\pm$	511 keV	1/2	4
	$\nu, \bar{\nu}$ (3 flavors)	< 0.1 eV	1/2	3×2
Gauge bosons	$W^\pm$	80 GeV	1	6
	Z	91 GeV	1	3
	γ	0	1	2
Higgs	H	125 GeV	0	1

$$\rho_{\rm tot} = \sum_{i=\text{species}} \frac{g_i}{2\pi^2} T_i^4 J_\pm(x_i)$$

For notational purposes, it's easier to define a relative temperature T_i/T(relative to the photon temperature $T_\gamma = T$) and define an **effective degree of freedom**

$$g_*(T) \equiv \sum_{i=\text{species}} g_i \frac{T_i^4 J_\pm(x)}{T^4 J_-(0)} \tag{3.33}$$

For a relativistic species, $J_\pm(x_i)/J_-(0)$ is 1 for a boson and 7/8 for a fermion, and for a non-relativistic one it's exponentially small. So if we only keep relativistic species, we can write it out explicitly as

$$g_*(T) \equiv \sum_{i=\text{boson}} g_i \frac{T_i^4}{T^4} + \frac{7}{8} \sum_{i=\text{fermion}} g_i \frac{T_i^4}{T^4}$$

And this will allow us to write the energy density of the whole universe as a single term

$$\boxed{\rho_r = \frac{\pi^2}{30} g_\star(T)\, T^4} \tag{3.34}$$

When every particle in the Standard Model is relativistic and in equilibrium (above about 200 GeV), all the T_i are equal and we can just count. The bosons give $16+6+3+2+1 = 28$ and the fermions give $72 + 12 + 6 = 90$, so

$$g_\star = 28 + \frac{7}{8} \times 90 = 106.75.$$

As the universe cools, each species drops out of the sum when T falls below its mass, and $g_\star$ decreases in steps (figure 3.2). Just above 1 MeV the only relativistic species left are photons, electrons, positrons and the three neutrinos, so $g_\star = 2 + \frac{7}{8}(4+6) = 10.75$.

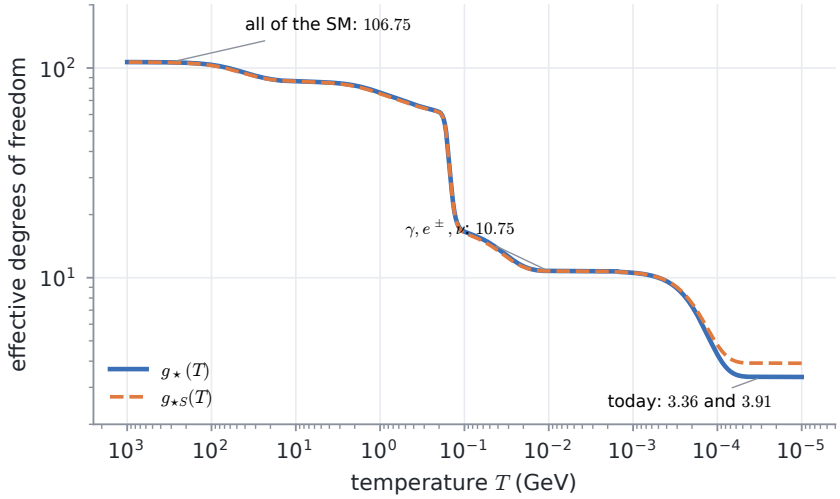


Figure 3.2: The effective number of relativistic degrees of freedom as the universe cools (temperature decreases to the right), computed from table 3.1. Each step is a species becoming non-relativistic and annihilating. The big drop around 150 MeV is the QCD transition, where quarks and gluons get confined into hadrons. Below the electron mass, $g_\star$ and $g_{\star S}$ differ because the neutrinos are colder than the photons.

3.2.2 Entropy

If we wanted to follow the evolution of the universe through statistical mechanics, the most useful quantity turns out to be the entropy. Sometimes we joke and call the universe an isolated system, due to the absurdity of treating the universe as some sort of experimental setup,

but a comoving volume of the universe really does behave like one: by homogeneity, nothing flows in or out of it on average. The entropy connects itself to all other thermodynamic observables through the first law

$$TdS = dU + PdV - \mu dN \tag{3.35}$$

Because the universe is expanding, we might be interested in the **entropy density**, which is just defined as $s \equiv S/V$. For similar reasons, we will be more interested in the energy density and the number density, denoted as ρ and n respectively. When we plug this into the first law, we get

$$Td(sV) = d(\rho V) + PdV - \mu d(nV)$$

when we expand this and use the chain rule, we get

$$Ts\ dV + TV\ ds = \rho dV + Vd\rho + PdV - \mu V\ dn - \mu n\ dV$$

Note that the densities s, ρ, and n only depend on the temperature T (in equilibrium, with μ/T fixed), not on the volume, which means we can write the above equation as

$$(Ts - \rho - P + \mu n)\, dV + \left(TV\frac{ds}{dT} - V\frac{d\rho}{dT} + \mu V\frac{dn}{dT}\right) dT = 0$$

Notice that the differentials dV and dT are independent of each other, so in order for the left hand side to be 0 for every change, both coefficients must vanish. This gives us

$$\boxed{s = \frac{\rho + P - \mu n}{T}} \tag{3.36}$$

and

$$\boxed{\frac{ds}{dT} = \frac{1}{T}\frac{d\rho}{dT} - \frac{\mu}{T}\frac{dn}{dT}} \tag{3.37}$$

Problem 6.

Equations (3.36) and (3.37) give us the evolution of the entropy density of the universe as the temperature cooled. However, we would like to relate the evolution of the entropy to time t. Using the continuity equation

$$\frac{d\rho}{dt} = -3H(\rho + P)$$

show that

$$\frac{d\left(sa^3\right)}{dt} = -\frac{\mu}{T}\frac{d\left(na^3\right)}{dt}$$

and show that in the limit that $\mu \ll T$, entropy is conserved.

Solution:

We can use equation (3.36) to say that $\rho + P = Ts + \mu n$, and thus

$$\frac{d\rho}{dt} = -3H(Ts + \mu n)$$

We can relate the evolution of the energy density to the entropy density via

$$\frac{ds}{dt} = \frac{ds}{dT}\frac{dT}{dt}$$

meaning if we multiply both sides of equation (3.37) by $\dot{T}$

$$\dot{s} = \frac{1}{T}\dot{\rho} - \frac{\mu}{T}\dot{n} \implies \dot{\rho} = T\dot{s} + \mu\dot{n}$$

Putting this back into our continuity equation

$$T\dot{s} + \mu\dot{n} = -3H(Ts + \mu n)$$

$$T\left(\frac{ds}{dt} + 3Hs\right) + \mu\left(\frac{dn}{dt} + 3Hn\right) = 0$$

Note that

$$\frac{d\left(sa^3\right)}{dt} = a^3\frac{ds}{dt} + 3a^2 s\frac{da}{dt} = a^3\left(\frac{ds}{dt} + 3Hs\right)$$

meaning if we divide every side by a^3 in the previous equation, we get

$$\frac{T}{a^3}\frac{d\left(sa^3\right)}{dt} + \frac{\mu}{a^3}\frac{d\left(na^3\right)}{dt} = 0$$

canceling out the a^3 term and rearranging gives

$$\boxed{\frac{d\left(sa^3\right)}{dt} = -\frac{\mu}{T}\frac{d\left(na^3\right)}{dt}}$$

Note that in our expanding universe, $V \propto a^3$, so sa^3 is the entropy in a comoving volume. In the early universe $\mu \ll T$ for all the relevant species (and anyway na^3 is conserved for conserved particle numbers), so the right hand side vanishes and

the comoving entropy is conserved. The expansion of a universe in equilibrium is **adiabatic**. Entropy is only produced when something falls out of equilibrium, and even the biggest such events (like electron-positron annihilation) turn out to happen in near-equilibrium, so sa^3 stays constant to a very good approximation all the way to today.

For relativistic particles with $\mu = 0$, (3.36) with $P = \rho/3$ gives $s = 4\rho/3T$, so each relativistic species contributes $\frac{2\pi^2}{45} g_i T_i^3$ (times 7/8 for fermions). We define the entropic degrees of freedom $g_{\star S}$ the same way as $g_\star$, but with T_i^3 instead of T_i^4,

$$s = \frac{2\pi^2}{45} g_{\star S}(T)\, T^3, \qquad g_{\star S}(T) \equiv \sum_{i=\text{bosons}} g_i \left(\frac{T_i}{T}\right)^3 + \frac{7}{8} \sum_{i=\text{fermions}} g_i \left(\frac{T_i}{T}\right)^3 . \tag{3.38}$$

When all species share the same temperature, $g_{\star S} = g_\star$. Conservation of comoving entropy now gives us the single most useful relation in this chapter,

$$\boxed{g_{\star S}(T)\, T^3 a^3 = \text{constant} \quad \Longrightarrow \quad T \propto g_{\star S}^{-1/3} a^{-1}} \tag{3.39}$$

Away from mass thresholds, $g_{\star S}$ is constant and $T \propto 1/a$, just as we guessed from redshifting. But every time a species annihilates, $g_{\star S}$ drops, and its entropy gets dumped into the particles that are still in equilibrium, so the temperature falls a little slower than $1/a$ for a while.

3.2.3 The Clock Of The Early Universe

With ρ as a function of T, we can connect spacetime to thermodynamics via the Friedmann equation (3.2). During radiation domination,

$$\boxed{H = \left(\frac{\pi^2}{90} g_\star(T)\right)^{1/2} \frac{T^2}{M_{\rm Pl}}} \tag{3.40}$$

Equation (3.40) is strikingly beautiful in its simplicity. If $g_\star$ is constant, then $a \propto t^{1/2}$ and $H = 1/2t$, so we can turn temperature into time,

$$t = \frac{1}{2H} \approx 2.4\, g_\star^{-1/2} \left(\frac{1\text{ MeV}}{T}\right)^2 \text{ s}. \tag{3.41}$$

This is the clock of the early universe. With $g_\star = 10.75$, the universe is about 0.74 seconds old when $T = 1$ MeV, and three minutes old when $T \approx 0.07$ MeV (after $e^\pm$ annihilation, $g_\star = 3.36$ and the coefficient becomes 1.32 s). The rule of thumb is "one MeV, one second."

3.2.4 Equilibrium Versus Expansion

Why should the early universe be in equilibrium at all? Particles stay in equilibrium as long as they interact often enough to share energy. If a particle has interaction rate $\Gamma = n\sigma v$ (number density of targets, times cross section, times relative velocity), then it interacts Γ times per unit time, while the universe changes appreciably on a timescale $1/H$. So the criterion is

$$\Gamma \gg H \quad \text{(equilibrium)}, \qquad \Gamma \ll H \quad \text{(decoupled)}. \tag{3.42}$$

For a relativistic species interacting through a force with a dimensionless coupling α, $n \propto T^3$ and $\sigma \sim \alpha^2/T^2$, so $\Gamma \sim \alpha^2 T$, while $H \sim T^2/M_{\rm Pl}$. The ratio $\Gamma/H \sim \alpha^2 M_{\rm Pl}/T$ is huge at "low" temperatures and only becomes small above about 10^{16} GeV. So below that, Standard Model particles are in equilibrium, at least until their interactions get weak in some other way. And that "other way" is exactly what happens for neutrinos, for dark matter, and eventually for photons, as we'll see. The moment when $\Gamma \approx H$ for a species is called **decoupling** (or **freeze-out**).

3.3 Leaving Equilibrium

The primordial plasma was in near thermal equilibrium and could be analyzed using equilibrium statistical mechanics. But equilibrium has no memory: if everything had stayed in equilibrium, the universe today would be a featureless gas at 2.7 K, with no matter left over, no nuclei, and no atoms. Every relic that we care about was left behind when some reaction stopped keeping up with the expansion. To make sense of what happened and why it happened, we need to describe non-equilibrium processes, and the central object for non-equilibrium statistical mechanics will be the **Boltzmann equation.**

3.3.1 The Boltzmann Equation

The evolution of matter in the universe can be described from just classical mechanics. If we have a Hamiltonian system, then evolution is described using the phase space function $f_i(\boldsymbol{x}, \boldsymbol{p}, t)$ for each particle species i

$$\frac{df_i(\boldsymbol{x}, \boldsymbol{p}, t)}{dt} = C_i[f] \tag{3.43}$$

where C_i on the right hand side is the collision (interactions) term for the particle species. The total derivative can be expanded

$$\frac{\partial f_i}{\partial t} + \frac{\partial f_i}{\partial x_j}\frac{dx_j}{dt} + \frac{\partial f_i}{\partial p_j}\frac{dp_j}{dt} = C_i[f]$$

In its full form, this is kind of ugly. However, we can reduce this down if we use the assumption that the universe is homogeneous and isotropic. Homogeneity means that f_i does not depend on $\boldsymbol{x}$, making the first term on the left vanish. Isotropy means that the momentum vector only depends on its magnitude p. We also know in an expanding universe, free particle momentum scales as $p \propto a^{-1}$, thus we can say

$$p = \frac{C}{a} \implies \frac{dp}{dt} = -\frac{C\dot{a}}{a^2} = -Hp$$

This allows us to write the Boltzmann equation as

$$\frac{\partial f_i}{\partial t} - Hp\frac{\partial f_i}{\partial p} = C_i[f] \tag{3.44}$$

We know that we can obtain the number density from $f(p,t)$ by integrating over momentum space.

$$n_i(t) = g_i \int \frac{d^3p}{(2\pi)^3} f_i(t,p) = g_i \int_0^\infty \frac{4\pi p^2 dp}{(2\pi)^3} f_i(t,p) \tag{3.45}$$

where g_i is the internal degrees of freedom for each particle species. Now let us differentiate with respect to time

$$\begin{aligned}
\frac{dn_i}{dt} &= g_i \frac{\partial}{\partial t} \int_0^\infty \frac{4\pi p^2 dp}{(2\pi)^3} f_i(t,p) \\
&= \frac{g_i}{2\pi^2} \int_0^\infty dp\, p^2 \frac{\partial f_i(t,p)}{\partial t} \\
&= \frac{g_i}{2\pi^2} \int_0^\infty dp\, p^2 \left[C_i[f] + Hp\frac{\partial f_i}{\partial p} \right]
\end{aligned}$$

The integral of the collision term we will just call $C_i[n]$ (it's the net number of particles created per unit volume per unit time). As for the momentum term, we can do integration by parts

$$\frac{g_i H}{2\pi^2} \int_0^\infty dp\, p^3 \frac{\partial f_i}{\partial p} = \frac{g_i H}{2\pi^2} \left[\left(p^3 f_i\right) \Big|_0^\infty - 3\int_0^\infty dp\, p^2 f_i \right]$$

The first term vanishes due to boundary conditions (phase space distribution is 0 at infinity) and the second integral is just 3 dimensional integral in momentum space. Thus we have

$$\frac{dn_i}{dt} = C_i[n] - 3H \underbrace{g_i \int_0^\infty \frac{4\pi p^2 dp}{(2\pi)^3} f_i}_{\equiv n_i}$$

simplifying down to

$$\boxed{\frac{dn_i}{dt} + 3Hn_i = C_i[n] \quad \Longleftrightarrow \quad \frac{1}{a^3}\frac{d(n_i a^3)}{dt} = C_i[n]} \tag{3.46}$$

which we call the **Boltzmann equation** for the number density. It says something very intuitive: without collisions, the number of particles in a comoving volume is conserved and $n \propto a^{-3}$; collisions are the only way to change na^3.

3.3.2 Detailed Balance

The Boltzmann equation depends on the interaction term on the right hand side. The most common interaction is a two-particle interaction

$$X_1 + X_2 \longleftrightarrow Y_1 + Y_2 \tag{3.47}$$

which says that particles X_1 and X_2 interact to produce Y_1 and Y_2. To use the Boltzmann equation, we will work with the number density of the particle species of, say, particle 1

$$\frac{dn_{X_1}}{dt} + 3Hn_{X_1} = -C_1 n_{X_1} n_{X_2} + C_2 n_{Y_1} n_{Y_2}$$

Note the negative sign on C_1 to signal the destruction of particle 1, while the reverse reaction creates it. Using (3.46), we can write this as

$$\boxed{\frac{1}{a^3}\frac{d\left(n_{X_1}a^3\right)}{dt} = -C_1 n_{X_1} n_{X_2} + C_2 n_{Y_1} n_{Y_2}} \tag{3.48}$$

We can relate C_2 to C_1 using equilibrium conditions. In equilibrium, the left hand side is 0, and we would have

$$C_2 = \left(\frac{n_{X_1} n_{X_2}}{n_{Y_1} n_{Y_2}}\right)_{eq} C_1 \tag{3.49}$$

This relation is called **detailed balance** (which just comes from the equal transition rates in stochastic processes). The rates themselves don't care whether we're in equilibrium, so this relation holds always. With detailed balance, we write

$$\boxed{\frac{1}{a^3}\frac{d\left(n_{X_1}a^3\right)}{dt} = C_1\left[\left(\frac{n_{X_1} n_{X_2}}{n_{Y_1} n_{Y_2}}\right)_{eq} n_{Y_1} n_{Y_2} - n_{X_1} n_{X_2}\right]} \tag{3.50}$$

In books like Baumann, C_1 is written as the **thermally averaged cross section** $\langle \sigma v \rangle$, where σ is the scattering cross section and v is the relative velocity of the particles. Physically,

$n_{X_2} \langle \sigma v \rangle$ is the rate Γ at which a given X_1 particle reacts. Notice what the equation says: if $\Gamma = n_{X_2} \langle \sigma v \rangle \gg H$, the right hand side is huge unless the bracket is nearly zero, so the densities are driven to their equilibrium values. If $\Gamma \ll H$, the right hand side is negligible and $n_{X_1} a^3$ freezes at whatever value it had.

Problem 7.

a). Consider massive particles and antiparticles with mass m and number densities $n(t)$ and $\bar{n}(t)$. If they can annihilate into a thermal gas of particles through an interaction with cross section σ at velocity v the evolution of $n(t)$ is described by a Boltzmann equation of the form

$$\frac{dn}{dt} = -3Hn - n\bar{n}\langle \sigma v \rangle + P(t),$$

where $P(t)$ is a positive function of time. Identify the physical origin of each of the terms of the right-hand-side, including $P(t)$.

b). For the remainder of the problem assume that all interactions are CP invariant (i.e. particle-antiparticle symmetric) and that the function $P(t)$ is CP invariant. The latter condition requires that the dependence of $P(t)$ on the number densities is also particle-antiparticle symmetric. By considering the evolution equations of particles and of antiparticles show that

$$(n - \bar{n})a^3 = \text{ const } .$$

Note that this implies that if the initial conditions for the number densities are CP invariant (i.e. $n = \bar{n}$), then $n = \bar{n}$ at all times.

c). Assuming CP invariance ($n = \bar{n}$), show that

$$\frac{1}{a^3} \frac{d\left(na^3\right)}{dt} = -\langle \sigma v \rangle \left[n^2 - n_{eq}^2\right],$$

where n_{eq} denotes the equilibrium number density.

d). Define $Y \equiv n/T^3$ and $x \equiv m/T$, and assume radiation domination (i.e. $H \propto T^2$) to show that

$$\frac{dY}{dx} = -\frac{\lambda}{x^2}\left[Y^2 - Y_{eq}^2\right]$$

where $\lambda \equiv m^3 \langle \sigma v \rangle / H(T = m)$.

e). Assuming that the cross-section $\langle \sigma v \rangle$ is x-independent it can be shown that at late times Y approaches a value which is approximately given by

$$Y_\infty \approx \frac{x_f}{\lambda},$$

where x_f is the value of m/T at freeze-out. Find an estimate for x_f by starting from the freeze-out condition $\Gamma = H$ and rewriting it as an equation for x_f. You should find

$$\frac{e^{x_f}}{\sqrt{x_f}} \propto \lambda$$

with a proportionality constant which you should determine. Argue that for large λ this equation is approximately solved by $x_f = \log \lambda$.

Solution:

a). The first term tells us that as the universe expands, the number density decreases (which makes sense because the number of particles is fixed but space grows). The second term tells us that particles and anti-particles annihilate each other. The third term is the production term: the inverse reaction, where particles in the thermal gas collide and produce a particle-antiparticle pair.

b). We can write this in the form of (3.46)

$$\frac{1}{a^3}\frac{d(na^3)}{dt} = -n\bar{n}\,\langle \sigma v \rangle + P(t)$$

Under CP invariance, we expect

$$\frac{1}{a^3}\frac{d(\bar{n}a^3)}{dt} = -n\bar{n}\,\langle \sigma v \rangle + P(t)$$

and then when we subtract the two equations, we get

$$\frac{1}{a^3}\frac{d(na^3)}{dt} - \frac{1}{a^3}\frac{d(\bar{n}a^3)}{dt} = 0$$
$$\frac{d(na^3)}{dt} - \frac{d(\bar{n}a^3)}{dt} = 0$$

Integrating both sides gives

$$\boxed{(n-\bar{n})\,a^3 = \text{Const.}}$$

c). In equilibrium, the left hand side vanishes and $n = \bar{n} = n_{eq}$, so detailed balance fixes $P(t) = \langle \sigma v \rangle\, n_{eq}^2$ (the production rate is set by the thermal bath, which stays in

equilibrium). With $n = \bar{n}$ we get

$$\frac{1}{a^3}\frac{d\left(na^3\right)}{dt} = -\langle\sigma v\rangle\left[n^2 - n_{eq}^2\right]$$

d). Let us assume $Ta = \text{Const.}$ (constant $g_{\star S}$). Then $na^3 = YT^3a^3$ is Y times a constant, and plugging this back into the previous equation, we have

$$\begin{aligned}\frac{1}{a^3}\frac{d(YT^3a^3)}{dt} &= -\langle\sigma v\rangle\left[Y^2T^6 - \left(YT^3\right)_{eq}^2\right]\\ C\frac{dY}{dt} &= -a^3\langle\sigma v\rangle\left[Y^2T^6 - Y_{eq}^2T^6\right]\\ \frac{dY}{dt} &= -\langle\sigma v\rangle T^3\left[Y^2 - Y_{eq}^2\right]\end{aligned}$$

Note that from the definition of x,

$$\frac{dx}{dt} = -\frac{m}{T^2}\frac{dT}{dt} \implies \frac{dt}{dx} = -\frac{T^2}{m}\frac{dt}{dT}$$

Also note that since $aT = \text{Const.}$

$$\frac{dT}{dt} = -HT$$

Now multiplying both sides of the original equation by dt/dx and simplifying gives

$$\frac{dY}{dx} = -\frac{\langle\sigma v\rangle T^4}{mH}\left[Y^2 - Y_{eq}^2\right]$$

At radiation domination, $H = H(T=m)T^2/m^2 = H(m)/x^2$, and then using $T = m/x$, $\frac{\langle\sigma v\rangle T^4}{mH} = \frac{\langle\sigma v\rangle m^3}{x^4}\cdot\frac{x^2}{H(m)}$, and we simplify to get

$$\boxed{\frac{dY}{dx} = -\frac{m^3\langle\sigma v\rangle}{H(T=m)}\frac{1}{x^2}\left[Y^2 - Y_{eq}^2\right]}$$

e). Freeze-out happens when the annihilation rate of an X particle drops to the expansion rate, $\Gamma = n_{eq}\langle\sigma v\rangle = H$, evaluated at $x = x_f$. Using the non-relativistic density (3.32), $n_{eq} = g(mT/2\pi)^{3/2}e^{-x}$, and $H = H(m)/x^2$,

$$g\left(\frac{m^2}{2\pi x_f}\right)^{3/2}e^{-x_f}\langle\sigma v\rangle = \frac{H(m)}{x_f^2} \implies \boxed{\frac{e^{x_f}}{\sqrt{x_f}} = \frac{g}{(2\pi)^{3/2}}\lambda}$$

where we recognized $\lambda = m^3 \langle\sigma v\rangle / H(m)$. For large λ, taking logs gives $x_f = \ln\left(\frac{g\lambda}{(2\pi)^{3/2}}\right) + \frac{1}{2}\ln x_f$. The exponential is so steep compared to $\sqrt{x_f}$ that, to leading logarithmic accuracy, $\boxed{x_f \approx \ln\lambda}$. For weak-scale particles λ is enormous, and x_f ends up between 10 and 30, with only logarithmic sensitivity to the details.

3.4 A Timeline Of The Early Universe

Now equipped with an understanding of equilibrium and non-equilibrium statistical mechanics, we can tell the story of the early universe in order. Table 3.2 is the map that we will follow. The temperatures and times are approximate, and the first row is speculative (we don't know the mechanism or the scale), but from about 1 MeV onwards, everything in the table is physics that we understand very well and that has been checked against observations.

Table 3.2: Key events in the thermal history of the universe.

Event	Temperature	Time	Redshift
Inflation and reheating	?	?	?
Baryogenesis	?	?	?
Electroweak transition	100 GeV	10^{-11} s	10^{15}
Dark matter freeze-out (WIMP)	~ 5 GeV	10^{-8} s	10^{13}
QCD transition	150 MeV	10^{-5} s	10^{12}
Neutrino decoupling	1 MeV	1 s	6×10^9
Electron-positron annihilation	500 keV	6 s	2×10^9
Big Bang nucleosynthesis	70 keV	3 min	3×10^8
Matter-radiation equality	0.8 eV	50,000 yr	3400
Recombination	0.3 eV	260,000 yr	1300
Photon decoupling (CMB)	0.25 eV	380,000 yr	1100

3.4.1 Baryogenesis

The first event on our list is also the least understood. What we know is that, at high temperatures, quarks and antiquarks were created in almost equal numbers. But as we can see, we were not all annihilated, which means that for some reason, the universe ended up with slightly more baryons than antibaryons. How slightly? Today there are about 6×10^{-10} baryons for every photon in the CMB. Since every photon (roughly) comes from an annihilation, this says that in the early universe, for every billion antiquarks there were a billion and one quarks. That one in a billion is all the matter we see. If the universe started out symmetric (and inflation would erase any initial asymmetry anyway), this asymmetry

has to be generated dynamically, and the mechanism is called **baryogenesis**. We don't know what it is, but we do know what it needs.

In 1967, Andrei Sakharov provided 3 conditions for a theory of barygenesis.

1). Baryon number can and must be violated. This feels a bit obvious because if we start with 0 baryons and end with nonzero baryons now, then this symmetry must be broken.

2). Interactions must violate C and CP symmetry. If C (charge conjugation, swapping particles and antiparticles) were a symmetry, every process making baryons would be matched by the conjugate process making antibaryons at the same rate. The same argument goes through for CP, which is the relevant symmetry because the weak interactions already violate C maximally.

3). The processes that created baryons must be out-of-equilibrium. We will spend some time discussing this.

Let us elaborate more on the third condition. Suppose we have a massive particle X that decays into 2 channels a and b. The particle's anti-matter equivalent also has the same channels for $\bar{a}$ and $\bar{b}$. The baryon numbers for all channels are denoted $B_{a/b}$ and $-B_{a/b}$ with the subscript denoting the channel and the minus sign to denote antimatter. The branching ratios into each channel are

$$r \equiv \frac{\Gamma(X \to a)}{\Gamma_X} \quad \bar{r} \equiv \frac{\Gamma(\bar{X} \to \bar{a})}{\Gamma_{\bar{X}}}$$

$$1 - r \equiv \frac{\Gamma(X \to b)}{\Gamma_X} \quad 1 - \bar{r} \equiv \frac{\Gamma(\bar{X} \to \bar{b})}{\Gamma_{\bar{X}}} \tag{3.51}$$

We can get total total baryon number by multiplying the baryon number of each channel by their decay rates. The net baryon number is given as

$$\begin{aligned} \Delta B &\equiv B_X + B_{\bar{X}} \\ &= (rB_a + (1-r)B_b) + (-\bar{r}B_a - (1-\bar{r})B_b) \\ &= (r - \bar{r})\,(B_a - B_b) \end{aligned}$$

If baryon number is preserved, then $B_a - B_b = 0$ and the net baryon number is 0, which is not possible. Similarly, if CP symmetry is preserved, then $r = \bar{r}$ and ΔB is also 0. So the first 2 Sakharov conditions are required for baryogenesis. But this algebra does not explain why it must be a non-equilibrium process. To understand this, we will need the Boltzmann equation for this process

$$\dot{n}_B + 3Hn_B = \Delta B \Gamma_X \,(n_X - n_X^{\text{eq}}) \tag{3.52}$$

where $\Delta B \equiv (r - \bar{r})(B_a - B_b)$ plays the role of asymmetry parameter. It is clear now why B, C, and CP violation is not enough to drive baryogenesis: if $n_X = n_X^{eq}$, then inverse decays wash out whatever the decays produce, and the right hand side vanishes. The baryon number density just dilutes with the expansion. Thus, a net change in baryon number must occur out of equilibrium, for example when the X particles decay late, after they've become non-relativistic but are still around in their relativistic abundance. The Standard Model has all three ingredients in principle (B is violated by non-perturbative "sphaleron" processes at high temperatures, CP is violated in the quark sector, and the electroweak transition could have been out of equilibrium), but the CP violation is far too small and the transition is too smooth. So baryogenesis is one of the strongest hints we have of physics beyond the Standard Model.

3.4.2 The Electroweak And QCD Transitions

Two more events happen before the ones that left measurable fossils. Around $T \sim 100$ GeV, the Higgs field gets its vacuum expectation value and the W and Z bosons, the quarks and the charged leptons become massive (the **electroweak transition**). Shortly after, the heaviest particles (top quarks, Higgs, W, Z) annihilate away. Around $T \sim 150$ MeV, the quarks and gluons, which were moving around freely in a quark-gluon plasma, get confined into hadrons (the **QCD transition**). The many colored degrees of freedom are replaced by a few pions, which is the big drop in $g_\star$ in figure 3.2, and the pions then annihilate too. In the Standard Model, both are smooth crossovers rather than sharp phase transitions, so they didn't produce much entropy or leave obvious relics. After the QCD transition, the baryons are protons and neutrons, which are already non-relativistic, and only the tiny asymmetry from baryogenesis survives.

3.4.3 Dark Matter Freeze-Out

Though we still do not know exactly what dark matter is, there's a very simple and natural way that it could have been produced, which is as a thermal relic. As the universe expands, particles have to travel larger distances to interact with each other, which just means that the interaction rate decreases. When $\Gamma \approx H$, a species gets left behind.

To begin, let us assume that we have a massive particle X and its antimatter particle $\bar{X}$, which annihilate each other into standard model particles. We assume that the standard model particles still couple to the plasma, and thus their number densities do not change. We will also assume no asymmetry between X and $\bar{X}$, so then the Boltzmann equation becomes

$$\frac{1}{a^3}\frac{\mathrm{d}\left(n_X a^3\right)}{\mathrm{d}t} = -\langle\sigma v\rangle\left[n_X^2 - \left(n_X^{\mathrm{eq}}\right)^2\right] \tag{3.53}$$

Having done problem 7, we know that we can define a new set of variables $Y \equiv n/T^3$ and $x = m/T$ to write

$$\frac{\mathrm{d}Y_X}{\mathrm{d}x} = -\frac{\lambda}{x^2}\left[Y_X^2 - \left(Y_X^{\mathrm{eq}}\right)^2\right] \tag{3.54}$$

This is a **Riccati equation**, with λ being defined as in problem 7. The Riccati equation cannot be solved analytically, but we can solve it numerically (figure 3.3) and understand the behavior in pieces. At early times ($x \lesssim 1$, high temperature), the interactions are fast and Y tracks Y_{eq}. As the universe cools past $T \sim m$, the equilibrium abundance is exponentially suppressed, $Y_{eq} \propto x^{3/2}e^{-x}$, and the annihilations try to keep up. Eventually, at x_f, the particles are too rare to find each other, annihilations stop, and Y freezes out at a constant value Y_∞, which then determines the final abundance of this dark matter particle. After freeze-out Y_{eq} is negligible, so the equation is $dY/dx \approx -\lambda Y^2/x^2$, which integrates to $1/Y_\infty - 1/Y_f = \lambda/x_f$. Since $Y_f \gg Y_\infty$, the abundance of dark matter (in terms of λ) is

$$Y_\infty \approx \frac{x_f}{\lambda}$$

and from problem 7, when λ is sufficiently large, we can approximate $x_f \approx \ln\lambda$, which tells us that the abundance is

$$Y_\infty \approx \frac{\ln\lambda}{\lambda}$$

and this makes sense: if λ is very large (remember that λ is proportional to the annihilation cross section), then the freeze-out abundance is very small. Physically, this is just saying that more interactions means a longer time in equilibrium and therefore more exponential suppression.

As the universe expands, the number density obviously decreases. We can predict the density of dark matter today relative to its freeze out abundance. We know that number density is proportional to a^{-3}

$$\frac{n_0}{n_1} = \frac{a_1^3}{a_0^3} \implies n_0 = Y_\infty T_0^3\left(\frac{a_1 T_1}{a_0 T_0}\right)^3$$

where we used the fact that $n_1 = Y_1 T_1^3$ and then multiplied top and bottom by T_0^3. Particle annihilations cause the temperature to deviate from $T \propto a^{-1}$, so we can't simplify the parentheses completely. However, we do know from entropy conservation (3.39) that $g_{\star S}a^3T^3 = \text{Const.}$, which means that (evaluating the freeze-out quantities at roughly $T \sim M_X$)

$$n_0 = Y_\infty T_0^3 \frac{g_{\star S}(T_0)}{g_{\star S}(M_X)}$$

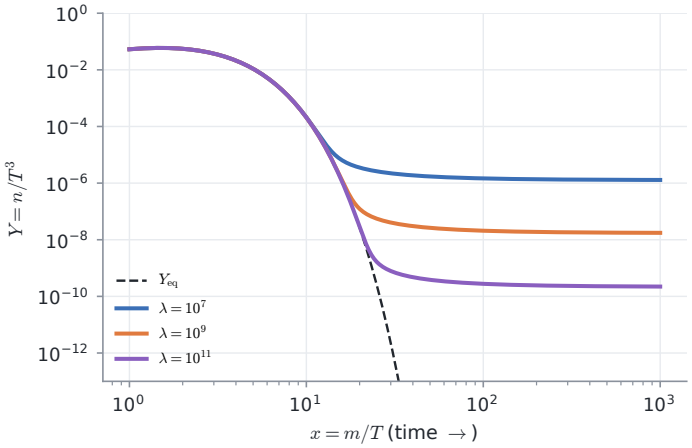


Figure 3.3: Numerical solutions of the freeze-out equation for three values of λ. The abundance follows equilibrium (dashed) until $x \sim 10$–25 and then freezes out. A bigger annihilation cross section keeps the particles in equilibrium longer and leaves fewer of them behind.

The normalized energy density for the dark matter particle X is

$$\begin{aligned}\Omega_X \equiv \frac{\rho_{X,0}}{\rho_{\text{crit},0}} &= \frac{M_X n_0}{3M_{\text{Pl}}^2 H_0^2} \\ &= \frac{M_X T_0^3}{3M_{\text{Pl}}^2 H_0^2}\frac{x_f}{\lambda}\frac{g_*(T_0)}{g_*(M_X)} \\ &= \frac{H(M_X)}{M_X^2}\frac{T_0^3}{3M_{\text{Pl}}^2 H_0^2}\frac{x_f}{\langle\sigma v\rangle}\frac{g_*(T_0)}{g_*(M_X)}\end{aligned}$$

where in the second equality we replaced Y_∞ with x_f/λ and then in the third equality substituted in the definition of λ. If we use equation (3.40) for $H(M_X)$, we get

$$\Omega_X = \frac{\pi}{\sqrt{90}}\frac{T_0^3}{3M_{\text{Pl}}^3 H_0^2}\frac{x_f}{\langle\sigma v\rangle}\sqrt{g_*(M_X)}\frac{g_{*S}(T_0)}{g_{*S}(M_X)} \tag{3.55}$$

At the time of freeze-out, $g_{\star S}(M_X) = g_\star(M_X)$, and today $g_{\star S}(T_0) = 3.91$. If we plug in numbers ($T_0 = 2.35 \times 10^{-13}$ GeV, $H_0 = 2.13 \times 10^{-42} h$ GeV), we get

$$\boxed{\Omega_X \sim 0.1\frac{x_f}{\sqrt{g_*(M_X)}}\frac{10^{-8}\text{GeV}^{-2}}{\langle\sigma v\rangle}} \tag{3.56}$$

Note the amazing result: it doesn't depend on the dark matter mass except logarithmically through x_f, only on the cross section. With $x_f \sim 20$ and $g_\star \sim 100$, matching the observed

$\Omega_X h^2 \approx 0.12$ requires $\langle\sigma v\rangle \sim 2 \times 10^{-9}$ GeV^{-2}, which is about what you'd get for a particle with a mass around 100 GeV and weak-scale couplings, $\langle\sigma v\rangle \sim G_F^2 m_X^2$. This coincidence is called the **WIMP miracle**, and it's why weakly interacting massive particles, or **WIMPs**, were the leading dark matter candidate for decades. (Direct detection experiments have not found them yet, which has made people a lot more open-minded.)

3.4.4 Neutrino Decoupling

Neutrinos only feel the weak force, through reactions like $\nu_e + \bar{\nu}_e \leftrightarrow e^+ + e^-$ and $\nu + e \leftrightarrow \nu + e$. At temperatures well below the W and Z masses, the weak cross section is $\sigma \sim G_F^2 T^2$, where $G_F \approx 1.17 \times 10^{-5}$ GeV^{-2} is the Fermi constant. With $n \sim T^3$ and $v \sim 1$, the interaction rate is

$$\Gamma_\nu \sim G_F^2 T^5, \tag{3.57}$$

which falls much faster than $H \sim T^2/M_{\rm Pl}$. The ratio is

$$\frac{\Gamma_\nu}{H} \sim \frac{G_F^2 T^5}{T^2/M_{\rm Pl}} = \left(\frac{T}{1\text{ MeV}}\right)^3, \tag{3.58}$$

where the numbers conspire (with $g_\star = 10.75$ and $M_{\rm Pl} \approx 2.4 \times 10^{18}$ GeV) to make the crossover land right around a temperature of 1 MeV, when the universe was about a second old. Below that, neutrinos stop interacting. They keep the Fermi-Dirac distribution that they had at decoupling, and since every momentum just redshifts as $p \propto 1/a$, the distribution keeps exactly the same shape with a temperature $T_\nu \propto 1/a$. These neutrinos are still here, about 336 per cubic centimeter, forming a **cosmic neutrino background** that we have never directly detected but whose gravitational effects we see in the CMB.

3.4.5 Electron-Positron Annihilation

Shortly after neutrinos decouple, the temperature drops below the electron mass, and electrons and positrons annihilate, $e^+ + e^- \to \gamma + \gamma$. Their entropy gets dumped into the photons, but not into the neutrinos, which have already decoupled. So the photons get heated relative to the neutrinos, and we can calculate by exactly how much using entropy conservation (3.39). Before annihilation, photons, electrons and positrons are in equilibrium with

$$g_{\star S}^{\rm before} = 2 + \frac{7}{8} \times 4 = \frac{11}{2}, \tag{3.59}$$

counting only the species coupled to the photons. After annihilation, only the photons are left, $g_{\star S}^{\text{after}} = 2$. Since $g_{\star S}(aT)^3$ is conserved for the photon-electron plasma,

$$\frac{(aT_\gamma)_{\text{after}}}{(aT_\gamma)_{\text{before}}} = \left(\frac{11}{4}\right)^{1/3}. \tag{3.60}$$

The neutrinos, meanwhile, just have aT_ν constant, and before annihilation $T_\nu = T_\gamma$. So after annihilation,

$$\boxed{T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma} \tag{3.61}$$

With $T_0 = 2.7255$ K for the photons today, the neutrino background should be at $T_{\nu,0} = 1.95$ K. This also lets us finish the count of degrees of freedom today. Photons plus three neutrino species at the lower temperature give

$$g_\star(T_0) = 2 + \frac{7}{8} \times 6 \times \left(\frac{4}{11}\right)^{4/3} = 3.36, \qquad g_{\star S}(T_0) = 2 + \frac{7}{8} \times 6 \times \frac{4}{11} = 3.91. \tag{3.62}$$

It's common to parametrize the energy density in neutrinos (or any other extra relativistic stuff) by the **effective number of neutrino species** N_{eff}, defined by $\rho_r = \rho_\gamma \left[1 + \frac{7}{8}\left(\frac{4}{11}\right)^{4/3} N_{\text{eff}}\right]$. Our calculation gives $N_{\text{eff}} = 3$. A more careful calculation, which accounts for the neutrinos not being completely decoupled when the electrons annihilate, gives $N_{\text{eff}} = 3.044$, and the CMB measures $N_{\text{eff}} = 2.99 \pm 0.17$ [2], a beautiful check of the whole story.

3.4.6 Big Bang Nucleosynthesis

With the neutrinos decoupled and the positrons gone, the universe is a plasma of photons, electrons, protons and neutrons, with about 1.6×10^9 photons per baryon. It is hot enough that any nucleus that forms is immediately blasted apart. The question is what happens as it cools. The answer is that, within about three minutes, about a quarter of the baryonic mass ends up in helium-4. This is **Big Bang nucleosynthesis** (BBN), and it's the oldest piece of the thermal history that we can directly test.

3.4.6.0.1 Step 1: neutron freeze-out. Protons and neutrons are kept in equilibrium by weak reactions like $n + \nu_e \leftrightarrow p + e^-$ and $n + e^+ \leftrightarrow p + \bar{\nu}_e$. While these are fast, the ratio of neutrons to protons follows the non-relativistic equilibrium (3.32). The masses differ by $Q = m_n - m_p = 1.293$ MeV, and the chemical potentials are negligible, so

$$\left(\frac{n_n}{n_p}\right)_{eq} = \left(\frac{m_n}{m_p}\right)^{3/2} e^{-Q/T} \approx e^{-Q/T}. \tag{3.63}$$

At high temperatures there are as many neutrons as protons. But these are weak interactions, with rates that fall like $G_F^2 T^5$ just like the neutrino rates, so they freeze out at almost the same moment, $T_f \approx 0.8$ MeV. At that point $n_n/n_p \approx e^{-1.293/0.8} \approx 1/5$. Solving the Boltzmann equation properly (figure 3.4) gives a slightly smaller frozen-out neutron fraction $X_n \equiv n_n/(n_n + n_p) \approx 0.15$.

3.4.6.0.2 Step 2: neutron decay. Free neutrons aren't stable. They decay with a lifetime $\tau_n \approx 879$ s, so after freeze-out the neutron fraction slowly decreases as $X_n(t) = X_n^{\rm f} e^{-t/\tau_n}$. This sets a clock: whatever happens has to happen within a few minutes.

3.4.6.0.3 Step 3: the deuterium bottleneck. To make helium, you first need deuterium, $n + p \leftrightarrow D + \gamma$, with binding energy $B_D = 2.22$ MeV. You might expect deuterium to form when T drops below 2.2 MeV, but it doesn't. The problem is that there are so many photons. A deuteron gets destroyed whenever it meets a photon with energy above B_D, and the fraction of photons in the high-energy tail of the Planck distribution is about $e^{-B_D/T}$. With $1/\eta \approx 1.6 \times 10^9$ photons per baryon, deuterium can only survive once $\eta^{-1} e^{-B_D/T} \lesssim 1$, which is

$$T_{\rm nuc} \approx \frac{B_D}{\ln(1/\eta)} \approx \frac{2.22 \text{ MeV}}{21} \approx 0.1 \text{ MeV}, \tag{3.64}$$

and a careful calculation with the Saha equation (which we'll derive for hydrogen in a moment) gives $T_{\rm nuc} \approx 0.07$ MeV. By (3.41) the universe is then about $t_{\rm nuc} \approx 250$ s old, and the neutrons have decayed down to $X_n \approx 0.15\, e^{-250/879} \approx 0.11$.

3.4.6.0.4 Step 4: helium. Once deuterium can survive, the reactions $D + D \to {}^3\text{He} + n$, ${}^3\text{He} + D \to {}^4\text{He} + p$ and so on run very quickly, and since helium-4 is by far the most tightly bound light nucleus, essentially every surviving neutron ends up in helium. Each helium nucleus has 2 neutrons and a mass of 4 nucleons, so the fraction of baryonic mass in helium is

$$\boxed{Y_P \equiv \frac{4 n_{\rm He}}{n_b} \approx 2 X_n(t_{\rm nuc}) \approx 0.22} \tag{3.65}$$

The precise calculation gives $Y_P \approx 0.247$, and observations of the most pristine gas in the universe find $Y_P = 0.245 \pm 0.003$. Small amounts of deuterium ($D/H \approx 2.5 \times 10^{-5}$), helium-3 and lithium-7 are also left over, and because these depend sensitively on η, they give a measurement of the baryon density that agrees with the one from the CMB. Notice how much physics went into a single number: the weak interaction, the neutron lifetime, the number of relativistic species (through H), and the baryon-to-photon ratio.

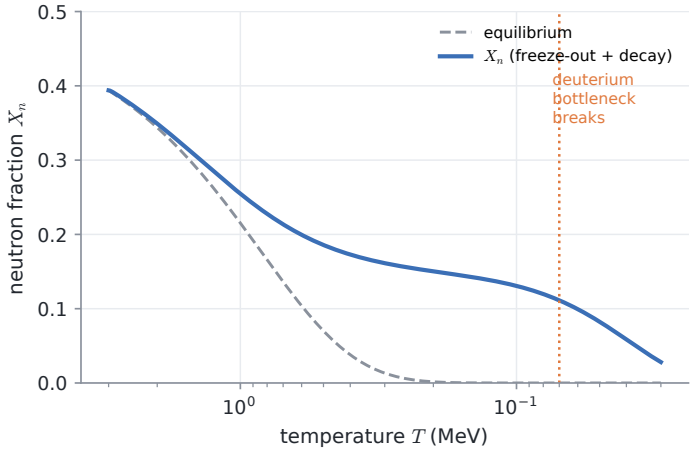


Figure 3.4: The neutron fraction as the universe cools. It tracks equilibrium (dashed) until about 1 MeV, freezes out near 0.15, and then slowly decays until deuterium forms at $T \approx 0.07$ MeV, leaving $X_n \approx 0.11$.

3.4.7 Matter-Radiation Equality

After BBN, nothing dramatic happens for a long time. The universe keeps expanding and cooling. Radiation dilutes like a^{-4} while matter dilutes like a^{-3}, so eventually matter wins. Using Ω_r/Ω_m from the last chapter, this happens at $a_{\rm eq} \approx 1/3400$, when $T \approx 0.8$ eV and the universe is about 50,000 years old. Equality doesn't change the particle content of the universe, but it changes how fast it expands (from $a \propto t^{1/2}$ to $a \propto t^{2/3}$), and we will see in the perturbation chapters that this is when density perturbations in the dark matter really start to grow.

3.4.8 Recombination

Last stop. The universe is now a plasma of protons, electrons, helium nuclei and photons (we will ignore helium for simplicity). The photons scatter constantly off the free electrons, so the universe is opaque, like the inside of a star. As it cools, electrons and protons combine into neutral hydrogen,

$$e^- + p^+ \leftrightarrow H + \gamma, \tag{3.66}$$

and once most of the electrons are bound up in atoms, the photons stop scattering and the universe becomes transparent. Somewhat confusingly, this is called **recombination**, even though the electrons and protons had never been combined before.

While the reaction is fast, the three species are in chemical equilibrium, $\mu_e + \mu_p = \mu_H$ (photons have $\mu_\gamma = 0$). Taking the ratio of the non-relativistic densities (3.32) with chemical

potentials, the chemical potentials cancel and we get

$$\left(\frac{n_H}{n_e n_p}\right)_{eq} = \frac{g_H}{g_e g_p}\left(\frac{m_H}{m_e m_p}\cdot\frac{2\pi}{T}\right)^{3/2} e^{(m_e+m_p-m_H)/T}. \tag{3.67}$$

The mass difference is the binding energy of hydrogen, $B_H = m_e + m_p - m_H = 13.6$ eV, the degeneracies are $g_p = g_e = 2$ and $g_H = 4$, and we can set $m_H \approx m_p$. Defining the free electron fraction $X_e \equiv n_e/n_b$ and using charge neutrality ($n_e = n_p$) and $n_b \approx n_p + n_H$, we get the **Saha equation**,

$$\boxed{\frac{1-X_e}{X_e^2} = \frac{2\zeta(3)}{\pi^2}\eta\left(\frac{2\pi T}{m_e}\right)^{3/2} e^{B_H/T}} \tag{3.68}$$

where we wrote $n_b = \eta n_\gamma = \eta\frac{2\zeta(3)}{\pi^2}T^3$. This is a quadratic equation for X_e at each temperature, and figure 3.5 shows the solution. Once again, the tiny baryon-to-photon ratio $\eta \approx 6\times10^{-10}$ delays everything: the transition happens not at $T \sim 13.6$ eV, but at $T_{\rm rec} \approx 0.3$ eV, or $z_{\rm rec} \approx 1300$, because there are so many photons in the high energy tail ready to re-ionize any atom that forms.

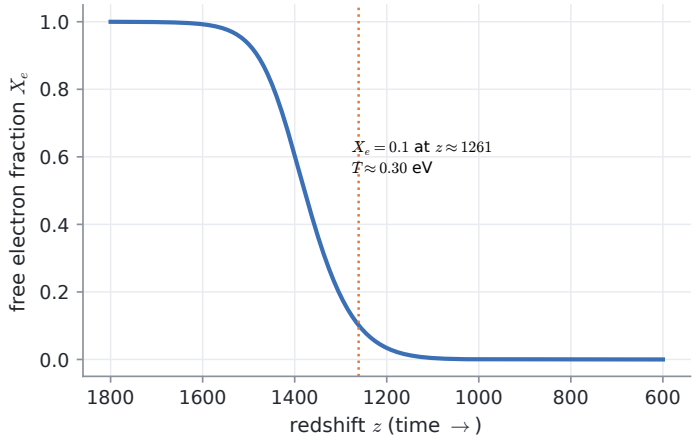


Figure 3.5: The free electron fraction from the Saha equation with $\eta = 6.1\times10^{-10}$. The universe goes from ionized to neutral over a narrow range of redshifts around $z \approx 1300$, at a temperature far below the 13.6 eV binding energy.

3.4.8.0.1 Photon decoupling. Photons scatter off free electrons through Thomson scattering, with cross section $\sigma_T \approx 6.65\times10^{-25}$ cm^2, so the scattering rate is $\Gamma_\gamma = n_e\sigma_T = X_e n_b \sigma_T$. As X_e plummets during recombination, Γ_γ drops below H, and the photons decouple at $T_{\rm dec} \approx 0.25$ eV, $z_{\rm dec} \approx 1100$, when the universe is about 380,000 years old. (The

Saha equation is a little too fast at the tail end, because the reactions themselves start to freeze out, and a proper Boltzmann treatment gives slightly more free electrons at late times.) The surface in space where the CMB photons last scattered is called the **surface of last scattering**, and it's what we see when we look at figure 1.2.

3.4.9 The Cosmic Microwave Background

Photons were not the first particles to decouple from the plasma. As we saw, the neutrinos beat them by about 380,000 years. But it could be argued that the decoupling of photons had the bigger impact. At around 0.25 eV, the universe cooled down enough for photons to become free streaming and the universe became transparent. For the first time in existence, you could see! As the universe expands, these free streaming photons get redshifted by a factor of 1100, from visible and infrared light into the microwave range, hence the name. They are relics of the early universe and they carry with them information of not just the thermodynamics of the universe but also the formation of matter (and possible dark matter).

Since photons are bosonic particles, we will need to use Bose-Einstein statistics. Remember from the beginning of the chapter that the average occupation number for bosons is

$$\langle N \rangle = \frac{1}{e^{\beta\hbar\omega} - 1} \tag{3.69}$$

where $\hbar\omega$ energy of the occupied state, with ω in our case being the angular frequency of the photons. From this, we can obtain the (number) density of photons within some frequency range $d\omega$ using equation (3.21), and all that we would need is the density of states. Because the dispersion relation for photons is linear, the density of states can be calculated simply

$$g(k)dk = \frac{V}{(2\pi)^3} d^3k \implies g(\omega)d\omega = \frac{4\pi V}{(2\pi c)^3}\, \omega^2 d\omega$$

where we have used spherical coordinates to reduce the 3 dimension k integral in a single integral in ω space using the linear dispersion relation. The density is then

$$n(\omega) = \frac{4\pi}{(2\pi c)^3} \frac{\omega^2}{e^{\beta\hbar\omega} - 1}$$

but remember that photons come in 2 polarizations, so the true number density per unit frequency is

$$\boxed{n(\omega) = \frac{1}{\pi^2 c^3} \frac{\omega^2}{e^{\beta\hbar\omega} - 1}} \tag{3.70}$$

and this distribution has a very special name; it is called the **blackbody spectrum** (multiplying by $\hbar\omega$ gives Planck's energy spectrum). So equilibrium statistical mechanics

predicts that the CMB should have a blackbody spectrum (figure 3.6). The funny thing

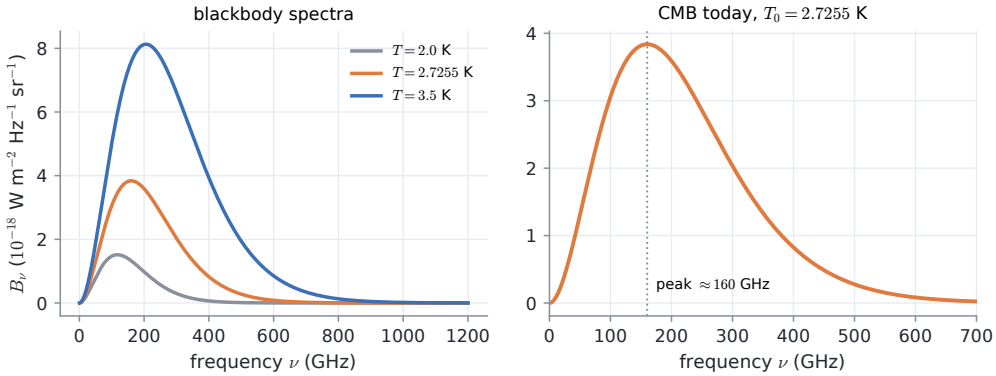


Figure 3.6: Left: blackbody spectra (spectral radiance) at three temperatures. Hotter bodies are brighter at every frequency and peak at higher frequencies. Right: the blackbody spectrum at the CMB temperature today, which peaks at about 160 GHz, in the microwave band.

here is that Bose-Einstein statistics assumes equilibrium, but after decoupling, photons are no longer in equilibrium with anything! So why should the CMB still be a blackbody? The answer is the expansion. A freely streaming photon has its frequency redshifted, $\omega \propto 1/a$, and the number density of photons falls like a^{-3}. If you plug $\omega \to \omega a_{\rm dec}/a$ into (3.70), you find that the spectrum at a later time is *exactly* a blackbody again, with temperature $T \propto 1/a$. Decoupling freezes the distribution, and the expansion preserves its shape. The COBE satellite's FIRAS instrument measured the CMB spectrum in 1990 and found a perfect blackbody at $T_0 = 2.7255$ K, with deviations smaller than 50 parts per million. It's the most perfect blackbody ever measured in nature.

Integrating the spectrum over all frequencies gives the number density of CMB photons today, $n_{\gamma,0} = \frac{2\zeta(3)}{\pi^2} T_0^3 \approx 411$ photons per cubic centimeter. The CMB is also not perfectly uniform. On top of the average temperature there are fluctuations of about one part in 10^5 (figure 1.2), and these are snapshots of the density perturbations in the plasma at the moment of decoupling. Where those perturbations came from (inflation), and how they evolve into the patterns we see in the CMB and in the distribution of galaxies (perturbation theory), is the subject of the rest of these notes.

3.5 Problems

1. Gravitinos

In supergravity theories, the superpartner of the graviton, the gravitino, is a spin- 3/2 dark

matter candidate. Assuming a negligible energy density of gravitinos after inflation, these particles would be produced most efficiently just after reheating. The rate at which Standard Model particles in the primordial plasma would convert into gravitinos is $\Gamma_g \sim T^3/M_{\rm P1}^2$. In this problem, we will estimate the relic density of gravitinos.

a). First, argue that the normalized number of gravitinos, $N_g \equiv n_g/s$, is given by

$$N_g \sim \int_0^{T_R} \frac{\mathrm{d}T}{T} \frac{\Gamma_g}{H} \frac{n_g}{s},$$

where T_R is the reheating temperature (temperature at the beginning of the universe), s is the total entropy density and n_g is the particle density. Using that $n_g/s \sim O(1)$ and $H \sim T^2/M_{\rm Pl}$, derive a simple expression for N_g in terms of T_R and $M_{\rm Pl}$.

b). Assuming one massless neutrino species, show that the total entropy density of the universe today is $s_0 \approx 2000\ \mathrm{cm}^{-3}$.

c). Show that the energy density of gravitinos today would be

$$\Omega_g h^2 \sim 0.1 \left(\frac{T_R}{10^9 \mathrm{GeV}}\right) \left(\frac{m_g}{1\mathrm{GeV}}\right).$$

For a reheating temperature of $T_R \sim 10^9$GeV, gravitinos with masses around $m_g \sim 1$GeV could hence constitute all of the dark matter. Note, however, that this scenario is severely constrained by its effects on BBN through the decay of heavier supersymmetric particles.

2. The Baryon-To-Photon Ratio

a). Show that the number density of CMB photons today is $n_{\gamma,0} \approx 411\ \mathrm{cm}^{-3}$.

b). Using $\Omega_b h^2 = 0.0224$ and $\rho_{\mathrm{crit},0} = 1.05 \times 10^{-5} h^2$ GeV cm^{-3}, compute the number density of baryons today and the ratio $\eta = n_b/n_\gamma$. Why is η constant once $e^\pm$ annihilation is over?

3. The Neutrino Background

a). Using (3.61) and (3.30), compute the number density of each neutrino species today (counting neutrinos and antineutrinos).

b). If one neutrino species has a mass $m_\nu \gg T_{\nu,0}$, show that its contribution to the density today is $\Omega_\nu h^2 = m_\nu/(94\ \mathrm{eV})$. What does the observed $\Omega_m h^2 \approx 0.14$ say about the neutrino masses?

4. Helium As A Probe Of New Physics

Suppose there were an additional relativistic species in equilibrium at BBN, increasing $g_\star$ from 10.75 to 12.5. Qualitatively, how would this change the freeze-out temperature of the neutron-to-proton ratio, the time of nucleosynthesis, and the helium fraction Y_P? Estimate the change in T_f using $\Gamma \propto G_F^2 T^5$ and $H \propto g_\star^{1/2} T^2$.

4

Inflation

"Inflation is as violent as a mugger, as frightening as an armed robber and as deadly as a hit man."

– Ronald Reagan

In the last two chapters, we studied the nature of spacetime and the thermal history of the early universe. The hot Big Bang works incredibly well. It predicts the expansion, the helium abundance, the neutrino background and the blackbody spectrum of the CMB. But it works so well that it is suspicious. It only works if the universe started out in a very, very special state: almost perfectly flat, and almost perfectly smooth over regions that, according to the hot Big Bang itself, could never have talked to each other. These are the **flatness problem** and the **horizon problem** from Chapter 1, and in this chapter we'll make them precise and then see how a short period of accelerated expansion before the hot Big Bang, called **inflation**, solves both of them at once. The bonus, and the real reason cosmologists take inflation seriously, is that inflation also explains where the small perturbations that seeded galaxies (and us) came from.

4.1 The Problems With The Hot Big Bang

4.1.1 The Flatness Problem

In chapter 2, we ended with using the framework of general relativity to find solutions for our universe, and we found that our universe is a flat expanding universe. Though we cannot

100% prove that the universe is flat, observations have put an upper bound on $|\Omega_k|$ of about 0.01, which is pretty indicative of a flat universe. Although a flat universe is an equilibrium solution, it is an unstable equilibrium. To see this, start with the FRW universe. The first Friedmann equation gives

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} \tag{4.1}$$

with $k = 0, \pm 1$. If we define the total density parameter at any time as $\Omega(t) \equiv \rho/\rho_{\rm crit}(t) = 8\pi G\rho/3H^2$, then dividing through by H^2 gives

$$\boxed{\Omega(t) - 1 = \frac{k}{(aH)^2}} \tag{4.2}$$

Flatness corresponds to $k = 0$, which means $\Omega = 1$ exactly, at all times. Now suppose that our universe is dominated by a single fluid with equation of state w. We know that $\rho \propto a^{-3(1+w)}$ and $a \propto t^{2/3(1+w)}$, so $aH = \dot{a} \propto t^{2/3(1+w)-1}$, and

$$|\Omega - 1| \propto \frac{1}{(aH)^2} \propto a^{1+3w} = \begin{cases} a^2 & \text{radiation} \\ a & \text{matter} \end{cases} \tag{4.3}$$

So in a decelerating universe ($1 + 3w > 0$), any deviation from flatness *grows*. Flatness is like a pencil balanced on its tip. Let's put in numbers. Today $|\Omega - 1| \lesssim 0.01$. Going back to matter-radiation equality ($a \approx 3 \times 10^{-4}$), the deviation was smaller by that factor, about 3×10^{-6}. Going further back to BBN at $T \sim 1$ MeV ($a \sim 2 \times 10^{-10}$), during radiation domination it shrinks like a^2, so

$$|\Omega(T_{\rm BBN}) - 1| \lesssim 10^{-18}, \tag{4.4}$$

and at the Planck scale it would have to be $\lesssim 10^{-61}$. If the early universe had been even slightly curved, say $\Omega - 1 = 10^{-10}$ at BBN, it would have either recollapsed in a big crunch long ago or expanded so fast that no structure could form. The hot Big Bang works, but only if we set the initial conditions with absurd precision, and nothing in the theory explains why.

4.1.2 The Horizon Problem

The second problem is about causality. In a time t, light can travel a comoving distance

$$\chi_p(t) = \int_0^t \frac{dt'}{a(t')} = \int_0^a \frac{da'}{a'^2 H(a')} = \int_0^{\ln a} (a'H)^{-1}\, d\ln a', \tag{4.5}$$

which is just the conformal time η from Chapter 2. This is the **comoving particle horizon**, the size of the region that could have been in causal contact since the Big Bang. The last way of writing it is the important one: the horizon is a logarithmic integral of the **comoving Hubble radius** $(aH)^{-1}$. The comoving Hubble radius is roughly the comoving distance light travels in one expansion time $1/H$, so it tells us which regions can talk to each other *right now*, while the particle horizon tells us which regions have *ever* talked to each other.

In a decelerating universe, $(aH)^{-1} \propto a^{(1+3w)/2}$ grows with time, so the integral in (4.5) is dominated by the latest times, and $\chi_p \propto a^{(1+3w)/2}$ (for example $\chi_p \propto a$ in radiation domination). The horizon at any time is only a bit bigger than the Hubble radius at that time. Now look at the CMB. The comoving horizon at decoupling was about 280 Mpc, while the CMB was emitted from a sphere at a comoving distance of about 14,000 Mpc. So the horizon at decoupling subtends an angle of roughly

$$\theta_{\rm hor} \approx \frac{280}{14{,}000} \approx 0.02 \text{ rad} \approx 1^\circ \tag{4.6}$$

on the sky. The CMB sky is made of about $4\pi/\theta_{\rm hor}^2 \sim 10^4$ patches that, according to the hot Big Bang, never had any causal contact before the CMB was released. And yet, they all have the same temperature to one part in 10^5. Maybe universal expansion explains it; if the universe is expanding, maybe at some point these regions were close enough to each other to equilibrate. But this can't be the answer, because going back in time doesn't help: the horizon shrinks even faster than the distances between the patches. So how is it possible that the temperature of the universe is so uniform?

Problem 1.

Consider a flat universe that is matter dominated all the way back to $a = 0$ (a decent approximation for the purpose of this estimate). Compute the conformal time $\eta(a)$, and use it to estimate the angle on the sky subtended by the comoving horizon at decoupling, $a_{\rm dec} = 1/1100$.

Solution: In matter domination, $H = H_0 a^{-3/2}$, so $(aH)^{-1} = a^{1/2}/H_0$ and

$$\eta(a) = \int_0^a \frac{da'}{a'^2 H(a')} = \frac{1}{H_0}\int_0^a a'^{-1/2} da' = \frac{2}{H_0} a^{1/2}.$$

Today, $\eta_0 = 2/H_0$, and at decoupling $\eta_{\rm dec} = \eta_0 a_{\rm dec}^{1/2} = \eta_0/\sqrt{1100} \approx 0.030\,\eta_0$. A light ray from the last scattering surface travelled a comoving distance $\eta_0 - \eta_{\rm dec}$, so the

horizon at decoupling subtends

$$\theta \approx \frac{\eta_{\rm dec}}{\eta_0 - \eta_{\rm dec}} = \frac{0.030}{0.970} \approx 0.031 \text{ rad} \approx 1.8°.$$

(Including radiation and dark energy changes this to about a degree.) This is roughly the angular scale of the first acoustic peak in the CMB, which is not a coincidence, as we will see in Chapter 6.

4.2 The Shrinking Hubble Sphere

Here's the idea. Both problems come from the comoving Hubble radius $(aH)^{-1}$ growing with time. Look at (4.2): if $(aH)^{-1}$ *shrinks*, then $|\Omega - 1|$ is driven towards zero, so flatness becomes an attractor instead of an unstable equilibrium. And look at (4.5): if there was a long period in the early universe when $(aH)^{-1}$ was large and shrinking, then the integral picks up a huge contribution from early times, and the particle horizon can be much, much bigger than the Hubble radius at decoupling. Regions that look causally disconnected today (because they're outside each other's Hubble radius) were actually in causal contact before (figure 4.1).

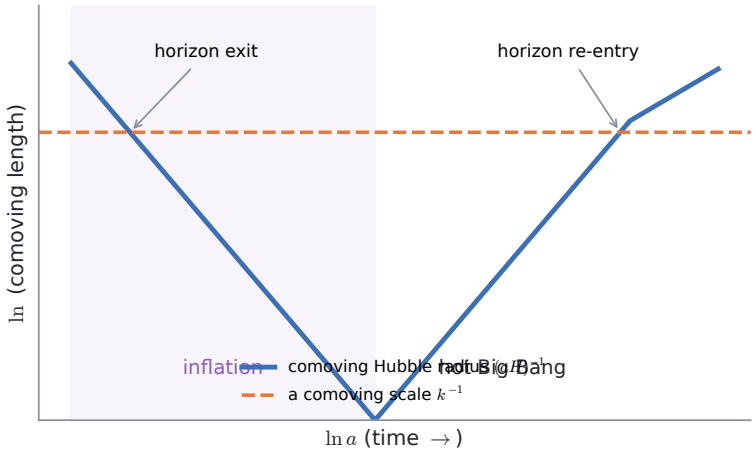


Figure 4.1: The comoving Hubble radius shrinks during inflation and grows afterwards. A comoving scale (dashed) that is inside the Hubble radius early on exits during inflation and re-enters much later, so scales that look causally disconnected at re-entry were in causal contact before inflation.

So we define **inflation** as a period with a shrinking comoving Hubble radius,

$$\frac{d}{dt}\left(\frac{1}{aH}\right) = \frac{d}{dt}\left(\frac{1}{\dot{a}}\right) = -\frac{\ddot{a}}{\dot{a}^2} < 0 \quad \Longleftrightarrow \quad \ddot{a} > 0. \tag{4.7}$$

Inflation is accelerated expansion! By the second Friedmann equation (2.91), $\ddot{a} > 0$ requires

$$\rho + 3P < 0 \quad \Longleftrightarrow \quad w < -\frac{1}{3}, \tag{4.8}$$

a negative pressure, like a cosmological constant. It is also useful to express this through the **Hubble slow-roll parameter**,

$$\epsilon \equiv -\frac{\dot{H}}{H^2} = -\frac{d\ln H}{dN}, \qquad dN \equiv H\,dt = d\ln a, \tag{4.9}$$

where N counts the number of **e-folds** of expansion. Since $\ddot{a}/a = \dot{H} + H^2 = H^2(1-\epsilon)$, inflation happens exactly when $\epsilon < 1$. The limit $\epsilon \to 0$ is a perfect de Sitter universe, with constant H and $a \propto e^{Ht}$, and successful inflation is something close to that, with H changing slowly.

4.2.1 How Much Inflation?

To solve the horizon problem, the comoving Hubble radius at the start of inflation has to be at least as big as the comoving Hubble radius today, so that everything we see today was once inside one Hubble patch. During inflation H is roughly constant, so $(aH)^{-1}$ shrinks by a factor of e^N. After inflation, it grows again. Let's estimate by how much, assuming that inflation ends at an energy scale $T_R \sim 10^{15}$ GeV and that the universe is radiation dominated right afterwards. During radiation domination $aH \propto a^{-1} \propto T$ (ignoring the changes in $g_\star$), so from the end of inflation to equality $(aH)^{-1}$ grows by a factor $T_R/T_{\rm eq} \approx 10^{15}$ GeV$/(0.8$ eV$) \approx 10^{24}$. During matter domination $aH \propto a^{-1/2}$, so from equality to today it grows by $a_{\rm eq}^{-1/2} \approx 60$. In total,

$$\frac{(a_0 H_0)^{-1}}{(a_{\rm end} H_{\rm end})^{-1}} \sim 10^{24} \times 60 \sim e^{59}, \tag{4.10}$$

so we need at least $N \approx 60$ e-folds of inflation. (If inflation happens at a lower energy, a few fewer e-folds are enough.) Sixty e-folds is a growth in the scale factor of 10^{26}, and it can happen in a tiny fraction of a second. The same 60 e-folds also fix the flatness problem: $|\Omega - 1| \propto (aH)^{-2}$ shrinks by $e^{-2N} \approx 10^{-52}$ during inflation, so whatever curvature there was before inflation gets flattened out. It's like blowing up a balloon to the size of the observable universe; any patch of its surface looks flat.

4.3 The Physics Of Inflation

What kind of stuff has negative pressure? The simplest candidate is a scalar field, called the **inflaton** $\phi(t, \boldsymbol{x})$, with some potential energy $V(\phi)$. In flat spacetime, a scalar field has Lagrangian density $\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)$, and by general covariance we just replace $\eta \to g$. The resulting energy-momentum tensor is

$$T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - g_{\mu\nu}\left(\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\,\partial_\beta\phi + V(\phi)\right). \tag{4.11}$$

By homogeneity the background field can only depend on time, $\phi = \phi(t)$. In the FRW metric, $g^{00} = -1$, so $\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi = -\frac{1}{2}\dot{\phi}^2$, and comparing with the perfect fluid (2.39),

$$\boxed{\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi), \qquad P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi)} \tag{4.12}$$

The kinetic energy contributes positive pressure, and the potential energy contributes *negative* pressure. So if the field is moving slowly compared to how much potential energy it has, $\frac{1}{2}\dot{\phi}^2 \ll V$, then $P \approx -\rho$ and $w \approx -1$, which is exactly what we need. A slowly rolling scalar field acts like a temporary cosmological constant.

4.3.1 The Equations Of Motion

The field obeys the continuity equation $\dot{\rho} + 3H(\rho + P) = 0$. Plugging in (4.12),

$$\dot{\phi}\ddot{\phi} + V'(\phi)\dot{\phi} + 3H\dot{\phi}^2 = 0 \quad \Longrightarrow \quad \boxed{\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0} \tag{4.13}$$

where $V' = dV/d\phi$. This is the **Klein-Gordon equation** in an expanding universe. It's just a ball rolling down a hill ($\ddot{\phi} = -V'$) with a friction term $3H\dot{\phi}$ coming from the expansion. Together with the Friedmann equation,

$$H^2 = \frac{1}{3M_{\rm Pl}^2}\left(\frac{1}{2}\dot{\phi}^2 + V(\phi)\right), \tag{4.14}$$

this is a closed system. Taking the time derivative of (4.14) and using (4.13) gives $\dot{H} = -\dot{\phi}^2/2M_{\rm Pl}^2$, so

$$\epsilon = -\frac{\dot{H}}{H^2} = \frac{\frac{1}{2}\dot{\phi}^2}{M_{\rm Pl}^2 H^2}. \tag{4.15}$$

Inflation ($\epsilon < 1$) happens when the kinetic energy is small compared to the total energy.

4.3.2 Slow Roll

For inflation to last 60 e-folds, it's not enough for ϵ to be small; it has to *stay* small. That means the field's acceleration also has to be small, so it doesn't pick up kinetic energy too fast. The **slow-roll approximation** drops the kinetic energy in the Friedmann equation and the acceleration in the Klein-Gordon equation,

$$H^2 \approx \frac{V}{3M_{\rm Pl}^2}, \qquad 3H\dot{\phi} \approx -V'. \tag{4.16}$$

This is the overdamped limit, where friction is so large that the field moves at its terminal velocity. Plugging these into (4.15), $\epsilon \approx \frac{\dot{\phi}^2}{2M_{\rm Pl}^2H^2} = \frac{V'^2}{18M_{\rm Pl}^2H^4} = \frac{M_{\rm Pl}^2}{2}\frac{V'^2}{V^2}$. This motivates the **potential slow-roll parameters**

$$\boxed{\epsilon_V \equiv \frac{M_{\rm Pl}^2}{2}\left(\frac{V'}{V}\right)^2, \qquad \eta_V \equiv M_{\rm Pl}^2\frac{V''}{V}} \tag{4.17}$$

Slow roll needs $\epsilon_V \ll 1$ (the potential is flat) and $|\eta_V| \ll 1$ (it stays flat), and inflation ends when $\epsilon_V \approx 1$. The number of e-folds between some field value ϕ and the end of inflation is

$$N(\phi) = \int_t^{t_{\rm end}} H\,dt = \int_{\phi_{\rm end}}^{\phi} \frac{H}{|\dot{\phi}|}d\phi \approx \int_{\phi_{\rm end}}^{\phi} \frac{V}{M_{\rm Pl}^2V'}d\phi = \int_{\phi_{\rm end}}^{\phi} \frac{d\phi}{M_{\rm Pl}\sqrt{2\epsilon_V}}. \tag{4.18}$$

The last form says something important: if ϵ_V is small, then to get 60 e-folds the field has to travel a distance of order $M_{\rm Pl}$ or more.

Problem 2.

Consider the simplest model of inflation, $V(\phi) = \frac{1}{2}m^2\phi^2$.
a). Compute ϵ_V and η_V, and find the field value $\phi_{\rm end}$ where inflation ends.
b). Find the field value ϕ_* that is $N_* = 60$ e-folds before the end of inflation.
c). Solve the slow-roll equations for $\phi(t)$ and $a(t)$.
Solution: a). With $V'/V = 2/\phi$ and $V''/V = 2/\phi^2$,

$$\epsilon_V = \eta_V = \frac{2M_{\rm Pl}^2}{\phi^2}.$$

Inflation ends when $\epsilon_V = 1$, so $\boxed{\phi_{\rm end} = \sqrt{2}M_{\rm Pl}}$.

b). Using (4.18),

$$N(\phi) = \int_{\phi_{\rm end}}^{\phi} \frac{\phi'}{2M_{\rm Pl}^2} d\phi' = \frac{\phi^2 - \phi_{\rm end}^2}{4M_{\rm Pl}^2} = \frac{\phi^2}{4M_{\rm Pl}^2} - \frac{1}{2},$$

so $\phi_*^2 = (4N_* + 2)M_{\rm Pl}^2$ and $\boxed{\phi_* \approx 15.6\, M_{\rm Pl}}$ for $N_* = 60$. The field has to start out much bigger than the Planck mass. That sounds alarming, but what matters for quantum gravity is the energy density $V \ll M_{\rm Pl}^4$, not the field value.

c). The slow-roll equations give $\dot\phi = -V'/3H = -m^2\phi/(3H)$, and $H = m\phi/(\sqrt{6}M_{\rm Pl})$, so

$$\dot\phi = -\sqrt{\frac{2}{3}} m M_{\rm Pl} \quad \Longrightarrow \quad \phi(t) = \phi_i - \sqrt{\frac{2}{3}} m M_{\rm Pl}\, t.$$

The field rolls down at a constant speed. Then $\dot a/a = H = m\phi/(\sqrt{6}M_{\rm Pl})$ integrates to

$$a(t) = a_i \exp\left[\frac{\phi_i^2 - \phi(t)^2}{4M_{\rm Pl}^2}\right],$$

which is quasi-exponential expansion. Figure 4.2 shows the full numerical solution of (4.13) and (4.14) (no slow-roll approximation): the field rolls at a constant speed, ϵ stays tiny for about 64 e-folds, and when $\phi \approx \sqrt{2}M_{\rm Pl}$ it shoots up past 1 and the field starts to oscillate.

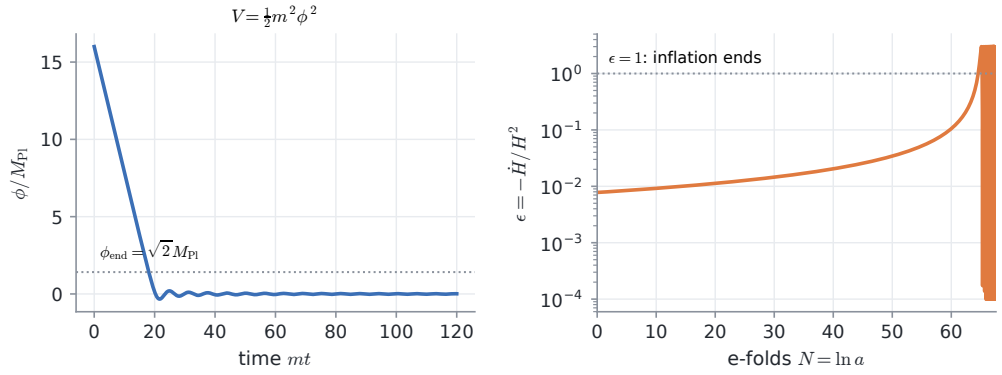


Figure 4.2: Numerical solution for $V = \frac{1}{2}m^2\phi^2$ starting from $\phi = 16M_{\rm Pl}$. Left: the field rolls slowly and then oscillates around the minimum. Right: the slow-roll parameter ϵ stays small for about 64 e-folds and crosses 1 when inflation ends.

4.3.3 Reheating

Inflation leaves behind a universe that is flat, smooth, and empty: any particles that existed before have been diluted by a factor of $e^{-3N} \approx 10^{-78}$. Somehow, the energy stored in the inflaton has to be converted into the hot plasma of Chapter 3. After inflation ends, the field oscillates around the minimum of its potential (figure 4.3). Near a quadratic minimum, averaging $\dot{\phi}^2/2$ and V over an oscillation gives $\langle P \rangle = 0$, so the oscillating field behaves like pressureless matter, $\rho_\phi \propto a^{-3}$. If the inflaton couples to other fields, it decays with some rate Γ, adding a friction term to its equation of motion, $\ddot{\phi} + (3H + \Gamma)\dot{\phi} + V' = 0$. The decay products thermalize into a radiation bath, and the universe is **reheated**. The decay becomes efficient once H drops to about Γ, so by (3.40), the **reheating temperature** is roughly

$$T_R \sim \left(\frac{90}{\pi^2 g_\star}\right)^{1/4} \sqrt{\Gamma M_{\rm Pl}}. \tag{4.19}$$

We don't know Γ, so we don't know T_R. It has to be above a few MeV (otherwise BBN would be ruined) and it can be as high as about 10^{15} GeV. From this point on, the hot Big Bang of Chapter 3 takes over.

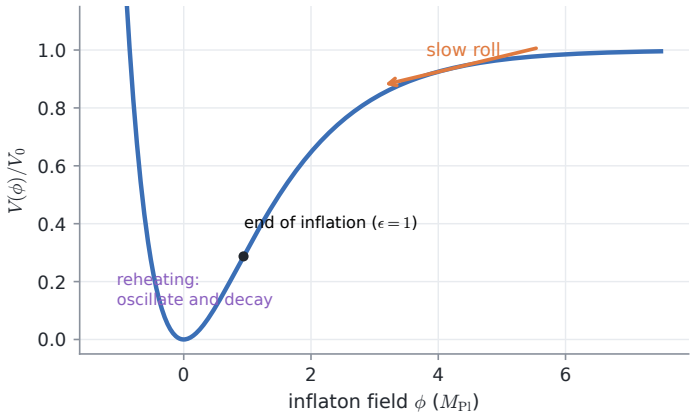


Figure 4.3: A typical inflationary potential (this one is the Starobinsky potential, which fits the data very well). The field slowly rolls down the flat plateau, inflation ends when the potential gets steep, and the field oscillates around the minimum and decays, reheating the universe.

4.4 Quantum Fluctuations And The Seeds Of Structure

So far inflation sounds like a clever fix for some fine-tuning problems. Here's where it becomes a real theory with predictions. The inflaton is a quantum field, so it can't be

perfectly homogeneous; it has quantum fluctuations $\delta\phi(t, \boldsymbol{x})$. Because the comoving Hubble radius is shrinking, a fluctuation with comoving wavenumber k starts out inside the Hubble radius ($k \gg aH$), where it behaves like an ordinary quantum oscillator, and then gets stretched outside ($k \ll aH$), where it freezes. Different parts of the universe end inflation at slightly different times, depending on the local value of $\delta\phi$, and this leaves behind small perturbations in the density. Much later, these re-enter the Hubble radius (figure 4.1) and become the seeds of the CMB anisotropies and of all structure in the universe. We will develop the theory of perturbations properly in the next two chapters; here is the sketch of the calculation.

4.4.1 A Scalar Field In De Sitter Space

Perturb the field around its background, $\phi(t, \boldsymbol{x}) = \bar{\phi}(t) + \delta\phi(t, \boldsymbol{x})$. During slow roll, the potential is so flat that we can treat $\delta\phi$ as a massless field, and the expansion is nearly de Sitter, $a(\eta) = -1/(H\eta)$ in conformal time (with η running from $-\infty$ to 0^-). Linearizing the Klein-Gordon equation (now with spatial gradients, ∇^2/a^2) and Fourier transforming gives

$$\delta\phi_k'' + 2\mathcal{H}\delta\phi_k' + k^2\delta\phi_k = 0, \tag{4.20}$$

where primes are conformal time derivatives and $\mathcal{H} = a'/a = -1/\eta$. The friction term can be removed by working with $v_k \equiv a\,\delta\phi_k$, which satisfies

$$\boxed{v_k'' + \left(k^2 - \frac{a''}{a}\right) v_k = 0, \qquad \frac{a''}{a} = \frac{2}{\eta^2}} \tag{4.21}$$

This is the **Mukhanov-Sasaki equation**, a harmonic oscillator with a time-dependent frequency. At early times, $|k\eta| \gg 1$, the mode is deep inside the Hubble radius, the a''/a term is negligible, and v_k is just a free oscillator, $v_k \propto e^{-ik\eta}$. Quantizing it like the harmonic oscillator in quantum mechanics and demanding that it start in its ground state (the **Bunch-Davies vacuum**) fixes the normalization to $v_k \to e^{-ik\eta}/\sqrt{2k}$. The exact solution with this behavior is

$$v_k(\eta) = \frac{e^{-ik\eta}}{\sqrt{2k}}\left(1 - \frac{i}{k\eta}\right). \tag{4.22}$$

Late in inflation, $|k\eta| \ll 1$, the second term wins, and $|v_k| \to 1/(\sqrt{2k}\,k|\eta|)$. Converting back, $|\delta\phi_k| = |v_k|/a = H/\sqrt{2k^3}$, which is constant: the fluctuation has frozen. The power spectrum (variance per logarithmic interval in k, which we will define carefully in the next chapter) is

$$\Delta^2_{\delta\phi}(k) = \frac{k^3}{2\pi^2}|\delta\phi_k|^2 = \left(\frac{H}{2\pi}\right)^2. \tag{4.23}$$

Quantum fluctuations during inflation have an amplitude set by the Hubble rate, and it's the same at every scale. This is the cosmological version of Hawking radiation: de Sitter space has a temperature $H/2\pi$.

4.4.2 The Primordial Power Spectrum

What we actually observe is not $\delta\phi$ but the curvature perturbation $\mathcal{R}$ (which we will define properly in Chapter 6). The connection is simple to state with the "δN" picture: a region where the field is ahead by $\delta\phi$ reaches the end of inflation earlier by $\delta t = -\delta\phi/\dot{\bar{\phi}}$, so it inflates by $\delta N = H\delta t$ fewer e-folds. A difference in the local amount of expansion is a perturbation in the spatial curvature, $\mathcal{R} = -H\delta\phi/\dot{\bar{\phi}}$. Using $\dot{\bar{\phi}}^2 = 2\epsilon M_{\rm Pl}^2 H^2$ from (4.15), the dimensionless power spectrum of curvature perturbations is

$$\boxed{\Delta^2_{\mathcal{R}}(k) = \frac{H^2}{\dot{\bar{\phi}}^2}\left(\frac{H}{2\pi}\right)^2 = \frac{1}{8\pi^2\epsilon}\frac{H^2}{M_{\rm Pl}^2}\bigg|_{k=aH}} \tag{4.24}$$

evaluated when the mode exits the Hubble radius, after which $\mathcal{R}$ is conserved (another result for Chapter 6). Since H and ϵ change slowly during inflation, the spectrum is *almost* scale-invariant, and we parametrize the small deviation as

$$\Delta^2_{\mathcal{R}}(k) = A_s\left(\frac{k}{k_*}\right)^{n_s-1}, \qquad n_s - 1 \equiv \frac{d\ln\Delta^2_{\mathcal{R}}}{d\ln k}. \tag{4.25}$$

Since $d\ln k = d\ln(aH) \approx dN$ at horizon exit, and $H^2/\epsilon \propto V/\epsilon_V$ in slow roll, a short calculation (Problem 3) gives

$$\boxed{n_s - 1 = 2\eta_V - 6\epsilon_V} \tag{4.26}$$

Gravitational waves are produced the same way. Each of the two polarizations of the tensor perturbation h_{ij} behaves like a massless scalar field with normalization $\frac{M_{\rm Pl}}{2}h$, so

$$\Delta_t^2 = 2\times\left(\frac{2}{M_{\rm Pl}}\right)^2\left(\frac{H}{2\pi}\right)^2 = \frac{2}{\pi^2}\frac{H^2}{M_{\rm Pl}^2}, \qquad r \equiv \frac{\Delta_t^2}{\Delta^2_{\mathcal{R}}} = 16\epsilon. \tag{4.27}$$

The **tensor-to-scalar ratio** r measures ϵ directly, and through $\Delta_t^2 \propto H^2 \propto V$ it measures the energy scale of inflation.

Problem 3.

a). Using the slow-roll approximation, show that $\Delta^2_{\mathcal{R}} = \frac{V}{24\pi^2 M_{\rm Pl}^4\epsilon_V}$.

b). Show that $n_s - 1 = 2\eta_V - 6\epsilon_V$, using $d\ln k \approx dN = -\frac{V}{M_{\rm Pl}^2 V'}d\phi$.
c). For $V \propto \phi^p$, show that $n_s = 1 - \frac{p+2}{2N_*}$ and $r = \frac{4p}{N_*}$, where N_* is the number of e-folds before the end of inflation when the CMB scales exited.
Solution: a). With $H^2 = V/3M_{\rm Pl}^2$ and $\epsilon = \epsilon_V$, (4.24) gives $\Delta_{\mathcal{R}}^2 = \frac{1}{8\pi^2\epsilon_V}\frac{V}{3M_{\rm Pl}^4}$.
b). Taking the log, $\ln\Delta_{\mathcal{R}}^2 = \ln V - \ln\epsilon_V +$ const. Using $\frac{d}{d\ln k} = -\frac{M_{\rm Pl}^2 V'}{V}\frac{d}{d\phi}$ (the minus sign is because the field rolls down the potential as k increases),

$$\frac{d\ln V}{d\ln k} = -M_{\rm Pl}^2\frac{V'^2}{V^2} = -2\epsilon_V, \qquad \frac{d\ln\epsilon_V}{d\ln k} = -\frac{M_{\rm Pl}^2 V'}{V}\left(\frac{2V''}{V'} - \frac{2V'}{V}\right) = -2\eta_V + 4\epsilon_V.$$

So $n_s - 1 = -2\epsilon_V - (-2\eta_V + 4\epsilon_V) = \boxed{2\eta_V - 6\epsilon_V}$.
c). For $V \propto \phi^p$, $\epsilon_V = \frac{p^2 M_{\rm Pl}^2}{2\phi^2}$, $\eta_V = \frac{p(p-1)M_{\rm Pl}^2}{\phi^2}$, and (4.18) gives $N \approx \phi^2/(2pM_{\rm Pl}^2)$ (ignoring $\phi_{\rm end}$). So $\epsilon_V = \frac{p}{4N}$ and $\eta_V = \frac{p-1}{2N}$, giving

$$n_s - 1 = \frac{p-1}{N} - \frac{3p}{2N} = -\frac{p+2}{2N}, \qquad r = 16\epsilon_V = \frac{4p}{N}.$$

For $p = 2$ and $N_* = 60$, $n_s = 0.967$ and $r = 0.13$.

4.4.3 Comparing With Observations

The CMB measures the amplitude and tilt of the primordial spectrum very precisely [2],

$$A_s \approx 2.1 \times 10^{-9}, \qquad n_s = 0.965 \pm 0.004, \tag{4.28}$$

and gravitational waves from inflation have not been detected yet, with the best bound being $r < 0.036$. Take a moment to appreciate this. The spectrum is nearly scale-invariant, as inflation predicts, but *not exactly* ($n_s \neq 1$ at about 8σ), which is also what slow-roll inflation predicts, since H and ϵ must change a little for inflation to end. The perturbations are also Gaussian and adiabatic to the precision we can measure, and the CMB shows correlations on scales that were outside the horizon at decoupling, both of which fit the inflationary picture. Figure 4.4 compares the predictions of a few models. The simple $m^2\phi^2$ model predicts too many gravitational waves and is now ruled out, while models with a flat plateau, like the Starobinsky model of figure 4.3, sit right in the middle of the allowed region.

The bound on r also bounds the energy scale of inflation. From part a) of Problem 3, $V = \frac{3\pi^2}{2}A_s r M_{\rm Pl}^4$, so

$$V^{1/4} = \left(\frac{r}{0.036}\right)^{1/4} 1.4 \times 10^{16}\ \text{GeV}. \tag{4.29}$$

If we ever detect primordial gravitational waves (through the B-mode polarization of the CMB), we will learn that something happened at energies about a trillion times higher than anything the LHC can reach.

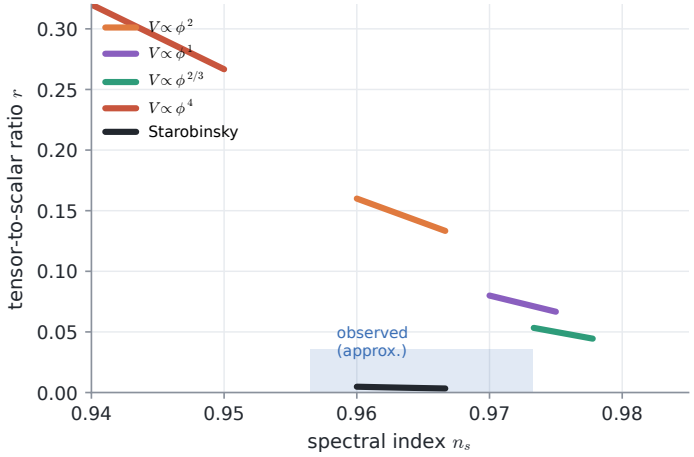


Figure 4.4: Predictions for n_s and r of a few inflationary models, for $N_* = 50$ to 60 e-folds, against an approximate version of the observed region ($n_s = 0.965 \pm 0.008$, $r < 0.036$). Simple monomial potentials predict too large an r, while plateau models like Starobinsky fit well.

4.5 Problems

1. Normalizing The Inflaton Mass

For $V = \frac{1}{2}m^2\phi^2$, use $\Delta^2_{\mathcal{R}} = 2.1\times10^{-9}$ at $N_* = 60$ to show that $m \approx 6\times10^{-6}M_{\rm Pl} \approx 1.4\times10^{13}$ GeV. What is the Hubble rate during inflation, and what is the energy density $V^{1/4}$ when the CMB scales exit the horizon?

2. The Starobinsky Model

The Starobinsky potential is $V = V_0\left(1 - e^{-\sqrt{2/3}\phi/M_{\rm Pl}}\right)^2$.

a). Show that for large ϕ, $\epsilon_V \approx \frac{4}{3}e^{-2\sqrt{2/3}\phi/M_{\rm Pl}}$ and $\eta_V \approx -\frac{4}{3}e^{-\sqrt{2/3}\phi/M_{\rm Pl}}$.

b). Show that $N \approx \frac{3}{4}e^{\sqrt{2/3}\phi/M_{\rm Pl}}$, and hence $n_s \approx 1 - \frac{2}{N}$ and $r \approx \frac{12}{N^2}$. Evaluate for $N = 55$.

3. Inflation Solves The Monopole Problem

Many grand unified theories predict that magnetic monopoles of mass $m_M \sim 10^{16}$ GeV were produced around $T \sim 10^{16}$ GeV, with a number density of about one per Hubble volume, $n_M \sim H^3$.

a). Without inflation, estimate n_M/s at production and Ω_M today. Why is this a disaster?

b). Show that N e-folds of inflation after the monopoles are produced reduce their abundance by e^{-3N}. How many e-folds are needed to make it less than one monopole in the observable universe?

4. Eternal De Sitter?

Show that a universe with only a cosmological constant has $\epsilon = 0$ exactly and never stops inflating. Then explain, using the comoving Hubble radius, why an exactly constant H cannot produce the hot Big Bang, and why the small deviation $n_s \neq 1$ is evidence that inflation ended.

5

Perturbation Theory

"The human mind treats a new idea the way the body treats a strange protein – it rejects it."

– P.B. Medawar

In the beginning of these notes, I stated that if we looked around us, we notice that we are not dead. Which is a bit surprising because for everything that we have done since the first chapter, we assumed that the universe was isotropic and homogeneous. We allowed it to be dynamical (through cosmological expansion) but even then, everything looked the same everywhere. In the last chapter we finally gave an explanation for *where* these *perturbations* (like you and me) come from: quantum fluctuations during inflation, stretched to cosmic scales, leave behind density perturbations of about one part in 10^5. In this chapter, we will explore how those tiny perturbations grow into the galaxies and clusters that we see today.

Perturbation theory is split into 2 "types" (if you will). One is Newtonian, and it's good for non-relativistic matter like dark matter and baryons on scales much smaller than the Hubble radius. The other is relativistic, and it's needed for radiation, for perturbations on scales comparable to the Hubble radius, and for perturbations of the metric itself (like gravitational waves). In this chapter, we will develop the Newtonian theory and use it to understand the growth of structure, and the next chapter does the relativistic theory. The reason for doing it in this order is that almost all of the intuition (gravity makes lumps grow, pressure fights back, expansion acts like friction) is already there in the Newtonian theory, without the headache of choosing coordinates in a perturbed spacetime.

5.1 Non-Relativistic Perturbation Theory

Most people when they think about Newtonian mechanics in its relation to relativity, they probably think that Newtonian gravity is this thing that is just fundamentally wrong and can't fully explain the motion of the stars. This is...actually incorrect. Newtonian mechanics, even on the scales of solar systems, works perfectly fine. In fact, the physics that got humans to the moon were all Newtonian mechanics. It is only in the extremes that general relativity becomes important, and even then, its correction to Newtonian gravity can often be considered small in magnitude. This is all to say that if we wanted to examine structure formation in the universe, it's safe enough to treat the system classically, as long as the velocities are small and we look at scales well inside the Hubble radius. How we will do this is through fluid mechanics; and this makes sense because on the largest of scales, objects like stars and planets behave like little atomic balls in a glass of water.

Because the contents of our universe behave like a fluid, we can use the results from the previous chapters. Recall that for a perfect fluid in curve space-time, the energy-momentum tensor takes the form

$$T^{\mu\nu} = (\rho + P)\, U^\mu U^\nu + P g^{\mu\nu} \tag{5.1}$$

where U^μ is the 4-velocity of our fluid (for some given mass). To warm up, let's turn gravity and the expansion off, so the metric will be the flat Minkowski metric. Classical mechanics also implies that matter is non-relativistic, and thus must have the relation $P \ll \rho$ (this was the equation of state condition we established for matter in chapter 2). This means that the energy momentum tensor takes the form

$$T^{\mu\nu} = \rho U^\mu U^\nu + P\eta^{\mu\nu}$$

Problem 1.

Show that

$$g_{\mu\nu} U^\mu U^\nu = -1$$

regardless of what the metric is.

Solution:

Remember that the role of the metric is to lower the index. This implies

$$g_{\mu\nu} U^\mu U^\nu = U_\nu U^\nu$$

Now, what is the velocity 4-vector by definition? It is $U^\mu = dx^\mu/d\tau$. In a local inertial

frame at any point, $g_{\mu\nu} = \eta_{\mu\nu}$ and $U^\mu = \gamma \langle 1, v_1, v_2, v_3 \rangle$ where γ is the Lorentz factor. Thus, the metric inner product gives

$$\boxed{U_\nu U^\nu = \gamma^2 \left(v^2 - 1\right) = -1}$$

Since $U_\nu U^\nu$ is a scalar, if it's -1 in one frame it's -1 in every frame and for every metric. (This is the same statement as (2.8), it's just the definition of proper time.)

With this, let us apply conservation of the energy-momentum tensor

$$\begin{aligned} \nabla_\mu T^{\mu\nu} = \nabla_\mu \left(\rho U^\mu U^\nu + P\eta^{\mu\nu}\right) &= 0 \\ \nabla_\mu \left(\rho U^\mu U^\nu\right) + \nabla_\mu \left(P\eta^{\mu\nu}\right) &= 0 \\ \left(\nabla_\mu \rho\right) U^\mu U^\nu + \rho \nabla_\mu \left(U^\mu U^\nu\right) + \left(\nabla_\mu P\right)\eta^{\mu\nu} + P\nabla_\mu \eta^{\mu\nu} &= 0 \end{aligned} \tag{5.2}$$

But we know that $\nabla_\mu g^{\mu\nu} = 0$ regardless of what the metric is. Next, we use the fact that for non-relativistic motion $\gamma \approx 1$, so $U^\mu \approx \langle 1, \boldsymbol{u} \rangle$, as well as the fact that in flat space, the covariant derivative ∇_μ reduces down to the regular 4 derivative ∂_μ. We consider 2 cases that will give us 2 sets of equations. The first is when $\nu = 0$. This reduces the equation above down to

$$\begin{aligned} \left(\nabla_\mu \rho\right) U^\mu + \rho \nabla_\mu \left(U^\mu\right) + \left(\nabla_\mu P\right)\eta^{\mu 0} &= 0 \\ \dot{\rho} + \left(\nabla\rho \cdot \boldsymbol{u}\right) + \rho \left(\nabla \cdot \boldsymbol{u}\right) - \dot{P} &= 0 \\ \dot{\rho} + \nabla \cdot \left(\rho \boldsymbol{u}\right) &= 0 \end{aligned} \tag{5.3}$$

where ∇ in this case is the usual Cartesian gradient. For non-relativistic matter, $\dot{P}$ is negligible compared to $\dot{\rho}$ because pressure is significantly smaller than ρ. So we get the familiar **continuity equation**! This also partially justifies the $\dot{P} \approx 0$ assumption because in the non-relativistic limit, the continuity equation is a statement about conservation. That is, non-relativistic matter is mostly rest-mass and rest mass is conserved as the fluid moves. The second case is when $\nu = i$. This case is a bit messier. Starting once again with the conservation equation

$$\begin{aligned} \left(\partial_\mu \rho\right) U^\mu U^i + \rho \left(\partial_\mu U^\mu\right) U^i + \rho U^\mu \left(\partial_\mu U^i\right) + \left(\partial_\mu P\right) \eta^{\mu i} &= 0 \\ \left(\dot{\rho} + \nabla\rho \cdot \boldsymbol{u}\right) \boldsymbol{u} + \left(\rho \nabla \cdot \boldsymbol{u}\right) \boldsymbol{u} + \rho \left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) + \nabla P &= 0 \\ \boldsymbol{u}(\dot{\rho} + \nabla\rho \cdot \boldsymbol{u} + \rho \nabla \cdot \boldsymbol{u}) + \rho \left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) + \nabla P &= 0 \\ \rho \left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) + \nabla P &= 0 \end{aligned} \tag{5.4}$$

$\dot{\rho} + \nabla\rho \cdot \boldsymbol{u} + \rho \nabla \cdot \boldsymbol{u} = 0$

the last equation is called the **Euler equation.** It is a statement about momentum conservation. Notice here that we don't take the gradient ∇P to be 0 even though we made the approximation that $\dot{P} \approx 0$. This is because pressure gradient is a force. Even a small pressure can produce an important force if it varies over short distances. Together, the 2 equations form the basis of the fluid mechanics of perturbation theory

$$\boxed{\begin{aligned} \dot{\rho} + \nabla \cdot (\rho \boldsymbol{u}) &= 0 \\ \rho \left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) &= -\nabla P \end{aligned}} \tag{5.5}$$

This set of equations is the equivalent of $F = ma$ but for a fluid element. Here $\boldsymbol{u}$ is the velocity field, and the combination $\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}$ tracks the movement of the fluid along its trajectory. That is, $\partial_t \boldsymbol{u}$ tells us how the velocity changes at a fixed position, while the whole combination tells us how the velocity of a fluid element changes as we move with it.

5.1.1 Density Perturbations

A homogeneous fluid at rest, with constant pressure $\bar{P}$ and density $\bar{\rho}$, trivially solves (5.5). What we want to do is study how the density and pressure evolves when it is pushed slightly out of equilibrium with the addition of $\delta\rho$ and δP

$$\begin{aligned} \rho(t, \boldsymbol{x}) &= \bar{\rho} + \delta\rho(t, \boldsymbol{x}) \\ P(t, \boldsymbol{x}) &= \bar{P} + \delta P(t, \boldsymbol{x}) \end{aligned} \tag{5.6}$$

Now let us plug these back into (5.5), treating $\delta\rho$, δP and $\boldsymbol{u}$ as small. The continuity equation gives

$$\begin{aligned} \frac{d}{dt}(\bar{\rho} + \delta\rho) + \nabla \cdot ((\bar{\rho} + \delta\rho)\boldsymbol{u}) &= 0 \\ \delta\dot{\rho} + \cancelto{0}{\nabla \cdot (\delta\rho \boldsymbol{u})} + \nabla \cdot (\bar{\rho}\boldsymbol{u}) &= 0 \\ \delta\dot{\rho} + \nabla \cdot (\bar{\rho}\boldsymbol{u}) &= 0 \end{aligned} \tag{5.7}$$

where in the second line we simplified the second term cancels to 0 since the term $\nabla \cdot (\delta\rho\boldsymbol{u})$ is a second order perturbation that we can approximate to be 0. Now for the Euler equations

$$\begin{aligned} (\bar{\rho} + \delta\rho)\left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) &= -\nabla(\bar{P} + \delta P) \\ \bar{\rho}\left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) + \delta\rho\left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) &= -\nabla\bar{P} - \nabla\delta P \\ \bar{\rho}\left(\dot{\boldsymbol{u}} + \cancelto{0}{\boldsymbol{u} \cdot \nabla \boldsymbol{u}}\right) + \cancelto{0}{\delta\rho\left(\dot{\boldsymbol{u}} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right)} &= \cancelto{0}{-\nabla\bar{P}} - \nabla\delta P \end{aligned}$$

$$\bar{\rho}\dot{\boldsymbol{u}} = -\nabla\delta P \tag{5.8}$$

Where in the the third line we recognize that $\delta\rho\,(\dot{\boldsymbol{u}} + \boldsymbol{u}\cdot\nabla\boldsymbol{u})$ is a second order perturbative term so we sent it to 0. Similarly, $\boldsymbol{u}\cdot\nabla\boldsymbol{u}$ is also a second order perturbative term (since we expect that $\boldsymbol{u}$ is to be very small in the non-relativistic limit, making $\boldsymbol{u}\cdot\nabla\boldsymbol{u}$ essentially a quadratically small term). The two equations can be combined by taking a time derivative of the continuity equation

$$\begin{aligned}
\frac{d}{dt}\left(\delta\dot{\rho} + \nabla\cdot(\bar{\rho}\boldsymbol{u})\right) &= 0 \\
\delta\ddot{\rho} + \nabla\cdot(\bar{\rho}\dot{\boldsymbol{u}}) &= 0 \\
\delta\ddot{\rho} - \nabla\cdot(\nabla\delta P) &= 0 \\
\frac{\partial^2\delta\rho}{\partial t^2} - \nabla^2\delta P &= 0
\end{aligned} \tag{5.9}$$

Which should be familiar to all of us as this is the wave equation! Well, it's almost the wave equation. It would be if δP was equal to $\delta\rho$ up to a constant. However, we can relate the 2 through the equation of state. Thermodynamics tells us that the pressure can be written in terms of the energy (density) and the temperature $P(\rho, T)$. This means we can write

$$\delta P = \frac{\partial P}{\partial\rho}\delta\rho \equiv c_s^2\delta\rho \tag{5.10}$$

where we defined $c_s^2 = \partial P/\partial\rho$ (at fixed entropy) as the square of the fluid's **speed of sound**. All together, we get the wave equation

$$\boxed{\frac{\partial^2\delta\rho}{\partial t^2} - c_s^2\nabla^2\delta\rho = 0} \tag{5.11}$$

This tells us that when we add small perturbations of the density, we get sound waves that propagate through space. Physically, this tells us that density perturbations do not grow or shrink; when we kick the fluid out of the equilibrium, its reaction is to compress and oscillate. This makes sense because in the absence of gravity (or universal expansion, as we'll see soon), nothing is pulling the lumps together or apart. An analogy would be like if you had a cubical block of jelly and poked it. By itself, all that it would do is jiggle back and forth forever (assume no friction).

Problem 2.

Here is an exercise in mathematical methods. Starting with the wave equation

$$\frac{\partial^2 \delta\rho}{\partial t^2} - c_s^2 \nabla^2 \delta\rho = 0$$

a). Take a Fourier transform of the PDE in $\boldsymbol{k}$ space.
b). Solve the now ordinary differential equation in k space.
c). Use an inverse Fourier transform to send the solutions back into real space $\boldsymbol{x}$.
Solution:
a). Here are some standard rules regarding the Fourier transform. We know that $\mathcal{F}[\partial_t f] = \partial_t \mathcal{F}[f]$, which is to say that the second derivative in time is basically un-affected by the spatial Fourier transform. Secondly, we know that the Fourier transform takes derivatives in real space and sends them to powers of $\boldsymbol{k}$ (with the convention $\delta\rho(\boldsymbol{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \delta\tilde{\rho}(\boldsymbol{k})$, $\nabla \to i\boldsymbol{k}$ and $\nabla^2 \to -k^2$). Thus, we have

$$\int d^3\boldsymbol{x}\; e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \left(\frac{\partial^2 \delta\rho}{\partial t^2} - c_s^2 \nabla^2 \delta\rho \right) = 0$$
$$\frac{\partial^2 \delta\tilde{\rho}}{\partial t^2} + c_s^2 k^2 \delta\tilde{\rho} = 0$$

where $\delta\tilde{\rho}$ is the $\boldsymbol{k}$-space Fourier transform of $\delta\rho$. Note here that the $\delta\tilde{\rho}(t, \boldsymbol{k})$ is still a function of k. This will become important in a second. For now, our reduced ordinary differential equation is

$$\boxed{\frac{\partial^2 \delta\tilde{\rho}}{\partial t^2} + c_s^2 k^2 \delta\tilde{\rho} = 0}$$

b). At this point, we should be completely familiar with how to solve the above differential equation; it's a harmonic oscillator. The solution is simply

$$\boxed{\delta\tilde{\rho}(t, \boldsymbol{k}) = A_{\boldsymbol{k}} e^{-i\omega_k t} + B_{\boldsymbol{k}} e^{i\omega_k t}}$$

where $\omega \equiv c_s k$. the reason that we put the $\boldsymbol{k}$ in the subscript of the coefficients A and B is to remind ourselves that the A and B are still functions of $\boldsymbol{k}$.
c). In general, we cannot specify the function $\delta\rho(t, \boldsymbol{x})$ without boundary conditions. This means that the general solution must be left as inverse Fourier transform

$$\boxed{\delta\rho(t, \boldsymbol{x}) = \int \frac{d^3\boldsymbol{k}}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \left(A_{\boldsymbol{k}} e^{-i\omega_k t} + B_{\boldsymbol{k}} e^{i\omega_k t} \right)}$$

a superposition of sound waves travelling in every direction at speed c_s, with the coefficients fixed by the initial conditions (and $B_{\boldsymbol{k}} = A^*_{-\boldsymbol{k}}$ for $\delta\rho$ to be real).

5.1.2 The Jeans Instability

At the end of the previous section, we noted that density perturbations by themselves do not grow. But we forgot something: if there is mass, there will be gravity. If we look back to the Euler equation, we made a comment saying that this equation was essentially $F = ma$ but for fluids, with $-\nabla P$ being the force term. Continuing from this comment, if we wanted to introduce gravity into the system, we know that classically the gravitational force is just the gradient of the gravitational potential $F_g = -\nabla\Phi$. This makes the updated Euler equation

$$\rho\left(\dot{\boldsymbol{u}} + \boldsymbol{u}\cdot\nabla\boldsymbol{u}\right) = -\nabla P - \rho\nabla\Phi \tag{5.12}$$

The gravitational potential tells our fluid how to move, and must be determined not by the Euler equation (as that would be circular) but from the Poisson equation for gravity

$$\nabla^2\Phi = 4\pi G_N\rho \tag{5.13}$$

In general, the gravitational potential is not something that we can solve easily (only in a handful of small models can this be true), but is often solved numerically.

When we do perturbation theory, we expand from a constant solution for the density and pressure, found by setting $\boldsymbol{u} = 0$. However, from the Euler equation, this would imply that $\nabla\bar{\Phi} = 0$, which cannot be possible from the Poisson equation with $\bar{\rho} \neq 0$. This is just a statement that there are no infinite, static, self-gravitating fluids (the universe can't be static, which is Einstein's lesson). The traditional fix, called the **Jeans swindle**, is to just ignore the background potential and only keep the perturbation $\delta\Phi$ sourced by $\delta\rho$. We'll see in a moment that the expanding universe justifies this properly. With the swindle, the linearized equations are

$$\begin{aligned}\frac{\partial\delta\rho}{\partial t} &= -\nabla\cdot(\bar{\rho}\boldsymbol{u})\\ \bar{\rho}\frac{\partial\boldsymbol{u}}{\partial t} &= -\nabla P - \bar{\rho}\nabla\delta\Phi\end{aligned} \tag{5.14}$$

subject to the 'constraint'

$$\nabla^2\delta\Phi = 4\pi G_N\delta\rho \tag{5.15}$$

Re-doing all of the analysis from the previous section, the wave equation picks up a new

term

$$\frac{\partial^2 \delta\rho}{\partial t^2} - c_s^2 \nabla^2 \delta\rho - 4\pi G_N \bar{\rho} \delta\rho = 0 \tag{5.16}$$

When we take the Fourier transform of this equation, we get the differential equation

$$\left(\frac{\partial^2}{\partial t^2} + c_s^2 k^2 - 4\pi G_N \bar{\rho} \right) \delta\tilde{\rho} = 0 \tag{5.17}$$

This is a harmonic oscillator with frequency $\omega^2 = c_s^2 k^2 - 4\pi G_N \bar{\rho}$, and we notice that there exists a value of k at which the frequency vanishes,

$$k_J^2 = \frac{4\pi G_N \bar{\rho}}{c_s^2}, \tag{5.18}$$

which we call the **Jeans scale**, with **Jeans length** $\lambda_J = 2\pi/k_J$. At exactly this scale, pressure and gravity balance, $\delta\tilde{\rho}'' = 0$, and the perturbation just drifts linearly in time. The Jeans scale marks a transition (figure 5.1). When $k \gg k_J$ (short wavelengths), $\omega^2 > 0$, pressure dominates gravity and we get back the sound waves seen before. When $k \ll k_J$ (long wavelengths), $\omega^2 < 0$, gravity dominates the pressure and (5.16) gives exponentially growing solutions $\delta\tilde{\rho} \propto e^{\pm|\omega|t}$, with a growth time of about $1/\sqrt{4\pi G_N \bar{\rho}}$. The exponential growth of these perturbations is called **the Jeans instability**.

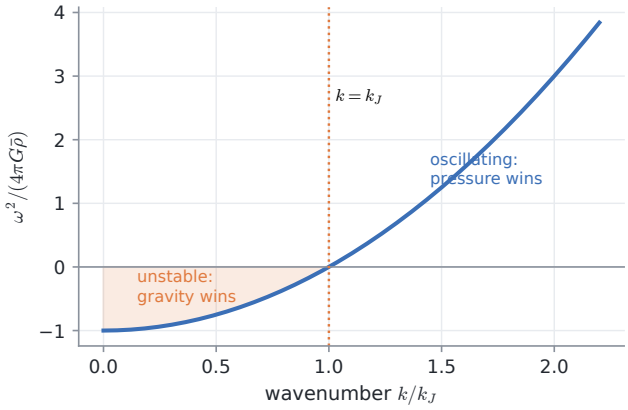


Figure 5.1: The dispersion relation $\omega^2 = c_s^2 k^2 - 4\pi G\bar{\rho}$ of a self-gravitating fluid. Perturbations larger than the Jeans length ($k < k_J$) have $\omega^2 < 0$ and grow, while smaller ones oscillate as sound waves.

We should take a second to consider what the word 'scale' means. We are imagining a situation where on the largest of length scales, the universe is a giant jelly cube that jiggles back and forth. The wavelength of oscillations is measured (inversely) through the wave number k. Thus, when we say scales $k \gg k_J$, what we're really saying is at very small length

scales. And this should intuitively make sense: the time it takes a sound wave to cross a region grows with its size, while the time it takes the region to collapse under its own gravity doesn't depend on the size at all. Small regions can respond with pressure before they collapse, and big regions can't. On the other hand, $k \ll k_J$ is a statement of large distances. On the scale of, say, galaxies, gravity is the dominating force that keeps the fluid in tact. At the scale of the Jeans instability $k \ll k_J$, the solutions are exponential. Physically, this means that any initial chunk of matter will attract other chunks and clump together. This creates a feedback loop where more chunks clump together and the perturbations grow exponentially.

There are two catches. Mathematically, we cheated with the Jeans swindle. Physically, exponential growth cannot continue forever. The Jeans derivation assumes the density perturbations are small $\delta\rho/\bar{\rho} \ll 1$, but the exponential mode from the Jeans instability suggests that there is a point in time where this is no longer valid. That is, the perturbations grow unbounded and are no longer 'perturbations.' Once it becomes large enough, the region collapses non-linearly, shocks may form, pressure heats up, etc. This means that the linearization of the density, pressure, and potential can not produce the final state.

5.1.3 Perturbations In Expanding Spacetime

The Jeans instability changes its behavior once we account for the fact that the universe is expanding. To do this, notice that the fluid equations track the velocity (and subsequently the position) of the fluid as it evolves. The coordinates of the velocity field $\boldsymbol{u}$ are the physical coordinates. The physical coordinates in terms of the co-moving coordinates are related through the expansion rate $a(t)$ via $\boldsymbol{r}(t) = a(t)\boldsymbol{x}$. This makes the velocity field

$$\boldsymbol{u}(t) = \dot{\boldsymbol{r}} = H\boldsymbol{r} + \boldsymbol{v} \tag{5.19}$$

where we remind ourselves that $H\boldsymbol{r}(t)$ is the Hubble flow and $\boldsymbol{v}$ is the peculiar velocity. Because the fluid equations were written in physical coordinates, we need to know how derivatives change when we switch to comoving coordinates. Spatial derivatives are easy, $\nabla_{\boldsymbol{r}} = a^{-1}\nabla_{\boldsymbol{x}}$. Time derivatives are trickier, because holding $\boldsymbol{r}$ fixed is not the same as holding $\boldsymbol{x}$ fixed. Consider any function f, written either as $f(t, \boldsymbol{r})$ or as $f(t, \boldsymbol{x})$ with $\boldsymbol{r} = a(t)\boldsymbol{x}$. By the chain rule, the time derivative at fixed $\boldsymbol{x}$ is

$$\left(\frac{\partial f}{\partial t}\right)_{\boldsymbol{x}} = \left(\frac{\partial f}{\partial t}\right)_{\boldsymbol{r}} + \frac{\partial \boldsymbol{r}}{\partial t}\Big|_{\boldsymbol{x}} \cdot \nabla_{\boldsymbol{r}} f = \left(\frac{\partial f}{\partial t}\right)_{\boldsymbol{r}} + H\boldsymbol{r} \cdot \nabla_{\boldsymbol{r}} f, \tag{5.20}$$

since $\partial \boldsymbol{r}/\partial t$ at fixed $\boldsymbol{x}$ is $\dot{a}\boldsymbol{x} = H\boldsymbol{r}$. Rearranging and writing $H\boldsymbol{r} \cdot \nabla_{\boldsymbol{r}} = H\boldsymbol{x} \cdot \nabla_{\boldsymbol{x}}$,

$$\left(\frac{\partial}{\partial t}\right)_{\boldsymbol{r}} = \left(\frac{\partial}{\partial t}\right)_{\boldsymbol{x}} - H\boldsymbol{x} \cdot \nabla_{\boldsymbol{x}}. \tag{5.21}$$

Now to introduce cosmic expansion into the density perturbations, we go back to the continuity and Euler equations and substitute in all the derivatives. Starting with the continuity equation, we have

$$\frac{\partial \rho(t, \boldsymbol{r})}{\partial t} + \nabla_{\boldsymbol{r}} \cdot (\rho \boldsymbol{u}) = 0$$

$$\frac{\partial \rho}{\partial t} - \nabla_{\boldsymbol{x}} \rho \cdot H\boldsymbol{x} + \frac{1}{a} \nabla_{\boldsymbol{x}} \cdot (\rho \boldsymbol{u}) = 0$$

Now substituting in the fluid field velocity expression in co-moving coordinates,

$$\begin{aligned}
\frac{\partial \rho}{\partial t} - \nabla_{\boldsymbol{x}} \rho \cdot H\boldsymbol{x} + \frac{1}{a} \nabla_{\boldsymbol{x}} \cdot (\rho(H\boldsymbol{r} + \boldsymbol{v})) &= 0 \\
\frac{\partial \rho}{\partial t} - \nabla_{\boldsymbol{x}} \rho \cdot H\boldsymbol{x} + \frac{1}{a} \nabla_{\boldsymbol{x}} \cdot (\rho(aH\boldsymbol{x} + \boldsymbol{v})) &= 0 \\
\frac{\partial \rho}{\partial t} - \nabla_{\boldsymbol{x}} \rho \cdot H\boldsymbol{x} + H \nabla_{\boldsymbol{x}} \cdot \rho \boldsymbol{x} + \frac{1}{a} \nabla_{\boldsymbol{x}} \cdot (\rho \boldsymbol{v}) &= 0 \\
\frac{\partial \rho}{\partial t} - \cancel{\nabla_{\boldsymbol{x}} \rho \cdot H\boldsymbol{x}} + \cancel{H \nabla_{\boldsymbol{x}} \rho \cdot \boldsymbol{x}} + H\rho \cancel{\nabla_{\boldsymbol{x}} \cdot \boldsymbol{x}}^{3} + \frac{1}{a} \nabla_{\boldsymbol{x}} (\rho \boldsymbol{v}) &= 0 \\
\frac{\partial \rho}{\partial t} + 3H\rho + \frac{1}{a} \nabla_{\boldsymbol{x}} (\rho \boldsymbol{v}) &= 0
\end{aligned} \tag{5.22}$$

where every time derivative is now at fixed $\boldsymbol{x}$. A similar analysis can be done for the Euler equation, and we will write it down in a second, but first, let's see what the continuity equation says about the background. From now on, we will write the comoving derivative $\nabla_{\boldsymbol{x}}$ as just ∇ so that we don't clutter notation. If the universe is perfectly homogeneous, then $\rho = \bar{\rho}(t)$ and $\boldsymbol{v} = 0$ (everything just moves with the Hubble flow), so the continuity equation reduces to

$$\dot{\bar{\rho}} + 3H\bar{\rho} = 0. \tag{5.23}$$

This is exactly what we found in Chapter 2 for non-relativistic matter. As the universe expands, the volume grows like a^3, and so the density falls like a^{-3}. Our job now is to introduce perturbations. We therefore write

$$\rho(t, \boldsymbol{x}) = \bar{\rho}(t) \left[1 + \delta(t, \boldsymbol{x})\right] \tag{5.24}$$

where

$$\delta(t, \boldsymbol{x}) \equiv \frac{\rho(t, \boldsymbol{x}) - \bar{\rho}(t)}{\bar{\rho}(t)} \tag{5.25}$$

is called the **density contrast**. The reason that we use δ instead of $\delta\rho$ is that δ is dimensionless, which is nice because it tells us immediately when perturbation theory is allowed. If $|\delta| \ll 1$, then the perturbation really is small. If $|\delta| \sim 1$, then the perturbation is not small anymore and the calculation is starting to lie to us. Now let us linearize the expanding fluid equations. Starting with the continuity equation,

$$\begin{aligned}
\dot{\rho} + 3H\rho + \frac{1}{a}\nabla\cdot(\rho\boldsymbol{v}) &= 0 \\
\frac{d}{dt}\left[\bar{\rho}(1+\delta)\right] + 3H\bar{\rho}(1+\delta) + \frac{1}{a}\nabla\cdot[\bar{\rho}(1+\delta)\boldsymbol{v}] &= 0 \\
\dot{\bar{\rho}}(1+\delta) + \bar{\rho}\dot{\delta} + 3H\bar{\rho}(1+\delta) + \frac{\bar{\rho}}{a}\nabla\cdot\boldsymbol{v} + \frac{\bar{\rho}}{a}\nabla\cdot(\delta\boldsymbol{v}) &= 0 \\
\cancelto{0}{[\dot{\bar{\rho}} + 3H\bar{\rho}]}(1+\delta) + \bar{\rho}\dot{\delta} + \frac{\bar{\rho}}{a}\nabla\cdot\boldsymbol{v} + \cancelto{0}{\frac{\bar{\rho}}{a}\nabla\cdot(\delta\boldsymbol{v})} &= 0 \\
\dot{\delta} + \frac{1}{a}\nabla\cdot\boldsymbol{v} &= 0
\end{aligned} \tag{5.26}$$

In the second to last line, we killed the term $\nabla\cdot(\delta\boldsymbol{v})$ because it is the product of two small perturbations, and linear perturbations mean we only keep things at first order. The first term is also killed by the background continuity equation. We turn our heads to the Euler equation. Before we begin, it will be slightly cleaner to write the Euler equation directly in terms of the peculiar velocity $\boldsymbol{v}$, rather than the full physical velocity $\boldsymbol{u} = H\boldsymbol{r} + \boldsymbol{v}$. The result is

$$\frac{\partial\boldsymbol{v}}{\partial t} + H\boldsymbol{v} + \frac{1}{a}(\boldsymbol{v}\cdot\nabla)\boldsymbol{v} = -\frac{1}{a\rho}\nabla P - \frac{1}{a}\nabla\phi$$

where ϕ is now the perturbation to the gravitational potential. Here is where the Jeans swindle gets justified. In physical coordinates, the background potential $\bar{\Phi} = \frac{2\pi G}{3}\bar{\rho}r^2$ produces an acceleration $-\frac{4\pi G}{3}\bar{\rho}\boldsymbol{r}$, which is exactly $\frac{\ddot{a}}{a}\boldsymbol{r}$ by the second Friedmann equation. So the background potential is completely used up in decelerating the Hubble flow, and when we subtract off the Hubble flow, only the perturbation is left. It should not be counted twice. The peculiar potential ϕ is sourced only by the density perturbation

$$\nabla^2\phi = 4\pi G_N a^2\bar{\rho}\,\delta$$

This is the comoving version of Poisson's equation. The factor of a^2 appears because ∇ is now the derivative with respect to co-moving coordinates. Assuming the fluid is barotropic, meaning $P = P(\rho)$, the pressure perturbation satisfies

$$\delta P = c_s^2\delta\rho = c_s^2\bar{\rho}\,\delta$$

Keeping only terms linear in perturbations, the Euler equation becomes

$$\dot{\boldsymbol{v}} + H\boldsymbol{v} + \frac{1}{a}\cancelto{0}{(\boldsymbol{v}\cdot\nabla)\boldsymbol{v}} = -\frac{1}{a\bar{\rho}}\nabla\delta P - \frac{1}{a}\nabla\phi$$
$$\dot{\boldsymbol{v}} + H\boldsymbol{v} = -\frac{c_s^2}{a}\nabla\delta - \frac{1}{a}\nabla\phi \tag{5.27}$$

Together, the linearized equations are

$$\boxed{\begin{aligned} \dot{\delta} + \frac{1}{a}\nabla\cdot\boldsymbol{v} &= 0, \\ \dot{\boldsymbol{v}} + H\boldsymbol{v} &= -\frac{c_s^2}{a}\nabla\delta - \frac{1}{a}\nabla\phi, \\ \nabla^2\phi &= 4\pi G_N a^2\bar{\rho}\,\delta \end{aligned}} \tag{5.28}$$

This is the main Newtonian system for perturbations in an expanding universe. It is the same physics as before: conservation of mass, conservation of momentum, and gravity. The only difference is that the universe is expanding, and this expansion enters the equations through $a(t)$ and $H(t)$. Notice the $H\boldsymbol{v}$ term in the Euler equation: without any forces, peculiar velocities decay as $\boldsymbol{v} \propto 1/a$, which is the same redshifting of momentum we saw for photons. We can, however, go a step further and combine these equations into 1. If we define the velocity divergence as $\theta \equiv \nabla\cdot\boldsymbol{v}$, then the continuity equation says $\dot{\delta} = -\theta/a$. Taking the divergence of the Euler equation gives

$$\begin{aligned} \dot{\theta} + H\theta &= -\frac{c_s^2}{a}\nabla^2\delta - \frac{1}{a}\nabla^2\phi \\ &= -\frac{c_s^2}{a}\nabla^2\delta - 4\pi G_N a\bar{\rho}\,\delta \end{aligned} \tag{5.29}$$

Now take a time derivative of $\dot{\delta} = -\theta/a$:

$$\begin{aligned} \ddot{\delta} &= -\frac{\dot{\theta}}{a} + \frac{\dot{a}}{a^2}\theta \\ &= -\frac{\dot{\theta}}{a} - H\dot{\delta} \end{aligned} \tag{5.30}$$

Using the equation for $\dot{\theta}$, we get

$$\begin{aligned} \ddot{\delta} &= -\frac{1}{a}\left[-H\theta - \frac{c_s^2}{a}\nabla^2\delta - 4\pi G_N a\bar{\rho}\,\delta\right] - H\dot{\delta} \\ &= \frac{H\theta}{a} + \frac{c_s^2}{a^2}\nabla^2\delta + 4\pi G_N\bar{\rho}\,\delta - H\dot{\delta} \end{aligned}$$

$$= -H\dot{\delta} + \frac{c_s^2}{a^2}\nabla^2\delta + 4\pi G_N \bar{\rho}\,\delta - H\dot{\delta}$$

Therefore,

$$\boxed{\ddot{\delta} + 2H\dot{\delta} - \frac{c_s^2}{a^2}\nabla^2\delta - 4\pi G_N\bar{\rho}\,\delta = 0} \tag{5.31}$$

This is the expanding-universe version of the Jeans equation! The only new term is $2H\dot{\delta}$, which we call the **Hubble friction**. It is friction from the fact that the coordinates are being stretched by the expansion of the universe; peculiar velocities get diluted by expansion, so perturbations do not grow as violently as they did in the static Jeans problem. Finally, Fourier transforming gives

$$\boxed{\ddot{\delta}_{\boldsymbol{k}} + 2H\dot{\delta}_{\boldsymbol{k}} + \left(\frac{c_s^2 k^2}{a^2} - 4\pi G_N\bar{\rho}\right)\delta_{\boldsymbol{k}} = 0} \tag{5.32}$$

This equation has the same two competing forces as before. The term $c_s^2k^2/a^2$ is pressure support and the term $4\pi G_N\bar{\rho}$ is gravitational attraction. The difference is that the physical wavelength of a mode is growing with the universe,

$$\lambda_{\text{phys}} = \frac{2\pi a}{k}$$

So the question of whether a mode is above or below the Jeans scale is time dependent. This is why cosmological structure formation is not just "the Jeans instability, but bigger." The expansion quite literally changes the clock.

Problem 3.

Above the Jeans scale, but below the Hubble radius, the evolution of matter perturbations is governed by

$$\ddot{\delta}_m + 2H\dot{\delta}_m = 4\pi G\bar{\rho}_m\delta_m$$

where $\bar{\rho}_m = \bar{\rho}_{m,0}a^{-3}$. We have ignored the fluctuations in the radiation and the dark energy on the right-hand side, but still include their homogeneous densities as a source for the Hubble rate.

a). Show that the above evolution equation can be written as

$$\frac{\mathrm{d}}{\mathrm{d}a}\left(a^3H\frac{\mathrm{d}\delta_m}{\mathrm{d}a}\right) = 4\pi G\bar{\rho}_{m,0}\frac{\delta_m}{Ha^2}$$

b). Assuming that the universe contains a combination of matter, curvature and a cosmological constant (no radiation), show that the solutions to the equation in a) are

$$\delta_m \propto \begin{cases} H \\ H \int \frac{\mathrm{d}a}{(aH)^3} \end{cases}$$

What is the scaling of the growing mode solution when the expansion of the universe is dominated by matter, by curvature, or by a cosmological constant?

c). At early times, the universe was dominated by radiation and pressureless matter. You may ignore baryons. Let $y \equiv a/a_{\rm eq}$, where $a_{\rm eq}$ is the scale factor at matter-radiation equality. Show that the Hubble parameter can be written as

$$H(y) = \frac{A}{y^2}\sqrt{1+y}$$

where the constant A should be determined in terms of the energy density at equality, $\rho_{\rm eq}$. Show that the evolution equation is equivalent to

$$\frac{\mathrm{d}^2\delta_m}{\mathrm{d}y^2} + \frac{2+3y}{2y(y+1)}\frac{\mathrm{d}\delta_m}{\mathrm{d}y} = \frac{3}{2}\frac{\delta_m}{y(y+1)}$$

Verify that the solutions of this equation are

$$\delta_m \propto \begin{cases} 1+\frac{3}{2}y \\ \left(1+\frac{3}{2}y\right)\ln\left(\frac{\sqrt{1+y}+1}{\sqrt{1+y}-1}\right) - 3\sqrt{1+y} \end{cases}$$

What are the growing and decaying modes at early and late times?

Solution: a). Using $\frac{d}{dt} = aH\frac{d}{da}$, we get $\dot\delta_m = aH\delta_m'$ (here primes are d/da) and $\ddot\delta_m = aH\frac{d}{da}(aH\delta_m')$. Then

$$\ddot\delta_m + 2H\dot\delta_m = \frac{H}{a^2}\left[a^3H\delta_m'' + a^2\frac{d(aH)}{da}\delta_m' + 2a^2H\delta_m'\right] = \frac{H}{a^2}\frac{d}{da}\left(a^3H\delta_m'\right),$$

since $\frac{d}{da}(a^3H) = 3a^2H + a^3\frac{dH}{da} = a^2\frac{d(aH)}{da} + 2a^2H$. Setting this equal to $4\pi G\bar\rho_{m,0}a^{-3}\delta_m$ and dividing by H/a^2 gives the result.

b). With matter, curvature and Λ, $H^2 = H_0^2(\Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda)$. Plug in $\delta_m = H$: we need $\frac{d}{da}(a^3HH')$ with $HH' = \frac{1}{2}\frac{dH^2}{da} = -\frac{H_0^2}{2}(3\Omega_m a^{-4} + 2\Omega_k a^{-3})$, so $a^3HH' = -\frac{H_0^2}{2}(3\Omega_m a^{-1} + 2\Omega_k)$ and its derivative is $\frac{3}{2}H_0^2\Omega_m a^{-2}$. Since $4\pi G\bar\rho_{m,0} = \frac{3}{2}H_0^2\Omega_m$, the right hand side is $\frac{3}{2}H_0^2\Omega_m\frac{H}{a^2H}$, and they match. So $\delta_m \propto H$ is a solution, and since H decreases, it's the decaying mode. The second solution follows from

reduction of order: writing $\delta_m = Hu$, the equation becomes $\frac{d}{da}(a^3H^3u') = 0$, so $u' \propto (aH)^{-3}$ and $\delta_m \propto H\int \frac{da}{(aH)^3}$. In matter domination $aH \propto a^{-1/2}$ and this gives $\delta_m \propto a^{-3/2}\cdot a^{5/2} = a$. In curvature domination $aH = \text{const}$, so $\delta_m \propto a^{-1}\cdot a = \text{const}$. In Λ domination H is constant and the integral converges, so $\delta_m \to \text{const}$. Growth stops once matter stops dominating.
c). With $\rho = \rho_m + \rho_r = \frac{\rho_{\rm eq}}{2}(y^{-3}+y^{-4})$, $H^2 = \frac{8\pi G}{3}\frac{\rho_{\rm eq}}{2}\frac{1+y}{y^4}$, so $A = \sqrt{4\pi G\rho_{\rm eq}/3}$. Changing variables from t to y in the growth equation and using this H gives the stated equation (it's a bit of algebra, but nothing new). Plugging in $\delta_m = 1+\frac{3}{2}y$: the left hand side is $0 + \frac{2+3y}{2y(1+y)}\cdot\frac{3}{2}$ and the right hand side is $\frac{3}{2}\frac{1+\frac{3}{2}y}{y(1+y)} = \frac{3(2+3y)}{4y(1+y)}$, which match. At early times ($y \ll 1$), the first solution is $\delta_m \approx 1$ (frozen) and the second goes like $\ln(4/y) - 3$, so the general solution is a constant plus a logarithm, $\delta_m \approx A + B\ln y$. A perturbation that enters the horizon with some velocity can therefore grow at most logarithmically. At late times ($y \gg 1$) the first solution grows like $y \propto a$ and the second decays like $y^{-3/2}$.

5.1.4 Growth Of Matter Perturbations

Most of the matter in the universe is cold dark matter, so let's focus on it. Cold dark matter has essentially no pressure, which means $c_s^2 \simeq 0$, and the Jeans length is zero. The evolution equation becomes

$$\boxed{\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_N\bar{\rho}_m\delta_m = 0} \tag{5.33}$$

This equation is one of the most important equations in structure formation. It says that matter perturbations grow because of gravity, but the growth is slowed by expansion. During matter domination, the scale factor behaves like

$$a(t) \propto t^{2/3} \qquad H = \frac{2}{3t}$$

The Friedmann equation also gives

$$4\pi G_N\bar{\rho}_m = \frac{2}{3t^2}$$

Therefore the growth equation becomes

$$\ddot{\delta}_m + \frac{4}{3t}\dot{\delta}_m - \frac{2}{3t^2}\delta_m = 0 \tag{5.34}$$

Since this equation is a power-law equation, we guess

$$\delta_m(t) = t^p$$

Plugging this in gives

$$\begin{aligned} p(p-1)t^{p-2} + \frac{4}{3t}pt^{p-1} - \frac{2}{3t^2}t^p &= 0 \\ p(p-1) + \frac{4}{3}p - \frac{2}{3} &= 0 \\ 3p^2 + p - 2 &= 0 \\ (3p-2)(p+1) &= 0 \end{aligned} \tag{5.35}$$

So the two solutions are

$$\boxed{\delta_m(t) = C_1 t^{2/3} + C_2 t^{-1}} \tag{5.36}$$

Because $a(t) \propto t^{2/3}$, this can be written as

$$\boxed{\delta_m(a) = C_1 a + C_2 a^{-3/2}} \tag{5.37}$$

The first term is the **growing mode** (basically governs the growth of perturbations); the second term is the **decaying mode**. Note here that structure formation in a matter-dominated universe is not exponential; It is power-law growth! The universe is trying to gravitationally collapse, but the expansion slows it down. This result also tells us *why* dark matter is so important. Baryons before recombination are tightly coupled to photons and photon pressure prevents them from clustering efficiently. Since dark matter is...dark (duh), it does not care about any interactions with photons, so dark matter can begin falling into gravitational wells earlier. Then, after recombination, baryons fall into the gravitational wells already created by the dark matter. This is why dark matter is not just extra mass in the universe, it is a background. A gravitational scaffold, if you will.

In the previous calculation, we assumed that we were in matter domination. However, the universe was radiation dominated before equality, and it is now becoming dark energy dominated. Since the growth equation doesn't depend on k (for pressureless matter on subhorizon scales), every mode grows by the same factor, and it is useful to define the **growth function** $D_+(a)$ by writing

$$\delta_m(a, \boldsymbol{k}) = D_+(a)\delta_m(a_i, \boldsymbol{k}) \tag{5.38}$$

For pressureless matter on subhorizon scales, the growth equation is

$$\ddot{\delta}_m + 2H\dot{\delta}_m - \frac{3}{2}H^2\Omega_m(a)\delta_m = 0 \tag{5.39}$$

where we used

$$4\pi G_N \bar{\rho}_m = \frac{3}{2}H^2\Omega_m(a)$$

This form is useful because it separates the perturbation physics from the background cosmology. If the universe contains lots of matter, $\Omega_m(a)$ is large and perturbations grow. If the universe is dominated by radiation or dark energy, matter perturbations grow more slowly. We can rewrite the equation using a rather than t via

$$\frac{d}{dt} = \dot{a}\frac{d}{da} = aH\frac{d}{da},$$

we find

$$\begin{aligned}
\dot{\delta}_m &= aH\frac{d\delta_m}{da}, \\
\ddot{\delta}_m &= aH\frac{d}{da}\left(aH\frac{d\delta_m}{da}\right) \\
&= aH\left[\frac{d(aH)}{da}\frac{d\delta_m}{da} + aH\frac{d^2\delta_m}{da^2}\right]
\end{aligned} \tag{5.40}$$

Substituting into the growth equation gives

$$\begin{aligned}
aH\left[\frac{d(aH)}{da}\frac{d\delta_m}{da} + aH\frac{d^2\delta_m}{da^2}\right] + 2H(aH)\frac{d\delta_m}{da} - 4\pi G_N\bar{\rho}_m\delta_m = 0 \\
a^2H^2\frac{d^2\delta_m}{da^2} + aH\frac{d(aH)}{da}\frac{d\delta_m}{da} + 2aH^2\frac{d\delta_m}{da} - 4\pi G_N\bar{\rho}_m\delta_m = 0.
\end{aligned} \tag{5.41}$$

There is an especially nice way to package the same equation (this is part a of Problem 3)

$$\boxed{\frac{d}{da}\left(a^3H\frac{d\delta_m}{da}\right) = \frac{4\pi G_N\bar{\rho}_{m,0}}{a^2H}\delta_m.} \tag{5.42}$$

This form is useful because it makes clear that changing the expansion history changes the growth of structure. Dark energy, curvature, and radiation all affect perturbations indirectly by changing $H(a)$.

5.1.5 Growth In Our Universe

Problem 3 hands us two very useful results. The first is about the radiation era. Before equality, a subhorizon dark matter perturbation feels its own gravity, but the universe is

expanding so fast (because of the radiation) that the Hubble friction wins, and from part c), the growing mode just sits at a constant plus a logarithm. Dark matter perturbations essentially freeze during radiation domination, and only start growing like a after equality (figure 5.2, left). This is called the **Mészáros effect**, and it will be the reason the matter power spectrum has a peak.

The second result is about the late universe. For matter plus a cosmological constant, part b) gives the growing mode in closed form,

$$\boxed{D_+(a) = \frac{5\Omega_m}{2}\frac{H(a)}{H_0}\int_0^a \frac{da'}{(a'H(a')/H_0)^3}} \tag{5.43}$$

normalized so that $D_+ = a$ deep in matter domination. Once dark energy takes over, H becomes constant, the integral converges, and growth stops (figure 5.2, right). With $\Omega_m = 0.315$, structures today are about 21% less developed than they would be in a matter-only universe, $D_+(1) \approx 0.79$. A handy approximation to the growth rate is

$$f \equiv \frac{d\ln D_+}{d\ln a} \approx \Omega_m(a)^{0.55}, \tag{5.44}$$

which galaxy surveys measure through the peculiar velocities of galaxies. It's one of the best ways we have of testing whether dark energy is really a cosmological constant and whether gravity is really GR on cosmological scales.

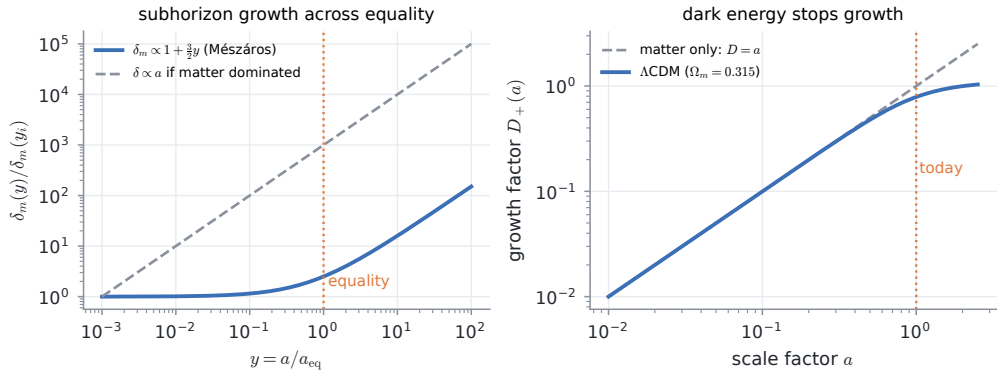


Figure 5.2: Left: the growing mode of subhorizon dark matter perturbations through matter-radiation equality. Perturbations barely grow during radiation domination and grow like a afterwards. Right: the growth factor in ΛCDM compared to a matter-only universe. Dark energy shuts off growth at late times.

5.1.6 The Transfer Function

The primordial universe gives us initial perturbations, but the late universe does not see those perturbations untouched. Different modes enter the horizon at different times. A small-scale mode has a large k, so it enters the horizon early. A large-scale mode has a small k, so it enters later. This matters because perturbations behave differently outside and inside the horizon. During radiation domination, subhorizon matter perturbations grow only very slowly (the Mészáros effect). The reason is not that gravity turns off. The reason is that the expansion rate is large and radiation dominates the gravitational bookkeeping of the universe. Matter is present, but it is not in charge yet. Once matter domination begins, the matter perturbations grow efficiently

$$\delta_m \sim \ln a \quad \text{during radiation domination} \qquad \delta_m \sim a \quad \text{during matter domination}$$

Outside the horizon, on the other hand, perturbations grow at the same rate whatever the universe is made of (we'll see this in the next chapter). This means that modes which entered the horizon early, during radiation domination, had their growth suppressed for a long time, while modes that entered after equality did not suffer at all. The dividing line is the mode that entered right at equality, $k_{\rm eq} = a_{\rm eq}H_{\rm eq} \approx 0.0104\ {\rm Mpc}^{-1}$ (or $0.015\,h\,{\rm Mpc}^{-1}$). A mode with $k \gg k_{\rm eq}$ entered when the scale factor was $a_k \approx a_{\rm eq}k_{\rm eq}/k$ (since $aH \propto a^{-1}$ in radiation domination), and it missed out on a factor of about $a_{\rm eq}/a_k = k/k_{\rm eq}$ of growth, up to the logarithm. We encode this scale-dependent processing by the **transfer function** $T(k)$

$$\boxed{\delta_m(a,k) = D_+(a)T(k)\delta_{\rm prim}(k)} \tag{5.45}$$

Equivalently, the matter power spectrum can be written as

$$\boxed{P_m(k,a) = D_+^2(a)T^2(k)P_{\rm prim}(k).} \tag{5.46}$$

The transfer function is basically the universe's memory. It remembers when each mode entered the horizon and what the universe was made of at that time. From the argument above, $T(k) \approx 1$ for $k \ll k_{\rm eq}$ and $T(k) \propto (k_{\rm eq}/k)^2 \ln(k/k_{\rm eq})$ for $k \gg k_{\rm eq}$ (the square is because the density is related to the potential by Poisson's equation, which brings in a k^2; we'll see this properly next chapter). A popular fitting formula, due to Bardeen, Bond, Kaiser and Szalay (BBKS), is

$$T(q) = \frac{\ln(1+2.34q)}{2.34q}\left[1 + 3.89q + (16.1q)^2 + (5.46q)^3 + (6.71q)^4\right]^{-1/4}, \tag{5.47}$$

with $q = k/(\Omega_m h^2\,\mathrm{Mpc}^{-1})$. Inflation gives $\Delta^2_{\mathcal{R}} \propto k^{n_s-1}$, so $P_{\mathcal{R}} \propto k^{n_s-4}$, and Poisson's equation ($\delta \propto k^2\Phi$ on subhorizon scales) multiplies the power by k^4, so the primordial density spectrum is $P_{\mathrm{prim}} \propto k^{n_s}$ (we'll see this properly next chapter). That means $P_m \propto k^{n_s}$ on large scales and $P_m \propto k^{n_s-4}\ln^2 k$ on small scales, with a turnover near k_{eq} (figure 5.3). Measuring the location of the peak measures $\Omega_m h^2$, which is one of the ways we know that most of the matter is dark.

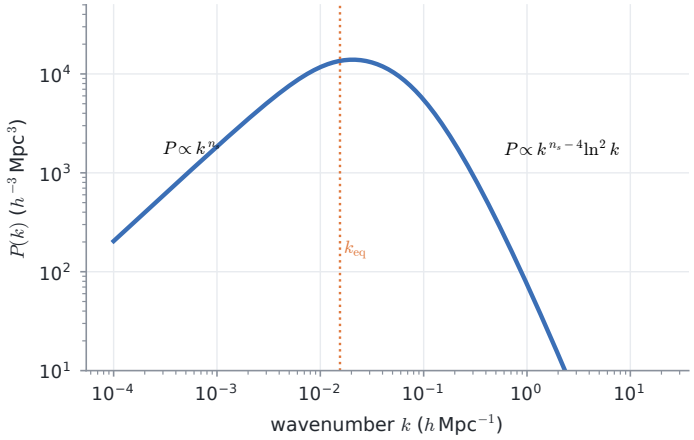


Figure 5.3: The linear matter power spectrum today from the BBKS transfer function, with $n_s = 0.965$ and normalized to $\sigma_8 = 0.81$. The turnover near k_{eq} is the Mészáros effect: modes that entered the horizon during radiation domination grew less.

5.1.7 Statistical Description Of Structure

We have been using the words "power spectrum" a lot, so let's define it. So far, we have talked about one perturbation mode $\delta_{\boldsymbol{k}}$. But the real universe has many modes, and we do not know their individual phases. Instead of predicting the exact value of $\delta(\boldsymbol{x})$ at every point, cosmology predicts statistics. This is the same attitude we take in statistical mechanics, where instead of tracking every molecule in a gas, we compute averages. The first useful statistic is the two-point correlation function

$$\boxed{\xi(r) \equiv \langle \delta(\boldsymbol{x})\delta(\boldsymbol{x}+\boldsymbol{r})\rangle\,.} \tag{5.48}$$

The brackets mean an ensemble average, the average over many universes drawn from the same random process. Since we only have one universe, this phrase is slightly sus. In practice, we use spatial averages over a large enough volume, assuming that the universe is statistically homogeneous and isotropic. In Fourier space, the same information is contained

in the power spectrum

$$\boxed{\langle \delta(\boldsymbol{k})\delta^*(\boldsymbol{k}')\rangle = (2\pi)^3\delta_{\rm D}(\boldsymbol{k}-\boldsymbol{k}')P(k)} \tag{5.49}$$

The Dirac delta function appears because translational invariance says that different Fourier modes are uncorrelated unless they have the same wavevector. Isotropy then says that P can only depend on $k = |\boldsymbol{k}|$. It is also common to define the dimensionless power spectrum

$$\boxed{\Delta^2(k) \equiv \frac{k^3}{2\pi^2}P(k)} \tag{5.50}$$

This quantity has a nice interpretation: $\Delta^2(k)$ measures the contribution to the variance of δ per logarithmic interval in k. More explicitly,

$$\begin{aligned}\big\langle \delta^2(\boldsymbol{x})\big\rangle &= \int \frac{d^3k}{(2\pi)^3}P(k) \\ &= \int_0^\infty \frac{4\pi k^2 dk}{(2\pi)^3}P(k) \\ &= \int_0^\infty d\ln k \frac{k^3}{2\pi^2}P(k) \\ &= \int_0^\infty d\ln k\, \Delta^2(k)\end{aligned} \tag{5.51}$$

This is why plots of the power spectrum often use $\Delta^2(k)$ rather than $P(k)$ itself. A particularly important class of perturbations are **Gaussian random fields**. A Gaussian random field is a random field whose statistics are completely determined by its two-point function. This is a miracle of Gaussian distributions. Once you know the mean and variance of a Gaussian, you know everything (peep my statistical field theory notes). In cosmology, this means that if primordial perturbations are Gaussian, then the power spectrum contains all the information. The simplest models of inflation predict Gaussian perturbations (they're the ground state fluctuations of a nearly free field, and the ground state of a harmonic oscillator is a Gaussian), and any deviation from Gaussianity would be a sign of more complicated early-universe physics.

In practice, the variance $\langle\delta^2\rangle$ is dominated by small scales, where linear theory breaks down, so we usually smooth the field first. The variance of δ smoothed with a spherical top-hat of radius R is

$$\sigma_R^2 = \int d\ln k\, \Delta^2(k)\, W^2(kR), \qquad W(x) = \frac{3(\sin x - x\cos x)}{x^3}, \tag{5.52}$$

where W is the Fourier transform of the top-hat. The amplitude of the matter power

spectrum is traditionally quoted as σ_8, the value of σ_R at $R = 8\,h^{-1}$Mpc in linear theory today, and it's measured to be $\sigma_8 \approx 0.81$. Since this is close to 1, it tells us that the universe is just becoming non-linear on the scales of galaxy clusters today.

5.2 Beyond Linear Theory: Spherical Collapse

Linear theory says δ grows like a forever, but once $\delta \sim 1$ it stops being a perturbation. To see what happens next, the simplest model is a spherical overdense region in a matter dominated universe. By Newton's shell theorem (Problem 1 of Chapter 1!), the sphere evolves as if the rest of the universe didn't exist, and each mass shell of radius R enclosing mass M obeys

$$\ddot{R} = -\frac{GM}{R^2}. \tag{5.53}$$

This is the same equation as a closed Friedmann universe, so the solution is the cycloid

$$R = A(1-\cos\theta), \qquad t = B(\theta - \sin\theta), \qquad A^3 = GMB^2. \tag{5.54}$$

The shell expands with the universe at first, slows down, turns around at $\theta = \pi$, and collapses to $R = 0$ at $\theta = 2\pi$. Comparing its density to the background $\bar{\rho} = 1/(6\pi G t^2)$ gives the exact nonlinear density contrast, and expanding it for small θ gives the linear one. The results (Problem 5) are that at turnaround, the true density contrast is $1 + \delta = 9\pi^2/16 \approx 5.55$ while linear theory says $\delta_{\rm lin} = 1.06$, and at collapse the true density is infinite while linear theory says

$$\delta_{\rm lin}(t_{\rm coll}) = \frac{3}{20}(12\pi)^{2/3} \approx 1.686. \tag{5.55}$$

In reality the collapse doesn't go to a point. The shells cross, and the matter settles into a **virialized** halo with about half the turnaround radius, which works out to a density about $18\pi^2 \approx 178$ times the background. The beautiful thing about the number 1.686 is that it gives a simple rule: *a region collapses into a halo when its linearly extrapolated density contrast reaches* $\delta_c \approx 1.686$. Combined with the Gaussian statistics of σ_R, this rule is the basis of the Press-Schechter theory of how many halos of each mass exist at each time, which is how we connect the power spectrum to counts of galaxy clusters.

5.3 Problems

1. A Polytropic Fluid

Consider a universe filled with a non-relativistic fluid with $P = \alpha\rho^{4/3}$, where $P \ll \rho$, so that the background evolution is well-approximated as matter dominated. Inside the Hubble

radius, a Fourier mode of the density contrast $\delta \equiv \delta\rho/\rho$ evolves as

$$\delta'' + \mathcal{H}\delta' + \left(c_s^2 k^2 - \frac{3}{2}\mathcal{H}^2\right)\delta = 0,$$

where $c_s^2 \equiv \partial P/\partial\rho$, primes denote derivatives with respect to conformal time η and $\mathcal{H} \equiv a'/a$.

a). Show that $c_s^2 = c_0^2\,(\eta_0/\eta)^2$, where c_0 and η_0 are constants.

b). Describe qualitatively how you expect the solution for the density contrast to behave on small and large scales.

c). Find power law solutions of the form $\delta \propto \eta^\beta$, where β depends on the wavenumber k. At what value of k does the solution change from monotonic to oscillatory? For what k is the growing mode a constant?

2. The Jeans Mass Of Baryons

a). Before recombination, the baryons are tightly coupled to photons, and the sound speed is $c_s \approx c/\sqrt{3}$. Show that the Jeans length is then comparable to the Hubble radius, so baryon perturbations can't grow inside the horizon.

b). After recombination, the baryons are a gas of hydrogen at temperature $T \approx T_{\rm CMB}$ (about 3000 K at $z = 1100$). Compute the Jeans mass $M_J = \frac{4\pi}{3}\bar{\rho}_m\left(\frac{\lambda_J}{2}\right)^3$ and compare it to the mass of a globular cluster ($\sim 10^5$–$10^6 M_\odot$).

3. Velocities From Densities

Using the linearized continuity equation, show that in linear theory the peculiar velocity field is

$$\boldsymbol{v}_{\boldsymbol{k}} = i\frac{aHf}{k^2}\boldsymbol{k}\,\delta_{\boldsymbol{k}},$$

where $f = d\ln D_+/d\ln a$. What is the typical peculiar velocity of a galaxy today on a scale where $\delta \sim 1$ and $k \sim 0.1\,h\,{\rm Mpc}^{-1}$?

4. Growth With Massive Neutrinos

Suppose a fraction f_ν of the matter is in neutrinos, which free-stream and don't cluster on small scales. On those scales, only $1 - f_\nu$ of the matter contributes to the source term, $\ddot{\delta} + 2H\dot{\delta} = 4\pi G\bar{\rho}_m(1 - f_\nu)\delta$. Show that in matter domination $\delta \propto a^p$ with $p \approx 1 - \frac{3}{5}f_\nu$. Over the roughly $\ln(a_0/a_{\rm eq}) \approx 8$ e-folds since equality, by how much is the small-scale power spectrum suppressed?

5. Spherical Collapse

Using the cycloid solution, show that the density contrast of the spherical region is

$$1 + \delta = \frac{9}{2} \frac{(\theta - \sin\theta)^2}{(1 - \cos\theta)^3},$$

expand for small θ to find $\delta_{\rm lin} = \frac{3}{20} [6(\theta - \sin\theta)]^{2/3}$, and verify the numbers 5.55, 1.06 and 1.686 quoted in the text.

6

Relativistic Perturbation Theory

"Spacetime tells matter how to move; matter tells spacetime how to curve."

– John Archibald Wheeler

In the previous chapter, we treated the universe like a Newtonian fluid. This is not a bad thing. In fact, it is a very good thing. If we are interested in dark matter perturbations well inside the Hubble radius, where velocities are small and gravity is weak, then Newtonian perturbation theory is basically the right tool. But there are several places where that story is not enough. The most important one is philosophical: in general relativity, gravity is not a force that lives on top of spacetime. Gravity *is* the geometry of spacetime. So if matter is perturbed, the metric must also be perturbed.

This is the purpose of relativistic perturbation theory. We are no longer just perturbing the density, pressure, and velocity of a fluid. We are perturbing spacetime itself. This sounds terrifying, but the idea is very close to what we have already done. Previously, we wrote

$$\rho(t, \boldsymbol{x}) = \bar{\rho}(t) + \delta\rho(t, \boldsymbol{x}).$$

Now we do the same thing to the metric,

$$g_{\mu\nu}(\eta, \boldsymbol{x}) = \bar{g}_{\mu\nu}(\eta) + \delta g_{\mu\nu}(\eta, \boldsymbol{x}).$$

The background metric $\bar{g}_{\mu\nu}$ is the FRW metric. The perturbation $\delta g_{\mu\nu}$ is the small ripple on top of the expanding universe. The game is then straightforward in principle, but

annoying in practice: expand the Einstein equations to first order in the perturbations. The zeroth-order equations give the Friedmann equations, and the first-order equations tell us how density perturbations, velocity perturbations, pressure perturbations, and metric perturbations evolve together.

6.1 Why Newtonian Perturbation Theory Is Not Enough

Let us start by being honest about what the Newtonian calculation did correctly. It gave us a clean equation for the growth of matter perturbations,

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G_N \bar{\rho}\delta = 0, \tag{6.1}$$

where $\delta \equiv \delta\rho/\bar{\rho}$. This equation is extremely useful. It explains why cold dark matter perturbations grow during matter domination. It also explains why the expansion of the universe acts like friction.

But this equation hides many things. First, it assumes that we are inside the Hubble radius. In other words, the physical wavelength of the perturbation must be smaller than the Hubble scale. In Fourier space this means, roughly,

$$k \gg \mathcal{H},$$

where $\mathcal{H} \equiv a'/a$ is the conformal Hubble parameter. Second, it assumes that the matter is non-relativistic. This immediately fails for photons and relativistic neutrinos. Third, it treats the gravitational potential as a Newtonian field satisfying a Poisson equation. But in GR, the gravitational potential is only part of the metric. It is not an independent field living on top of spacetime.

So here is the real list of reasons we need relativistic perturbation theory:

(1) We need to understand perturbations whose wavelengths are comparable to, or larger than, the Hubble radius.
(2) We need to include relativistic species, especially photons and neutrinos.
(3) We need to understand the metric perturbations themselves.
(4) We need to be careful about coordinates, because a fake perturbation can be created just by choosing a bad slicing of spacetime.
(5) We need a language that connects inflationary initial conditions to the CMB and large-scale structure.

The last point is the most important one. Inflation does not directly give us galaxies. It gives us primordial curvature perturbations, on scales that were far outside the Hubble radius at the end of inflation (figure 4.1). Newtonian mechanics has no concept of the horizon, and so it breaks down for modes with $k \lesssim \mathcal{H}$. Relativistic perturbation theory is the bridge between those primordial curvature perturbations and the messy late-time universe.

The fourth point deserves one more sentence now, because it's the thing that makes this subject feel slippery at first. To say "the energy density at position $\boldsymbol{x}$ is perturbed by an amount $\delta\rho$ relative to the background" requires specifying which point of the smooth background universe we are comparing to, and in general relativity there is no preferred way to do that. That choice is called a choice of **gauge**. In one coordinate system, the surfaces of constant time might coincide with the surfaces of constant density, so $\delta\rho = 0$. In another, the constant-time surfaces are tilted, giving $\delta\rho \neq 0$, even for *exactly the same physical spacetime*. We will have to learn to tell the physical perturbations apart from the coordinate artifacts.

6.2 Conformal Time And The Background Metric

It will be convenient to work in conformal time η, defined by

$$d\eta = \frac{dt}{a(t)}. \tag{6.2}$$

The reason conformal time is nice is that light rays in a flat FRW universe look like they are moving in ordinary Minkowski space, except for an overall factor of $a^2(\eta)$. For a spatially flat universe, the background metric is

$$d\bar{s}^2 = a^2(\eta)\left[-d\eta^2 + \delta_{ij}dx^i dx^j\right]. \tag{6.3}$$

We will use a prime to denote derivatives with respect to conformal time,

$$f' \equiv \frac{df}{d\eta},$$

and define

$$\mathcal{H} \equiv \frac{a'}{a} = aH. \tag{6.4}$$

This is not a new expansion rate. It is just the Hubble parameter written in conformal time, the same $\mathcal{H}$ that we met at the end of Chapter 2. In radiation domination $a \propto \eta$ and $\mathcal{H} = 1/\eta$, and in matter domination $a \propto \eta^2$ and $\mathcal{H} = 2/\eta$.

Converting the Friedmann equations to conformal time (using $\dot{a} = a'/a$ and $\ddot{a} = (a''a -$

$a'^2)/a^3$, as we did in Chapter 2), the background Einstein equations become

$$\boxed{\mathcal{H}^2 = \frac{8\pi G_N}{3} a^2 \bar{\rho}} \tag{6.5}$$

for a spatially flat universe, and

$$\boxed{\mathcal{H}' - \mathcal{H}^2 = -4\pi G_N a^2 (\bar{\rho} + \bar{P}).} \tag{6.6}$$

The background continuity equation is

$$\boxed{\bar{\rho}' + 3\mathcal{H}(\bar{\rho} + \bar{P}) = 0.} \tag{6.7}$$

These background equations will be used constantly. In perturbation theory, we are always subtracting off the background equations so that only the first-order physics remains.

6.3 Perturbing The Metric

The full metric is

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}.$$

The perturbation $\delta g_{\mu\nu}$ has ten independent components because the metric is symmetric. But not all ten components describe the same kind of physics. Under spatial rotations, they can be decomposed into scalar, vector, and tensor perturbations. This decomposition is extremely useful because at linear order these three sectors do not mix.

The scalar perturbations are the ones that couple to density perturbations. These are the perturbations that eventually become galaxies, clusters, and the temperature anisotropies of the CMB. The vector perturbations are like rotational velocity perturbations. They usually decay in an expanding universe and are often ignored. The tensor perturbations are gravitational waves.

In general, we can parametrize the ten components of $\delta g_{\mu\nu}$ as

$$\begin{aligned}
\delta g_{00} &= -2a^2 A \\
\delta g_{0i} &= a^2 \left(\partial_i B + S_i\right) \\
\delta g_{ij} &= a^2 \left[-2\psi\, \delta_{ij} + 2\partial_i\partial_j E + \partial_i F_j + \partial_j F_i + h_{ij}\right]
\end{aligned} \tag{6.8}$$

where all functions depend on $(\eta, \boldsymbol{x})$. The fields A, B, ψ, E are *scalars* under spatial rotations; the vectors S_i, F_i satisfy $\partial^i S_i = \partial^i F_i = 0$ (transverse vectors); and the tensor h_{ij} satisfies

$h^i{}_i = 0$ (traceless) and $\partial^i h_{ij} = 0$ (transverse). Now let us account for all ten components. The four scalar functions contribute four degrees of freedom. The transverse vectors S_i and F_i each have two independent components (three minus one divergence-free constraint), giving $2 + 2 = 4$. The symmetric transverse traceless tensor h_{ij} has $6 - 3 - 1 = 2$. Total: $4 + 4 + 2 = 10$, which matches. This is the **scalar-vector-tensor** (SVT) decomposition. The theorem underlying it is that at *linear* order, the three sectors decouple, because of rotational symmetry: in Fourier space, each mode with wavevector $\boldsymbol{k}$ can be classified by how it behaves under rotations about $\boldsymbol{k}$ (helicity 0, ± 1 and ± 2), and a linear equation can't mix different helicities.

Since we mostly care about density perturbations, we will start with scalar perturbations. A general scalar perturbation of the metric can be written as

$$ds^2 = a^2(\eta)\left[-(1+2A)d\eta^2 + 2\partial_i B\, d\eta dx^i + ((1-2\psi)\delta_{ij} + 2\partial_i\partial_j E)\, dx^i dx^j\right]. \tag{6.9}$$

There are four scalar functions here: A, B, ψ, and E. But this does not mean that there are four physical scalar degrees of freedom in the metric. Some of these are just coordinate artifacts. This is the first annoying lesson of relativistic perturbation theory: in GR, not every perturbation you write down is physical. For later use, the inverse metric to first order is

$$g^{00} = -a^{-2}(1-2A), \qquad g^{0i} = a^{-2}\partial^i B, \qquad g^{ij} = a^{-2}\left[(1+2\psi)\delta^{ij} - 2\partial^i\partial^j E\right], \tag{6.10}$$

which you can check satisfies $g^{\mu\rho}g_{\rho\nu} = \delta^\mu{}_\nu$ up to second order terms.

Problem 1.

A symmetric spatial tensor h_{ij} has six independent components. Show that it can be decomposed into scalar, vector, and tensor pieces as

$$h_{ij} = -2\psi\delta_{ij} + 2\partial_i\partial_j E + \partial_i F_j + \partial_j F_i + h^{\mathrm{TT}}_{ij},$$

where $\partial_i F_i = 0$, $\partial_i h^{\mathrm{TT}}_{ij} = 0$, and $h^{\mathrm{TT\,i}}{}_i = 0$. Count the number of independent components in each part.

Solution:

A symmetric 3×3 matrix has

$$\frac{3(3+1)}{2} = 6$$

independent components. The scalar part contains two scalar functions, ψ and E, so it contributes 2 degrees of freedom. The vector F_i starts with three components, but the constraint $\partial_i F_i = 0$ removes one, leaving 2 degrees of freedom. Finally, the transverse-traceless tensor $h_{ij}^{\rm TT}$ starts with six components. The transversality condition removes three and the trace condition removes one, leaving 2 degrees of freedom. Hence,

$$6 = 2_{\rm scalar} + 2_{\rm vector} + 2_{\rm tensor}.$$

The important physics is that the two tensor degrees of freedom are the two polarizations of gravitational waves.

6.4 Gauge Freedom

Before deriving equations, we need to pause and discuss a conceptual disease that infects all of relativistic perturbation theory: **gauge freedom**. In Newtonian physics, a density perturbation means what you think it means. There is a background density and then there is a small local excess. In GR, the statement is more subtle because choosing a perturbation means choosing how to identify points in the perturbed universe with points in the background universe.

Here is the problem in its simplest form. Suppose I shift the time coordinate by a tiny amount,

$$\eta \to \tilde{\eta} = \eta + T(\eta, \boldsymbol{x}).$$

Even if the background density is perfectly homogeneous, the density evaluated at the shifted time will change:

$$\begin{aligned} \bar{\rho}(\eta) &\to \bar{\rho}(\tilde{\eta} - T) \\ &= \bar{\rho}(\tilde{\eta}) - \bar{\rho}' T. \end{aligned} \tag{6.11}$$

This means the density perturbation transforms as

$$\widetilde{\delta\rho} = \delta\rho - \bar{\rho}' T. \tag{6.12}$$

So a density perturbation can be changed just by changing the time slicing. This is not a small technicality. On superhorizon scales, where gradients are weak, gauge artifacts can look like physical perturbations if we are not careful.

In general, a gauge transformation is a small change of coordinates $x^\mu \to \tilde{x}^\mu = x^\mu + \xi^\mu$.

For the scalar sector, we write $\xi^0 = T$ and $\xi^i = \partial^i L$ for two scalar functions. Every scalar quantity f (like ρ or P) transforms the way ρ did above, $\widetilde{\delta f} = \delta f - \bar{f}' T$. The metric is a tensor, so it transforms by the tensor transformation law (2.25), and expanding to first order in ξ one finds that its perturbation shifts by minus the **Lie derivative** of the background metric along ξ, $\delta\tilde{g}_{\mu\nu} = \delta g_{\mu\nu} - \left(\bar{g}_{\mu\lambda}\partial_\nu\xi^\lambda + \bar{g}_{\lambda\nu}\partial_\mu\xi^\lambda + \xi^\lambda\partial_\lambda\bar{g}_{\mu\nu}\right)$. Working out each component (Problem 3), the scalar metric perturbations transform as

$$\tilde{A} = A - T' - \mathcal{H}T, \qquad \tilde{B} = B + T - L', \qquad \tilde{\psi} = \psi + \mathcal{H}T, \qquad \tilde{E} = E - L, \tag{6.13}$$

and the velocity potential, defined by $v^i = dx^i/d\eta = \partial^i v$, transforms as $\tilde{v} = v + L'$.

There are two ways to deal with this. The first is to fix the gauge, which means using the freedom in T and L to set two of the four metric functions to zero. The second is to build combinations of variables that do not change under small coordinate transformations. These are called **gauge-invariant variables**. Two famous ones, found by Bardeen in 1980, are the **Bardeen potentials**

$$\Psi_B \equiv A + \mathcal{H}(B - E') + (B - E')', \qquad \Phi_B \equiv \psi - \mathcal{H}(B - E'). \tag{6.14}$$

To check that they're invariant, notice that $\sigma \equiv B - E'$ transforms as $\tilde{\sigma} = \sigma + T$ (the L' cancels). Then $\tilde{\Phi}_B = \psi + \mathcal{H}T - \mathcal{H}(\sigma + T) = \Phi_B$, and similarly for Ψ_B. They represent genuine physical content, independent of any coordinate choice.

Problem 2.

Suppose the background density is $\bar{\rho}(\eta)$ and we make a small time shift

$$\eta \to \tilde{\eta} = \eta + T(\eta, \boldsymbol{x}).$$

Show that the density perturbation transforms as

$$\widetilde{\delta\rho} = \delta\rho - \bar{\rho}' T.$$

Then explain why this means $\delta\rho$ alone is not a gauge-invariant object.

Solution:

The total density is a scalar, so it does not care what coordinates we use. What changes is how we split the total density into a background piece and a perturbation.

We write

$$\rho(\eta, \boldsymbol{x}) = \bar{\rho}(\eta) + \delta\rho(\eta, \boldsymbol{x}),$$
$$\rho(\tilde{\eta}, \boldsymbol{x}) = \bar{\rho}(\tilde{\eta}) + \widetilde{\delta\rho}(\tilde{\eta}, \boldsymbol{x}).$$

Since $\eta = \tilde{\eta} - T$, the old background becomes

$$\begin{aligned}\bar{\rho}(\eta) &= \bar{\rho}(\tilde{\eta} - T) \\ &= \bar{\rho}(\tilde{\eta}) - \bar{\rho}' T.\end{aligned}$$

Keeping only first-order terms, the total density is therefore

$$\rho = \bar{\rho}(\tilde{\eta}) - \bar{\rho}' T + \delta\rho.$$

Comparing with $\rho = \bar{\rho}(\tilde{\eta}) + \widetilde{\delta\rho}$ gives

$$\boxed{\widetilde{\delta\rho} = \delta\rho - \bar{\rho}' T.}$$

So $\delta\rho$ changes under a coordinate transformation. This is why we either fix the gauge or use gauge-invariant combinations.

Problem 3.

Derive the transformation rules (6.13) by computing $\bar{g}_{\mu\lambda}\partial_\nu\xi^\lambda + \bar{g}_{\lambda\nu}\partial_\mu\xi^\lambda + \xi^\lambda\partial_\lambda\bar{g}_{\mu\nu}$ for $\bar{g}_{\mu\nu} = a^2\eta_{\mu\nu}$ and $\xi^\mu = (T, \partial^i L)$.

Solution: The only coordinate the background metric depends on is η, so $\xi^\lambda\partial_\lambda\bar{g}_{\mu\nu} = T\partial_\eta(a^2)\eta_{\mu\nu} = 2a^2\mathcal{H}T\eta_{\mu\nu}$. Now the components.

00: $\bar{g}_{00}\partial_0\xi^0 + \bar{g}_{00}\partial_0\xi^0 + 2a^2\mathcal{H}T\eta_{00} = -2a^2(T' + \mathcal{H}T)$. Since $\delta g_{00} = -2a^2 A$, subtracting gives $-2a^2\tilde{A} = -2a^2 A + 2a^2(T' + \mathcal{H}T)$, so $\tilde{A} = A - T' - \mathcal{H}T$.

$0i$: $\bar{g}_{00}\partial_i\xi^0 + \bar{g}_{ij}\partial_0\xi^j = -a^2\partial_i T + a^2\partial_i L'$. Since $\delta g_{0i} = a^2\partial_i B$, $\tilde{B} = B + T - L'$.

ij: $\bar{g}_{ik}\partial_j\xi^k + \bar{g}_{kj}\partial_i\xi^k + 2a^2\mathcal{H}T\delta_{ij} = 2a^2\partial_i\partial_j L + 2a^2\mathcal{H}T\delta_{ij}$. Comparing with $\delta g_{ij} = a^2(-2\psi\delta_{ij} + 2\partial_i\partial_j E)$, the δ_{ij} part gives $-2\tilde{\psi} = -2\psi - 2\mathcal{H}T$ and the $\partial_i\partial_j$ part gives $\tilde{E} = E - L$.

6.5 Newtonian Gauge

We will not use the fully general scalar metric for most calculations. Instead, we will make a convenient choice of coordinates called **Newtonian gauge** (also called the longitudinal gauge). In this gauge, we set

$$B = 0, \qquad E = 0.$$

These two conditions fix the gauge completely: from (6.13), we need $L = E$ and then $T = L' - B = E' - B$, so given any starting coordinates there is exactly one choice of (T, L) that gets us here, with no leftover freedom. We rename the remaining two functions $A \to \Psi$ and $\psi \to \Phi$, and the scalar metric becomes

$$\boxed{ds^2 = a^2(\eta)\left[-(1+2\Psi)d\eta^2 + (1-2\Phi)\delta_{ij}dx^i dx^j\right].} \tag{6.15}$$

The two functions Ψ and Φ are the metric perturbations. Comparing with (6.14), with $B = E = 0$ we get $\Psi = \Psi_B$ and $\Phi = \Phi_B$: the Newtonian gauge potentials *are* the Bardeen gauge-invariant variables. So any equation written in Newtonian gauge in terms of Φ and Ψ is secretly gauge-invariant. The reason this is called Newtonian gauge is that Ψ behaves like the Newtonian gravitational potential (compare with $g_{00} = -(1+2\Phi)$ in (2.66)). If we take a non-relativistic particle and compute its geodesic equation, we will find that it is accelerated by $-\nabla\Psi$, just like in Newtonian gravity. Overdense regions have $\Psi < 0$, potential wells.

The potential Φ has a slightly different interpretation. It perturbs the spatial curvature: the proper volume element is $a^3(1-3\Phi)d^3x$, and the Ricci scalar of the constant-time slices is $\frac{4}{a^2}\nabla^2\Phi$. For light, both potentials matter, and the deflection of light by a mass is proportional to $\Phi + \Psi$, which is the origin of the famous factor of two between the GR and Newtonian predictions for the bending of starlight by the Sun. If there is no anisotropic stress, then the Einstein equations imply

$$\boxed{\Phi = \Psi \qquad \text{when anisotropic stress is negligible.}} \tag{6.16}$$

This is true for a perfect fluid, but it is not exactly true when free-streaming particles like neutrinos are important.

From the metric, we can read off

$$\begin{aligned} g_{00} &= -a^2(1+2\Psi), \\ g_{0i} &= 0, \\ g_{ij} &= a^2(1-2\Phi)\delta_{ij}. \end{aligned} \tag{6.17}$$

The inverse metric is found by demanding $g^{\mu\alpha}g_{\alpha\nu} = \delta^{\mu}{}_{\nu}$. Keeping only first order terms, we get

$$\begin{aligned} g^{00} &= -a^{-2}(1-2\Psi), \\ g^{0i} &= 0, \\ g^{ij} &= a^{-2}(1+2\Phi)\delta^{ij}. \end{aligned} \tag{6.18}$$

The logic is the same as the elementary identity

$$\frac{1}{1+\epsilon} = 1 - \epsilon + O(\epsilon^2).$$

This is what we do over and over again in perturbation theory. We expand, keep the terms linear in perturbations, and throw away anything quadratic.

6.5.1 Some Christoffel Symbols

The Christoffel symbols are defined as usual by

$$\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2}g^{\mu\lambda}\left(\partial_\alpha g_{\beta\lambda} + \partial_\beta g_{\alpha\lambda} - \partial_\lambda g_{\alpha\beta}\right). \tag{6.19}$$

Let us compute one explicitly, because this is where people tend to get lost. Consider Γ^0_{00}. We have

$$\begin{aligned} \Gamma^0_{00} &= \frac{1}{2}g^{0\lambda}\left(\partial_0 g_{0\lambda} + \partial_0 g_{0\lambda} - \partial_\lambda g_{00}\right) \\ &= \frac{1}{2}g^{00}\partial_0 g_{00} \\ &= \frac{1}{2}\left[-a^{-2}(1-2\Psi)\right]\left[-2aa'(1+2\Psi) - 2a^2\Psi'\right] \\ &= \frac{a'}{a} + \Psi' \\ &= \mathcal{H} + \Psi'. \end{aligned} \tag{6.20}$$

In the second-to-last line, every term like $\Psi\Psi'$ or Ψ^2 was dropped. We only keep first order terms.

The Christoffel symbols that we will need most often are

$$\boxed{\begin{aligned}
\Gamma^0_{00} &= \mathcal{H} + \Psi', \\
\Gamma^0_{0i} &= \partial_i \Psi, \\
\Gamma^0_{ij} &= \left[\mathcal{H}(1 - 2\Phi - 2\Psi) - \Phi'\right] \delta_{ij}, \\
\Gamma^i_{00} &= \partial^i \Psi, \\
\Gamma^i_{0j} &= (\mathcal{H} - \Phi')\delta^i{}_j, \\
\Gamma^i_{jk} &= -\partial_j \Phi \delta^i{}_k - \partial_k \Phi \delta^i{}_j + \partial^i \Phi \delta_{jk}.
\end{aligned}} \tag{6.21}$$

These are not beautiful, but they are manageable. The entire subject is basically bookkeeping.

6.6 Matter Perturbations

Now that we have perturbed the metric, we need to perturb the matter. For a perfect fluid,

$$T^\mu{}_\nu = (\rho + P) U^\mu U_\nu + P \delta^\mu{}_\nu. \tag{6.22}$$

We expand the density and pressure as

$$\begin{aligned}
\rho(\eta, \boldsymbol{x}) &= \bar{\rho}(\eta) + \delta\rho(\eta, \boldsymbol{x}), \\
P(\eta, \boldsymbol{x}) &= \bar{P}(\eta) + \delta P(\eta, \boldsymbol{x}).
\end{aligned} \tag{6.23}$$

It is also useful to define the density contrast

$$\delta \equiv \frac{\delta\rho}{\bar{\rho}}. \tag{6.24}$$

For the velocity, we write

$$U^\mu = \frac{1}{a}(1 - \Psi, v^i),$$

to first order, where $v^i = dx^i/d\eta$ is the peculiar velocity. For scalar perturbations it's a gradient, $v_i = \partial_i v$, and its divergence $\theta \equiv \partial_i v^i$ will be the variable we evolve. This normalization is chosen so that

$$g_{\mu\nu} U^\mu U^\nu = -1$$

to first order in perturbations.

Let us check this quickly:

$$g_{\mu\nu} U^\mu U^\nu = g_{00} (U^0)^2 + g_{ij} U^i U^j$$

$$
\begin{aligned}
&= -a^2(1+2\Psi)\frac{1}{a^2}(1-\Psi)^2 + a^2(1-2\Phi)\delta_{ij}\frac{v^i}{a}\frac{v^j}{a} \\
&= -(1+2\Psi)(1-2\Psi) + O(v^2) \\
&= -1 + O(\Psi^2, v^2).
\end{aligned} \tag{6.25}
$$

Since v^2 is second order, it is dropped. The spatial velocity is itself a first-order perturbation.

The perturbed stress-energy tensor becomes

$$
\boxed{
\begin{aligned}
T^0{}_0 &= -\bar{\rho} - \delta\rho, \\
T^0{}_i &= (\bar{\rho} + \bar{P})v_i, \\
T^i{}_0 &= -(\bar{\rho} + \bar{P})v^i, \\
T^i{}_j &= (\bar{P} + \delta P)\delta^i{}_j + \Pi^i{}_j.
\end{aligned}
} \tag{6.26}
$$

The extra term $\Pi^i{}_j$ is the anisotropic stress, a traceless part of the stress that picks out directions. A perfect fluid has $\Pi^i{}_j = 0$. Photons and neutrinos can have anisotropic stress when they free-stream, because then the momentum distribution at a point is not isotropic. This will matter later.

6.7 Conservation Equations

The equations of motion for matter perturbations follow from

$$\nabla_\mu T^\mu{}_\nu = 0. \tag{6.27}$$

This one equation contains both the continuity equation and the Euler equation. The calculation is long, but the logic is the same as what we did in the non-relativistic section. We expand everything to first order and use the background equations to remove zeroth-order terms.

For a fluid with equation of state

$$w \equiv \frac{\bar{P}}{\bar{\rho}},$$

and sound speed

$$c_s^2 \equiv \frac{\delta P}{\delta\rho},$$

the linearized conservation equations in Newtonian gauge, written in Fourier space (so

$\nabla^2 \to -k^2$, and θ is the Fourier transform of $\partial_i v^i$), are

$$\boxed{\delta' = -(1+w)(\theta - 3\Phi') - 3\mathcal{H}(c_s^2 - w)\delta,} \tag{6.28}$$

where

$$\theta \equiv \partial_i v^i,$$

and

$$\boxed{\theta' = -\mathcal{H}(1-3w)\theta - \frac{w'}{1+w}\theta + \frac{c_s^2}{1+w}k^2\delta + k^2\Psi - k^2\sigma.} \tag{6.29}$$

Here σ is the scalar anisotropic stress. If the fluid is perfect, then $\sigma = 0$.

Let us unpack this. The term θ is the relativistic version of $\nabla \cdot \boldsymbol{v}$. The term $3\Phi'$ appears because the spatial metric is changing. In Newtonian theory, density changes because the fluid flows into and out of a region. In GR, the volume of the region itself is also perturbed by the metric. That is why Φ' appears.

Now consider cold dark matter. For cold dark matter,

$$w = 0, \qquad c_s^2 = 0, \qquad \sigma = 0.$$

The conservation equations reduce to (with θ in Fourier space)

$$\boxed{\delta_c' = -\theta_c + 3\Phi',} \tag{6.30}$$

and

$$\boxed{\theta_c' = -\mathcal{H}\theta_c + k^2\Psi.} \tag{6.31}$$

Inside the Hubble radius, the potential is often slowly varying, so $\Phi' \approx 0$. Then the continuity equation becomes

$$\delta_c' \approx -\theta_c,$$

which is the same structure as the Newtonian continuity equation.

Problem 4.

Using the perturbed stress tensor in Newtonian gauge, show that the energy conservation equation can be written as

$$\delta\rho' = -3\mathcal{H}(\delta\rho + \delta P) + 3\Phi'(\bar{\rho} + \bar{P}) - \partial_i q^i,$$

where

$$q^i \equiv (\bar{\rho} + \bar{P})v^i.$$

Then show that this is equivalent to equation (6.28).

Solution:

Start from

$$\nabla_\mu T^\mu{}_0 = 0.$$

Expanding the covariant derivative gives

$$\nabla_\mu T^\mu{}_0 = \partial_\mu T^\mu{}_0 + \Gamma^\mu_{\mu\lambda} T^\lambda{}_0 - \Gamma^\lambda_{\mu 0} T^\mu{}_\lambda.$$

Now we insert the stress tensor components. To first order,

$$T^0{}_0 = -\bar{\rho} - \delta\rho, \qquad T^i{}_0 = -q^i, \qquad T^i{}_j = (\bar{P} + \delta P)\delta^i{}_j.$$

The partial derivative term gives

$$\begin{aligned}\partial_\mu T^\mu{}_0 &= \partial_0 T^0{}_0 + \partial_i T^i{}_0 \\ &= -\bar{\rho}' - \delta\rho' - \partial_i q^i.\end{aligned}$$

For the Christoffel terms, $\Gamma^\mu_{\mu 0} = \Gamma^0_{00} + \Gamma^i_{i0} = \mathcal{H} + \Psi' + 3(\mathcal{H} - \Phi')$, and the last term only gets contributions from $\lambda = \mu$ (the off-diagonal parts of T are first order and multiply first-order or vanishing Christoffels), $\Gamma^0_{00} T^0{}_0 + \Gamma^i_{i0} T^i{}_i$ summed over i. Putting it together,

$$\Gamma^\mu_{\mu 0} T^0{}_0 - \Gamma^0_{00} T^0{}_0 - \Gamma^i_{i0} T^i{}_i = 3(\mathcal{H} - \Phi')\left(-\bar{\rho} - \delta\rho - \bar{P} - \delta P\right).$$

The zeroth-order part is $-3\mathcal{H}(\bar{\rho} + \bar{P})$, which cancels against $-\bar{\rho}'$ by the background continuity equation. The first-order part leaves

$$-3\mathcal{H}(\delta\rho + \delta P) + 3\Phi'(\bar{\rho} + \bar{P}).$$

Therefore,

$$\delta\rho' = -3\mathcal{H}(\delta\rho + \delta P) + 3\Phi'(\bar{\rho} + \bar{P}) - \partial_i q^i.$$

Now divide by $\bar{\rho}$ and use

$$\delta \equiv \frac{\delta\rho}{\bar{\rho}}, \qquad \delta P = c_s^2 \delta\rho, \qquad \bar{P} = w\bar{\rho},$$

together with

$$\bar{\rho}' = -3\mathcal{H}(1+w)\bar{\rho}.$$

After a little algebra, we get

$$\boxed{\delta' = -(1+w)(\theta - 3\Phi') - 3\mathcal{H}(c_s^2 - w)\delta.}$$

This is the relativistic continuity equation.

6.8 The Linearized Einstein Equations

Now we need the gravitational equations. The full Einstein equation is

$$G^{\mu}{}_{\nu} = 8\pi G_N T^{\mu}{}_{\nu}. \tag{6.32}$$

Perturbing both sides gives

$$\bar{G}^{\mu}{}_{\nu} + \delta G^{\mu}{}_{\nu} = 8\pi G_N \left(\bar{T}^{\mu}{}_{\nu} + \delta T^{\mu}{}_{\nu}\right).$$

The background part gives the Friedmann equations. The first-order part gives

$$\delta G^{\mu}{}_{\nu} = 8\pi G_N \delta T^{\mu}{}_{\nu}. \tag{6.33}$$

In Newtonian gauge, the scalar Einstein equations can be written as (I'll spare you the computation of $\delta G^{\mu}{}_{\nu}$, which is a page of Christoffel symbols, and is a good exercise to do with EinsteinPy)

$$\boxed{\nabla^2\Phi - 3\mathcal{H}(\Phi' + \mathcal{H}\Psi) = 4\pi G_N a^2 \delta\rho.} \tag{6.34}$$

This is the relativistic version of the Poisson equation. Notice the extra term involving $\mathcal{H}(\Phi' + \mathcal{H}\Psi)$. Newtonian gravity only sees the spatial Laplacian $\nabla^2\Phi$. General relativity also knows that spacetime is expanding and that the potential can evolve in time.

The $0i$ equation is the momentum constraint,

$$\boxed{\partial_i(\Phi' + \mathcal{H}\Psi) = -4\pi G_N a^2(\bar{\rho} + \bar{P})v_i.} \tag{6.35}$$

The pressure equation is

$$\boxed{\Phi'' + \mathcal{H}(\Psi' + 2\Phi') + (2\mathcal{H}' + \mathcal{H}^2)\Psi - \frac{1}{3}\nabla^2(\Phi - \Psi) = 4\pi G_N a^2 \delta P.} \tag{6.36}$$

Finally, the traceless spatial equation is

$$\boxed{\left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right)(\Phi - \Psi) = 8\pi G_N a^2 \Pi_{ij}.} \tag{6.37}$$

If $\Pi_{ij} = 0$, then $\Phi = \Psi$ up to boundary conditions. This is why the two metric potentials are often identified in simple cosmological calculations.

6.8.1 The Newtonian Limit

The relativistic Poisson equation (6.34) becomes the Newtonian Poisson equation on small scales. To see this, go to Fourier space, where

$$\nabla^2 \to -k^2.$$

Equation (6.34) becomes

$$-k^2\Phi - 3\mathcal{H}(\Phi' + \mathcal{H}\Psi) = 4\pi G_N a^2 \delta\rho.$$

If $k \gg \mathcal{H}$, then the $k^2\Phi$ term dominates over the terms with $\mathcal{H}$. Therefore,

$$\boxed{k^2\Phi \simeq -4\pi G_N a^2 \delta\rho, \qquad k \gg \mathcal{H}.} \tag{6.38}$$

Equivalently, in real space,

$$\boxed{\nabla^2\Phi \simeq 4\pi G_N a^2 \delta\rho.} \tag{6.39}$$

This is exactly the Poisson equation we used in Newtonian perturbation theory. So Newtonian perturbation theory was not wrong. It was the subhorizon limit of the relativistic theory.

Problem 5.

Assume cold dark matter, negligible anisotropic stress, and subhorizon modes $k \gg \mathcal{H}$. Starting from

$$\delta_c' = -\theta_c + 3\Phi', \qquad \theta_c' = -\mathcal{H}\theta_c + k^2\Psi,$$

and the subhorizon Poisson equation

$$k^2\Phi = -4\pi G_N a^2 \bar{\rho}_c \delta_c,$$

show that

$$\delta_c'' + \mathcal{H}\delta_c' - 4\pi G_N a^2 \bar{\rho}_c \delta_c = 0.$$

Solution:
On subhorizon scales, we may neglect Φ' in the continuity equation. Hence

$$\delta_c' = -\theta_c.$$

Taking another derivative gives

$$\delta_c'' = -\theta_c'.$$

Using the Euler equation,

$$\begin{aligned}\delta_c'' &= -\left(-\mathcal{H}\theta_c + k^2\Psi\right) \\ &= \mathcal{H}\theta_c - k^2\Psi.\end{aligned}$$

Since $\theta_c = -\delta_c'$ and $\Psi = \Phi$, this becomes

$$\delta_c'' = -\mathcal{H}\delta_c' - k^2\Phi.$$

Using the Poisson equation,

$$-k^2\Phi = 4\pi G_N a^2 \bar{\rho}_c \delta_c,$$

we get

$$\boxed{\delta_c'' + \mathcal{H}\delta_c' - 4\pi G_N a^2 \bar{\rho}_c \delta_c = 0.}$$

This is the conformal-time version of the Newtonian growth equation (5.31) (with $c_s = 0$), since $\ddot{\delta} = (\delta'' - \mathcal{H}\delta')/a^2$ and $\dot{\delta} = \delta'/a$.

6.9 Evolution Of The Potential

Let us now ask what happens to the gravitational potential itself. Assume a perfect fluid with no anisotropic stress, so that $\Phi = \Psi$. Also assume an adiabatic perturbation, so that

$$\delta P = c_s^2 \delta\rho.$$

Combining the Einstein equations gives an evolution equation for Φ:

$$\boxed{\Phi'' + 3(1 + c_s^2)\mathcal{H}\Phi' + \left[2\mathcal{H}' + (1 + 3c_s^2)\mathcal{H}^2 - c_s^2\nabla^2\right]\Phi = 0.} \tag{6.40}$$

This equation looks a little scary, but it contains a lot of physics.

During matter domination, $w = 0$ and $c_s^2 = 0$. Also $a \propto \eta^2$, so $\mathcal{H} = 2/\eta$. Equation (6.40) becomes

$$\begin{aligned} \Phi'' + 3\mathcal{H}\Phi' &= 0 \\ \Phi'' + \frac{6}{\eta}\Phi' &= 0. \end{aligned} \tag{6.41}$$

The solutions are

$$\boxed{\Phi(\eta) = C_1 + C_2\eta^{-5}.} \tag{6.42}$$

So the gravitational potential is constant during matter domination, plus a rapidly decaying mode. This is why matter domination is so simple: the wells of the gravitational potential are basically frozen.

During radiation domination, $w = c_s^2 = 1/3$, $\mathcal{H} = 1/\eta$ and $2\mathcal{H}' + 2\mathcal{H}^2 = 0$, so in Fourier space

$$\Phi'' + \frac{4}{\eta}\Phi' + \frac{k^2}{3}\Phi = 0. \tag{6.43}$$

With $x \equiv k\eta/\sqrt{3}$, the solution that stays finite as $\eta \to 0$ is

$$\boxed{\Phi(\eta) = 3\Phi_p \frac{\sin x - x\cos x}{x^3}} \tag{6.44}$$

where Φ_p is the primordial value (you can check by plugging in, or by noticing it's $3j_1(x)/x$ with j_1 a spherical Bessel function). Outside the horizon ($x \ll 1$) the potential is constant, $\Phi \approx \Phi_p$. Once the mode enters the horizon, the potential decays like η^{-2} and oscillates (figure 6.1). Physically, radiation pressure resists gravitational collapse, so the potential wells can't remain fixed. And since dark matter only feels gravity through these potentials, this is the relativistic reason that dark matter perturbations only grow logarithmically during radiation domination (the Mészáros effect from the last chapter).

6.10 Conserved Curvature Perturbations

Outside the horizon, gradients are negligible and the physics in different places doesn't communicate. You might expect something to be conserved there, and something is. Using the gauge transformations, we can build two important gauge-invariant curvature perturbations,

$$\boxed{\zeta \equiv -\psi - \mathcal{H}\frac{\delta\rho}{\bar{\rho}'}, \qquad \mathcal{R} \equiv -\psi + \mathcal{H}(v + B)} \tag{6.45}$$

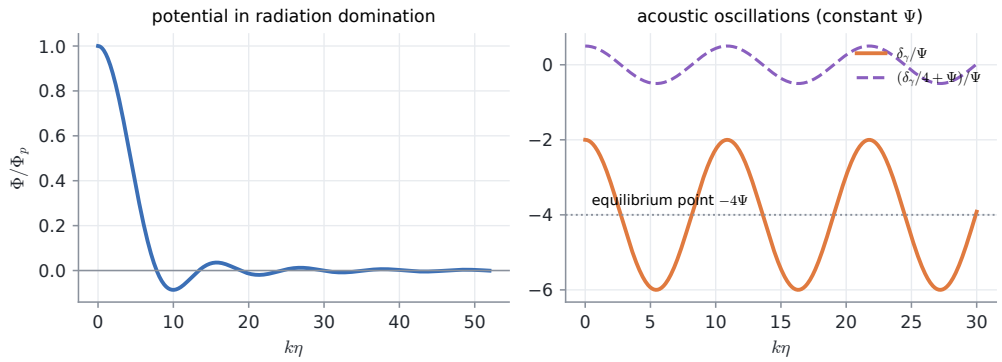


Figure 6.1: Left: the gravitational potential in the radiation era. It's frozen outside the horizon and decays while oscillating once the mode enters. Right: the photon density in a constant potential, which oscillates around the equilibrium point -4Ψ. The effective temperature $\delta_\gamma/4+\Psi$ oscillates around zero.

Using (6.13), $\delta\rho \to \delta\rho - \bar{\rho}'T$ and $\psi \to \psi + \mathcal{H}T$, so the T's cancel in ζ, and $v+B \to v+B+T$, so they cancel in $\mathcal{R}$ too. The variable ζ is (minus) the curvature perturbation ψ on slices of uniform density (where $\delta\rho = 0$), and $\mathcal{R}$ is (minus) the curvature perturbation on **comoving** slices, where the fluid is at rest ($v + B = 0$). In Newtonian gauge, using $\bar{\rho}' = -3\mathcal{H}(\bar{\rho} + \bar{P})$ and the momentum constraint (6.35) to eliminate the velocity,

$$\zeta = -\Phi + \frac{\delta}{3(1+w)}, \qquad \mathcal{R} = -\Phi - \frac{2}{3(1+w)}\left(\frac{\Phi'}{\mathcal{H}} + \Psi\right). \tag{6.46}$$

On superhorizon scales the relativistic Poisson equation (6.34) without the gradient term says $\delta = -\frac{2}{\mathcal{H}}(\Phi' + \mathcal{H}\Psi)$, which makes $\zeta = \mathcal{R}$, so the two agree there.

Now the payoff. Take $\Phi = \Psi$ and a single fluid with constant $w = c_s^2$ (so $2\mathcal{H}' + (1+3w)\mathcal{H}^2 = 0$). Differentiating (6.46) and using the potential equation (6.40), almost everything cancels and we are left with

$$\boxed{\mathcal{R}' = \frac{2c_s^2}{3(1+w)\mathcal{H}} k^2 \Phi} \tag{6.47}$$

For $k \ll \mathcal{H}$, $\mathcal{R}'/(\mathcal{H}\mathcal{R}) \sim (k/\mathcal{H})^2 \to 0$: *the curvature perturbation is conserved outside the horizon.* The same is true for any mixture of fluids, as long as the perturbations are adiabatic (more on that in a second). This is one of the deepest facts in cosmology. It means that once a mode is far outside the Hubble radius, we can forget about everything that happens to the universe (reheating, phase transitions, annihilations) and just carry $\mathcal{R}$ along. Inflation sets the value of $\mathcal{R}$ when the mode exits the horizon, and the hot Big Bang inherits it unchanged until the mode re-enters. That's why we could compute $\Delta^2_{\mathcal{R}}$ in the last chapter and compare it directly with the CMB.

Problem 6.

a). For a constant superhorizon potential in an era with constant w, show that $\mathcal{R} = -\frac{5+3w}{3+3w}\Phi$.

b). A superhorizon mode passes from radiation domination to matter domination. Using the conservation of $\mathcal{R}$, show that the potential drops by a factor of 9/10.

Solution: a). With $\Phi' = 0$ and $\Psi = \Phi$, (6.46) gives $\mathcal{R} = -\Phi - \frac{2}{3(1+w)}\Phi = -\frac{3+3w+2}{3+3w}\Phi = \boxed{-\frac{5+3w}{3+3w}\Phi}$.

b). In radiation domination ($w = 1/3$), $\mathcal{R} = -\frac{6}{4}\Phi_{\rm RD} = -\frac{3}{2}\Phi_{\rm RD}$. In matter domination ($w = 0$), $\mathcal{R} = -\frac{5}{3}\Phi_{\rm MD}$. Since $\mathcal{R}$ is conserved,

$$\Phi_{\rm MD} = \frac{3}{5}\cdot\frac{3}{2}\Phi_{\rm RD} = \boxed{\frac{9}{10}\Phi_{\rm RD}}$$

Thus, even modes outside the horizon remember the radiation-to-matter transition. The potential does not change wildly, but it is reduced by a factor of 9/10. The exact superhorizon solution through equality (found by Kodama and Sasaki) is

$$\Phi(y) = \frac{\Phi_{\rm RD}}{10y^3}\left[16\sqrt{1+y} + 9y^3 + 2y^2 - 8y - 16\right], \qquad y = \frac{a}{a_{\rm eq}},$$

which goes smoothly from $\Phi_{\rm RD}$ to $\frac{9}{10}\Phi_{\rm RD}$ (figure 6.2).

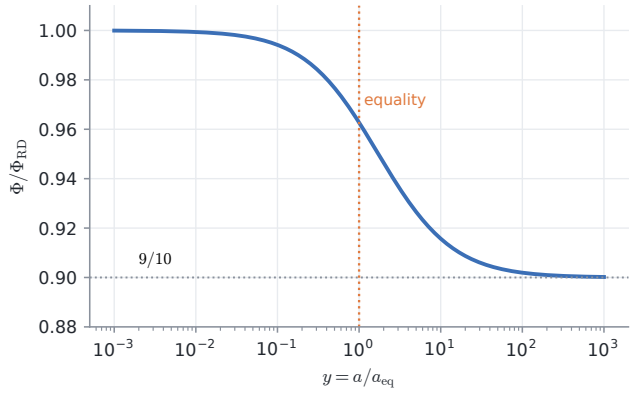


Figure 6.2: The superhorizon potential across matter-radiation equality. It drops smoothly by 10%, which is what conservation of $\mathcal{R}$ demands.

6.11 Initial Conditions

Relativistic perturbation theory is not just about evolution. It is also about initial conditions. Inflation predicts a spectrum of primordial curvature perturbations $\mathcal{R}$. These perturbations are laid down when modes exit the Hubble radius during inflation. Much later, they re-enter the Hubble radius during radiation or matter domination, and before they do, conservation of $\mathcal{R}$ tells us everything. Deep in the radiation era and outside the horizon, $\Phi = -\frac{2}{3}\mathcal{R}$ (Problem 6), and the superhorizon Poisson equation gives $\delta_\gamma = -2\Psi$ for the radiation.

There are two broad classes of initial conditions: adiabatic and isocurvature.

6.11.1 Adiabatic Perturbations

An adiabatic perturbation is one where all species are perturbed in the same local way. More precisely, the relative number densities are not perturbed. For two species a and b, the entropy perturbation is

$$S_{ab} \equiv \frac{\delta_a}{1+w_a} - \frac{\delta_b}{1+w_b}. \tag{6.48}$$

Adiabatic initial conditions mean

$$\boxed{S_{ab} = 0 \qquad \text{for all species } a, b.} \tag{6.49}$$

For matter and radiation, this gives

$$\boxed{\delta_m = \frac{3}{4}\delta_r.} \tag{6.50}$$

The factor 3/4 comes from the fact that matter has $w = 0$ while radiation has $w = 1/3$.

Adiabatic perturbations are the simplest and most important initial conditions. They are what we expect from the simplest models of inflation, where a single scalar field controls the local clock of the universe: every region goes through exactly the same history, just slightly ahead of or behind its neighbours. Shifting a uniform universe in time by δt changes each density by $\delta\rho_a = \dot{\bar{\rho}}_a \delta t = -3H(1+w_a)\bar{\rho}_a \delta t$, so $\delta_a/(1+w_a)$ is the same for every species, which is exactly $S_{ab} = 0$. It is also the condition under which $\mathcal{R}$ is conserved on superhorizon scales.

6.11.2 Isocurvature Perturbations

An isocurvature perturbation is different. It is a perturbation in the relative composition of the universe without an initial perturbation in the total curvature. For example, one could

imagine a region with slightly more dark matter and slightly less radiation in such a way that the total density is initially unchanged. The geometry is initially unperturbed, but the composition is not.

Isocurvature modes are observationally constrained because they produce a different pattern in the CMB (they're out of phase with the adiabatic acoustic oscillations). The CMB says that they are at most a few percent of the total. Their existence would be a clue that more than one field was active in the early universe.

6.11.3 The Primordial Power Spectrum

The most useful statistical object is the power spectrum of the curvature perturbation,

$$\langle \mathcal{R}(\boldsymbol{k})\mathcal{R}^*(\boldsymbol{k}')\rangle = (2\pi)^3\delta_D(\boldsymbol{k}-\boldsymbol{k}')P_{\mathcal{R}}(k). \tag{6.51}$$

The dimensionless power spectrum is

$$\Delta^2_{\mathcal{R}}(k) \equiv \frac{k^3}{2\pi^2}P_{\mathcal{R}}(k). \tag{6.52}$$

Observationally, it is usually parameterized as

$$\boxed{\Delta^2_{\mathcal{R}}(k) = A_s\left(\frac{k}{k_*}\right)^{n_s-1}}. \tag{6.53}$$

Here $A_s \approx 2.1\times 10^{-9}$ is the amplitude, $n_s \approx 0.965$ is the scalar spectral index, and $k_* = 0.05$ Mpc^{-1} is a chosen pivot scale. A perfectly scale-invariant spectrum would have $n_s = 1$. The observed value is close to, but slightly below, one, just as the slow-roll calculation of Chapter 4 predicts. This is also where the density power spectrum of the last chapter comes from. Inside the horizon during matter domination, Poisson's equation says $\delta_m = -\frac{2k^2}{3\mathcal{H}^2}\Phi$ with $\Phi = -\frac{3}{5}\mathcal{R}$, so $P_m \propto k^4 P_{\mathcal{R}} \propto k^{n_s}$, before the transfer function bends it over.

Problem 7.

For matter and radiation, show that adiabatic initial conditions imply

$$\delta_m = \frac{3}{4}\delta_r.$$

Then construct a simple matter-radiation isocurvature perturbation by requiring that the total density perturbation initially vanishes:

$$\delta\rho_m + \delta\rho_r = 0.$$

Solution:
The entropy perturbation between matter and radiation is

$$S_{mr} = \frac{\delta_m}{1+w_m} - \frac{\delta_r}{1+w_r}.$$

For matter $w_m = 0$, and for radiation $w_r = 1/3$. Therefore,

$$S_{mr} = \delta_m - \frac{3}{4}\delta_r.$$

Adiabaticity means $S_{mr} = 0$, so

$$\boxed{\delta_m = \frac{3}{4}\delta_r.}$$

For an isocurvature perturbation, we can instead demand that the total density perturbation vanishes:

$$\bar{\rho}_m\delta_m + \bar{\rho}_r\delta_r = 0.$$

Solving for δ_r gives

$$\delta_r = -\frac{\bar{\rho}_m}{\bar{\rho}_r}\delta_m.$$

This has no initial total density perturbation, but the relative composition is perturbed. That is why it is called isocurvature.

Problem 8.

(Neutrinos and the difference between Φ and Ψ.) This problem is an adapted version of the standard superhorizon initial-condition calculation with neutrino anisotropic stress. During the radiation era, let

$$f_\nu \equiv \frac{\bar{\rho}_\nu}{\bar{\rho}_\gamma + \bar{\rho}_\nu}.$$

For adiabatic initial conditions on superhorizon scales, assume the growing mode where Φ is constant. Free-streaming neutrinos produce anisotropic stress, so Φ and Ψ are no longer exactly equal. Show that the potentials are related by

$$\Psi = \frac{\Phi}{1 + \frac{2}{5} f_\nu}.$$

Then show that the photon density perturbation satisfies

$$\delta_\gamma = -2\Psi.$$

Solution:
If neutrinos behaved like a perfect fluid, then their anisotropic stress would vanish and the traceless Einstein equation would give $\Phi = \Psi$. Free-streaming neutrinos are different. They move fast enough to develop directional stress, and this stress appears in the traceless spatial Einstein equation. Schematically,

$$k^2(\Phi - \Psi) \propto a^2(\bar{\rho}_\nu + \bar{P}_\nu)\Pi_\nu.$$

On superhorizon scales, $k\eta \ll 1$, the neutrino anisotropic stress is suppressed by powers of $k\eta$, but the equation also has the explicit k^2 on the left. The ratio remains finite. Solving the leading-order Einstein-Boltzmann system gives

$$\Phi - \Psi = \frac{2}{5} f_\nu \Psi.$$

Rearranging,

$$\Phi = \left(1 + \frac{2}{5} f_\nu\right) \Psi,$$
$$\Psi = \frac{\Phi}{1 + \frac{2}{5} f_\nu}.$$

This is the desired relation.
For the density perturbation, use the superhorizon 00 Einstein equation in the radiation era. Since the growing mode has constant potentials, $\Phi' = \Psi' = 0$, and the gradient term is negligible, $-3\mathcal{H}^2\Psi = 4\pi G a^2 \delta\rho = \frac{3}{2}\mathcal{H}^2\delta$. This gives the standard radiation-era relation

$$\delta_\gamma = \delta_\nu = -2\Psi.$$

The main lesson is not the numerical factor. The main lesson is that relativistic species with anisotropic stress change the relation between the two metric potentials. This is a purely relativistic effect with no Newtonian analogue.

6.12 Photon And Baryon Perturbations

Before recombination, photons and baryons were tightly coupled by Thomson scattering. Photons scatter off free electrons, and the electrons are electromagnetically coupled to the baryons. So the photons and baryons behave approximately like a single fluid. This is one of the most important physical systems in cosmology.

The photons provide pressure. The baryons provide extra inertia. Gravity tries to compress the fluid. Photon pressure pushes back. The result is a sound wave. These sound waves are the origin of the acoustic peaks in the CMB and the baryon acoustic oscillation feature in the matter distribution.

For photons alone, $w = 1/3$ and $c_s^2 = 1/3$. Ignoring anisotropic stress, the continuity and Euler equations are

$$\delta_\gamma' = -\frac{4}{3}\theta_\gamma + 4\Phi', \tag{6.54}$$

and

$$\theta_\gamma' = \frac{k^2}{4}\delta_\gamma + k^2\Psi. \tag{6.55}$$

Combining them gives

$$\begin{aligned}\delta_\gamma'' &= -\frac{4}{3}\theta_\gamma' + 4\Phi'' \\ &= -\frac{4}{3}\left(\frac{k^2}{4}\delta_\gamma + k^2\Psi\right) + 4\Phi'' \\ &= -\frac{k^2}{3}\delta_\gamma - \frac{4}{3}k^2\Psi + 4\Phi''.\end{aligned} \tag{6.56}$$

Therefore,

$$\boxed{\delta_\gamma'' + \frac{k^2}{3}\delta_\gamma = -\frac{4}{3}k^2\Psi + 4\Phi''.} \tag{6.57}$$

This is a driven harmonic oscillator. If the gravitational potentials were zero, the photon perturbations would simply oscillate with frequency $k/\sqrt{3}$.

When baryons are included, the oscillator changes because baryons add mass to the fluid but

do not add much pressure. Define (not to be confused with the curvature perturbation $\mathcal{R}$)

$$R_b \equiv \frac{3\bar{\rho}_b}{4\bar{\rho}_\gamma} \propto a, \tag{6.58}$$

which is about 0.6 at recombination. The sound speed of the photon-baryon fluid is

$$\boxed{c_s^2 = \frac{\delta P_\gamma}{\delta\rho_\gamma + \delta\rho_b} = \frac{1}{3(1+R_b)}} \tag{6.59}$$

(using $\delta_b = \frac{3}{4}\delta_\gamma$ for adiabatic perturbations). The full oscillator equation can be written as

$$\boxed{\delta_\gamma'' + \frac{R_b'}{1+R_b}\delta_\gamma' - \frac{1}{3(1+R_b)}\nabla^2\delta_\gamma = \frac{4}{3}\nabla^2\Psi + 4\Phi'' + \frac{4R_b'}{1+R_b}\Phi'.} \tag{6.60}$$

This equation is one of the crown jewels of cosmological perturbation theory. It is the mathematical origin of the statement: the early universe was ringing. The baryons do two things. They slow the sound waves down, and they shift the equilibrium point of the oscillator deeper into the potential wells, so compressions (odd peaks) get enhanced relative to rarefactions (even peaks). That's how the relative heights of the CMB peaks measure the baryon density.

Problem 9.

Ignore baryons and assume the gravitational potentials are constant. Starting from

$$\delta_\gamma'' + \frac{k^2}{3}\delta_\gamma = -\frac{4}{3}k^2\Psi,$$

solve for $\delta_\gamma(\eta)$. Interpret the result physically.

Solution:

The equation is a forced harmonic oscillator:

$$\delta_\gamma'' + \omega^2\delta_\gamma = -4\omega^2\Psi,$$

where

$$\omega \equiv \frac{k}{\sqrt{3}}.$$

A constant particular solution is

$$\delta_{\gamma,p} = -4\Psi.$$

The homogeneous solution is

$$\delta_{\gamma,h} = A\cos(k\eta/\sqrt{3}) + B\sin(k\eta/\sqrt{3}).$$

Therefore,

$$\boxed{\delta_\gamma(\eta) = -4\Psi + A\cos\left(\frac{k\eta}{\sqrt{3}}\right) + B\sin\left(\frac{k\eta}{\sqrt{3}}\right).}$$

For adiabatic initial conditions, the mode starts outside the horizon with $\delta_\gamma = -2\Psi$ and $\delta'_\gamma = 0$, which fixes $A = 2\Psi$ and $B = 0$:

$$\delta_\gamma(\eta) = 2\Psi\cos\left(\frac{k\eta}{\sqrt{3}}\right) - 4\Psi.$$

(Strictly, the potential changes at equality, and in the matter era the starting point becomes $\delta_\gamma = -\frac{8}{3}\Psi$, but the idea is the same.) The photon perturbation oscillates because gravity compresses the fluid and radiation pressure pushes it back. What a CMB photon actually carries to us is its temperature perturbation $\Theta = \delta_\gamma/4$, plus the redshift Ψ it picks up climbing out of the potential well at last scattering, so the observed temperature fluctuation is $\Theta + \Psi = \frac{\Psi}{2}\cos(k\eta_*/\sqrt{3})$ (figure 6.1, right). Modes caught at an extremum when the photons decoupled, $kr_s = n\pi$ with sound horizon $r_s = \eta_*/\sqrt{3}$, give the largest fluctuations: these are the **acoustic peaks** of the CMB. On the largest scales, which hadn't entered the horizon yet, the matter-era version gives $\Theta + \Psi = \Psi/3$, the **Sachs-Wolfe effect**.

6.13 Matter Perturbations Revisited

We can now understand the evolution of matter perturbations more honestly. Dark matter starts clustering early because it does not feel photon pressure. Baryons, however, are tied to the photons before recombination. They participate in acoustic oscillations instead of freely falling into dark matter wells.

After recombination, photons decouple and free stream. The baryons lose their pressure support and begin falling into the dark matter potential wells. This is why dark matter matters so much. Without dark matter, baryons would only start growing efficiently after

recombination, which is far too late to form the structures we observe today.

The final matter power spectrum therefore carries the memory of several events:

(1) modes entering the horizon during radiation domination grow slowly;
(2) modes entering during matter domination grow more efficiently;
(3) baryons oscillate with photons before recombination;
(4) after recombination, baryons fall into the dark matter wells;
(5) dark energy suppresses growth at late times.

This is why the matter power spectrum is not a simple power law, even if the primordial spectrum nearly is.

6.14 Gravitational Waves

We now turn to tensor perturbations. These are gravitational waves. The metric is

$$ds^2 = a^2(\eta)\left[-d\eta^2 + (\delta_{ij} + h_{ij})dx^i dx^j\right], \tag{6.61}$$

where h_{ij} is transverse and traceless:

$$\boxed{\partial_i h_{ij} = 0, \qquad h^i{}_i = 0.} \tag{6.62}$$

These two conditions remove the unphysical parts of the tensor perturbation. What remains are the two physical polarizations of a gravitational wave.

The perturbed Einstein equation gives

$$\boxed{h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = 16\pi G_N a^2 \hat{\Pi}_{ij}.} \tag{6.63}$$

Here $\hat{\Pi}_{ij}$ is the transverse-traceless part of the anisotropic stress. If there is no anisotropic stress, then

$$\boxed{h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = 0.} \tag{6.64}$$

In Fourier space, this becomes

$$\boxed{h''_k + 2\mathcal{H}h'_k + k^2 h_k = 0.} \tag{6.65}$$

This looks like a damped harmonic oscillator. The damping comes from the expansion of the universe.

There are two important limits. Outside the horizon, $k \ll \mathcal{H}$, the gradient term is small

and the gravitational wave is approximately frozen. Inside the horizon, $k \gg \mathcal{H}$, the wave oscillates and its amplitude decays due to expansion like $1/a$, just like any other radiation (its energy density $\propto h'^2/a^2 \propto a^{-4}$). In radiation domination, with $\mathcal{H} = 1/\eta$, the substitution $h_k = u_k/\eta$ turns the equation into $u_k'' + k^2 u_k = 0$, so the solution that is regular at $\eta = 0$ is exactly

$$h_k(\eta) = h_k^{\rm prim} \frac{\sin(k\eta)}{k\eta}, \tag{6.66}$$

frozen at $h_k^{\rm prim}$ (set by inflation) until $k\eta \sim 1$ and then decaying like $1/\eta \propto 1/a$ (figure 6.3).

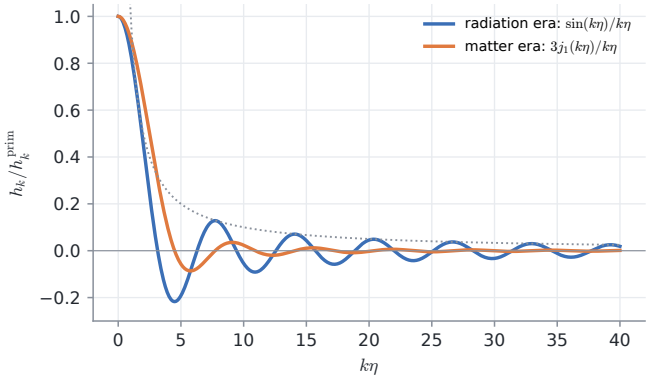


Figure 6.3: Primordial gravitational waves stay frozen outside the horizon and decay as damped oscillations once they enter. The dotted line is the $1/(k\eta)$ envelope of the radiation era solution.

Problem 10.

Consider the tensor-perturbed metric

$$ds^2 = a^2(\eta)\left[-d\eta^2 + (\delta_{ij} + h_{ij})dx^i dx^j\right],$$

where $\partial_i h_{ij} = 0$ and $h^i{}_i = 0$. Suppose the transverse-traceless anisotropic stress vanishes. Show that the tensor perturbation obeys

$$h_{ij}'' + 2\mathcal{H}h_{ij}' - \nabla^2 h_{ij} = 0.$$

Then solve this equation outside the horizon.

Solution:

The tensor part of the linearized Einstein equation gives

$$\delta G^i{}_j\big|_{\rm TT} = 8\pi G_N \delta T^i{}_j\big|_{\rm TT}.$$

If the transverse-traceless stress vanishes, then the right-hand side is zero. The left-hand side, computed from the tensor-perturbed metric (Christoffels, then the Ricci tensor, keeping only the transverse traceless part), is

$$\delta G^i{}_j = \frac{1}{2a^2}\left(h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij}\right),$$

so the equation of motion is

$$\boxed{h''_{ij} + 2\mathcal{H}h'_{ij} - \nabla^2 h_{ij} = 0.}$$

In Fourier space,

$$h''_k + 2\mathcal{H}h'_k + k^2 h_k = 0.$$

Outside the horizon, $k \ll \mathcal{H}$, so we drop the $k^2 h_k$ term:

$$h''_k + 2\mathcal{H}h'_k = 0.$$

This can be written as

$$(a^2 h'_k)' = 0.$$

Therefore,

$$h'_k = \frac{C}{a^2}.$$

Integrating,

$$h_k(\eta) = C_1 + C_2 \int^\eta \frac{d\tilde{\eta}}{a^2(\tilde{\eta})}.$$

The second term decays in an expanding universe (for example, in radiation domination $a \propto \eta$ and it's $\propto 1/\eta$), so

$$\boxed{h_k \simeq \text{constant}, \qquad k \ll \mathcal{H}.}$$

Superhorizon gravitational waves freeze, just like curvature perturbations.

6.15 Putting The Story Together

Let us end by stepping back. The Newtonian calculation told us that density perturbations can grow because gravity amplifies overdensities. Relativistic perturbation theory tells us what that statement really means in an expanding spacetime.

The essential objects are:

- Ψ, the perturbation to time-time part of the metric, which behaves like the Newtonian gravitational potential.
- Φ, the perturbation to the spatial curvature.
- δ, the density contrast.
- θ, the velocity divergence.
- ζ and $\mathcal{R}$, the curvature perturbations that are conserved on superhorizon scales for adiabatic modes.
- h_{ij}, the tensor perturbation describing gravitational waves.

The conservation equations tell matter how to move in the perturbed geometry. The Einstein equations tell the geometry how to respond to matter. Together, they form the coupled system of cosmological perturbation theory.

This is the machinery that lets us connect the early universe to observations. Inflation gives us an initial spectrum of curvature perturbations. Relativistic perturbation theory evolves those perturbations through radiation domination, matter domination, recombination, and eventually to the structures we observe today. The CMB is therefore not just a pretty picture. It is a map of the metric perturbations of the early universe, and the fact that a theory with half a dozen parameters (the densities of baryons, dark matter and dark energy, the amplitude and tilt of the primordial spectrum, and when the universe got reionized) fits thousands of measured numbers is, to me, the most remarkable thing in all of cosmology. There's plenty left to understand, starting with what the dark matter and dark energy actually are, but we now have the tools to ask those questions properly.

6.16 Problems

1. Tensors Are Gauge Invariant

Consider a transverse-traceless perturbation h_{ij} of the spatial metric.

a). Under a gauge transformation with $\xi^\mu = (T, \xi^i)$, show that δg_{ij} shifts by $-a^2(\partial_i \xi_j + \partial_j \xi_i + 2\mathcal{H} T \delta_{ij})$. Decompose $\xi^i = \partial^i L + L^i_\perp$ with $\partial_i L^i_\perp = 0$.

b). Show that neither T, L nor $L^i_\perp$ can generate a transverse traceless contribution. Conclude that h_{ij} is gauge-invariant on its own, without needing to form any combination.

2. Gravitational Waves In The Matter Era

a). Show that during matter domination ($\mathcal{H} = 2/\eta$), the tensor equation $h_k'' + 2\mathcal{H}h_k' + k^2 h_k = 0$ has the regular solution $h_k = 3h_k^{\rm prim}\, j_1(k\eta)/(k\eta)$, where $j_1(x) = (\sin x - x\cos x)/x^2$.

b). At what frequency today would we observe a gravitational wave that re-entered the Hubble radius at matter-radiation equality? (Use $k_{\rm eq} \approx 0.01$ Mpc^{-1} and $f = ck/2\pi$.) Why do CMB B-mode experiments probe such long wavelengths, while LIGO probes very short ones?

3. Synchronous Gauge

Instead of Newtonian gauge, one can choose $A = B = 0$, so that the coordinates are attached to freely falling observers (this is what the CMB code CAMB uses). Using (6.13), show that this does not fix the gauge completely: there is a residual freedom $T = C(\boldsymbol{x})/a$ with C an arbitrary function of space. What does this residual freedom correspond to physically?

4. The Integrated Sachs-Wolfe Effect

A CMB photon travelling through a time-dependent potential picks up an extra temperature shift $\Delta\Theta = \int (\Phi' + \Psi')d\eta$ along its path. Explain why this vanishes during matter domination, and why it does not vanish at late times when dark energy dominates (use the growth equation of the last chapter and Poisson's equation, $\Phi \propto D_+(a)/a$ on subhorizon scales). This late-time effect is one of the ways the CMB detects dark energy.

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